



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

COMMON SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

I/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

GROUP – B (ECE, EEE & Mechanical) Branches

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1101	English *	4	3	1	--	4	30	70	100
B16 ENG 1102	Mathematics-I *	4	3	1	--	4	30	70	100
B16 ENG 1103	Mathematics-II *	4	3	1	--	4	30	70	100
B16 ENG 1105	Physics	4	3	1	--	4	30	70	100
B16 ENG 1107	Engineering Graphics	4	2	--	3	5	30	70	100
B16 ENG 1109	Professional Ethics & Moral Values	2	2	--	--	2	30	70	100
B16 ENG 1111	Physics Lab	2	--	--	3	3	50	50	100
B16 ENG 1113	Workshop	2	--	--	3	3	50	50	100
B16 ENG 1114	NCC/NSS (Audit)	--	--	--	--	3	--	---	--
Total		26	16	4	9	32	280	520	800

Note: Add-On Course: Technical English *

* Common to both Group - A and Group - B

ENGLISH
(Common to All Branches)

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

COURSE OBJECTIVES:

Reading Skills

1. Addressing explicit and implicit meanings of a text on current topics.
2. Understanding the context.
3. Learning new words and phrases.
4. Using words and phrases in different contexts.

Writing Skills

1. Using the basic structure of a sentence.
2. Applying relevant writing formats to create paragraphs, essays, letters, emails, reports and presentations.
3. Retaining a logical flow while writing.
4. Planning and executing an assignment creatively.

Interactive Skills

1. Analyzing a topic of discussion and relating to it.
2. Participating in discussions and influencing them.
3. Communicating ideas effectively.
4. Presenting ideas coherently within a stipulated time.

Life Skills and Core Skills

1. Examining self-attributes and identifying areas that require improvement: self-diagnosis and self-motivation.
2. Adapting to a given situation and developing a functional approach to finding solutions: adaptability and problem solving.
3. Understanding the importance of helping others: community services and enthusiasm.

COURSE OUTCOMES:

1. The overall performance of the students will be enhanced after the course; they will be in a position to make presentations on topics of current interests – politics, famous personalities, science and technology, tourism, work and business environment, with increased public speaking skills.
2. Students will be able to read, listen, speak and write effectively in both academic and non-academic environment.
3. The students will be updated with certain real life situations, which they can handle when come face to face.

SYLLABUS

Listening Skills

Conversations: Life in a Hostel – Eating Away those Blues – Meeting Carl Jung – A Documentary on the Big Cat – A Consultant Interviewing Employees – A Conversation about a Business Idea.

Speaking Skills

Your Favorite Holiday Destination – Describe Yourself – Why we need to save our Tiger – A Dialogue – Your First Interview – Pair Work: Setting up a New Business.

Reading Skills

Reading Comprehension: Famous People – What is Personality, Personality based on Blood Groups – News Report, Magazine Article, Mobile Towers and Health – An Excerpt from a Short Story, An Excerpt from a Biography – Open Letter to Prime Minister, Business Dilemmas: An Email Exchange – A Review of IPL: The Inside Story, Mark Zuckerberg: World's Youngest Billionaire.

Writing Skills

Letter Writing, Essay Writing, Email Writing, Report Writing, Paragraph Writing, Drafting a Pamphlet, Argument Writing, Dialogue Writing.

Grammar

Types of Sentences, Articles, Prepositions, Gerunds and Infinitives, Conjunctions, Tense, Quantifiers, Punctuations, Correction of Sentences, Fill-in the Blanks.

Vocabulary

Synonyms, Antonyms, Idioms, One Word Substitution.

Life Skills and Core Skills

Self Awareness and Self Motivation – Communication, Adaptability – Motivation, Problem Solving – Personal Presentation Skills, Stress Management – Professionalism – Ethics.

TEXT BOOK:

1. Life Through Language: A Holistic Approach to Language Learning. Board of Editors, Pearson Publishers, India 2013.

REFERENCE BOOKS:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008
3. Know Your English (Volume 1 & 2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Maathukutty Monipally. Tata Mc Grahill P. 2009.

MATHEMATICS – I
(Common to All Branches)**Theory : 3 Periods**
Tutorial : 1 Period
Exam : 3 Hrs.**Sessionals : 30**
Ext. Marks : 70
Credits : 4

COURSE OBJECTIVES:

Students learn

1. How to find partial derivatives , Jacobians, how to change from one set of variables to another set of variables
2. Know Geometrical interpretation of partial derivatives
3. Learn applications to errors and approximations, maxima and minima
4. How to form ordinary differential equations (ODE) and solve equations of first order and first degree.
5. Applications of first order ODE to orthogonal trajectories, simple electrical circuits, Newton's law of cooling, Law of natural growth and decay
6. How to solve higher order linear differential equations with constant coefficients, Cauchy's linear equation, Legendre's linear equation and simultaneous first order ODEs
7. How to obtain Fourier Series for a periodic function of period 2π and Period $2C$ (C is a constant)
8. How to obtain Half range Cosine & Sine series and also Parseval's formula

COURSE OUTCOMES:

Students will be able to

1. Find partial derivatives, expand a function of more than one variable in a Taylor series and utilize them for errors and approximations, maxima and minima.
2. Solve a first order ODE and also find orthogonal trajectories and solve problems related to simple applications.
3. Solve a given higher order ODE, an equation with constant coefficients, a Cauchy's equation or a Legendre's equation.
4. Utilize knowledge of Fourier series for solving partial differential equations and also in understanding courses like Signals & Systems

SYLLABUS

Partial Differentiation

Functions of two or more variables – Partial derivatives – Homogeneous Functions – Euler’s Theorem – Total Derivative – Change of Variables – Jacobians – Geometrical Interpretation: Tangent Plane and Normal to a Surface.

Applications of Partial Differentiation

Taylor’s Theorem for functions of two variables – Errors and Approximations – Total Differential – Maxima and Minima of functions of two variables – Lagrange’s Method of Undetermined Multipliers – Differentiation Under the Integral Sign – Liebnitz’s Rules.

Ordinary Differential Equations of First Order and First Degree

Formation of ordinary differential equations (ODEs) – Solution of an ordinary differential equation – Equations of the First Order and First Degree – Linear Differential Equation – Bernoulli’s Equation – Exact Differential Equations – Equations Reducible to exact equations.

Applications of Differential Equations of First Order

Orthogonal Trajectories – Simple electric (LR & CR) Circuits – Newton’s Law of Cooling – Law of Natural growth and decay.

Linear Differential Equations of Higher Order

Solutions of Linear Ordinary Differential Equations With Constant Coefficients – Rules for finding Complimentary Function – Rules for finding the particular integral – Method of variation of parameters – Cauchy’s linear equation – Legendre’s Linear Equation – Simultaneous linear equations.

Fourier series

Introduction - Euler’s Formulae - Conditions for a Fourier Expansion - Functions having points of discontinuity - Change of Interval - Odd and Even Functions - Expansions of Odd or Even Periodic Functions, Half-Range Series - Parseval’s Formula.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

MATHEMATICS – II
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. The concept of rank, Normal form of a matrix, consistency of a system of linear algebraic equations
2. Eigen values and Eigen vectors of a matrix, the Cayley Hamilton theorem
3. Orthogonal reduction of a quadratic form.
4. Complex matrices and their properties
5. Laplace transform, existence, properties.
6. Inverse Laplace transform, convolution theorem, how to solve Ordinary Differential Equations (ODE) and simultaneous ODE
7. Difference equations and their methods of solution.
8. Z-transform, its properties, important results and its use to solve difference equations.

COURSE OUTCOMES:

Students will be capable of

1. Utilizing the knowledge of matrices for solving linear simultaneous equations, find Eigen values and Eigen vectors and handle quadratic forms
2. Utilizing the knowledge of Laplace Transforms to find transforms of important functions that arise in applications and also solve ODE
3. Utilizing the knowledge of Laplace Transforms in courses like Net Works, Signals & Systems and Control Systems
4. Utilizing the knowledge of difference equations and Z-transforms in understanding courses like Discrete Mathematical Structures and also Signals & Systems.

SYLLABUS

Matrices – I

Rank of a matrix - Normal Form – Solutions of Linear System of Equations- Consistency of Linear System of Equations – Rouche's Theorem(statement) - Direct Methods: Gauss Elimination Method, LU Factorization Method – Eigen Values and Eigen Vectors of a Matrix – Properties - Cayley – Hamilton Theorem – Inverse and Powers of a Matrix using Cayley – Hamilton Theorem.

Matrices – II

Diagonalization of a Matrix – Quadratic Forms – Reduction of Quadratic Form to Canonical Form – Nature of a Quadratic Form – Complex Matrices: Hermitian, Skew-Hermitian and Unitary Matrices and their Properties.

Laplace Transforms-I

Introduction – Existence Conditions – Transforms of Elementary Functions – Properties of Laplace Transforms – Transforms of Derivatives – Transforms of Integrals – Multiplication by 't' – Division by t – Evaluation of Integrals by Laplace Transforms – Laplace Transforms of Unit Step Function, Unit Impulse Function and Periodic Functions.

Laplace Transforms-II

Inverse Laplace Transform – different methods - Convolution Theorem – Applications of Laplace Transforms to Ordinary Differential Equations, Simultaneous Linear Differential Equations with Constant Coefficients.

Difference Equations

Definition - order and solution of a difference equation - Formation of difference equations - Linear difference equations - Rules for finding C.F. - Rules for finding P.I.- Simultaneous difference equations with constant coefficients. Application to deflection of a loaded string.

Z-transforms

Z-transforms – definition - some standard Z-transforms - Linear property, Damping rule - some standard results - shifting rules - initial and final value theorems - convolution theorem - Evaluation of inverse transforms - Applications of Z-transform to difference equation.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company

PHYSICS
(Common to ECE, EEE & Mechanical)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To understand the basic and advanced concepts of classical and modern physics – Identify, analyze, formulate and solve a wide variety of real world problems.
2. To develop scientific reasoning and problem solving methods.
3. To bring awareness of latest technologies (physics based) – lasers and fiber optics.

COURSE OUTCOMES:

1. Students learn in depth about the topics of Lasers, fiber optics, quantum mechanical theory and classical theories of thermodynamics and electromagnetism.
2. Students understand the classical and modern concepts.

SYLLABUS**Thermodynamics**

Introduction, Heat and Work, First Law of Thermodynamics and applications, Reversible and Irreversible Process, Carnot Cycle and Efficiency, Second Law of Thermodynamics, Carnot's Theorem, Entropy, Second Law in terms of entropy, Entropy and disorder, Third Law of Thermodynamics (Statement Only).

Electromagnetism

Effect of Magnetic Field on – moving charges, current in long straight wire and rectangular Current Loop, Hall Effect, Biot-Savart's Law, B near a Long Wire, B for a Circular Current Loop, Ampere's Law, B for a Solenoid, Faraday's Law of electromagnetic induction, Lenz's law, Inductance of a solenoid, L-R Circuit, Displacement Current, Maxwell's Equations (integral form) and their significance (without derivation).

Interference

Principle of Super Position – Young's Experiment – Coherence – Inference in thin transparent films, Newton's Rings, Michelson Interferometer and its applications.

Lasers

Introduction, spontaneous and stimulated emissions, requirements of laser device, Ruby Laser, Gas Laser (He-Ne Laser), Semiconductor diode Laser, Characteristics and applications of Lasers.

Optical Fibers

Introduction, Principles of light propagation in optical fiber, Acceptance angle, Numerical aperture, types of fiber, Applications of Optical Fibers, Optical Fiber in Communications, advantages.

Ultrasonics

Definition, Production of Ultrasonics by Magnetostriction and Piezoelectric methods, detection methods, acoustic grating, characteristics and applications of Ultrasonics.

Modern Physics

Introduction, de Broglie concept of matter waves, Properties of matter waves, experimental verification (Davisson - Germer experiment), Heisenberg uncertainty principle, Wave function and its physical significance, Schrodinger time independent wave equation, application to a particle in a box, Band theory of Solids, Kronig - Penney model (qualitative treatment), Origin of energy band formation in solids, Classification of materials into conductors, semi conductors and insulators .

Nanophase Materials

Definition, Synthesis – Synthesis methods, Condensation and Ball milling, Chemical vapour deposition method – sol-gel methods, Characterisation, analysis and applications of nano materials.

TEXT BOOKS

1. Physics by David Halliday and Robert Resnick – Part I and Part II, Wiley Eastern India(pvt.) Ltd.
2. Engineering Physics by M.N. Avadhanulu & P.G. Kshirasagar; S. Chand & Company Ltd

REFERENCE BOOKS

1. Modern Engineering Physics by A.S. Vasudeva, S. Chand & Company Ltd
2. University Physics by Young and Freedman, Addison-Wesley
3. Engineering Physics by R.K. Gaur and S.L. Gupta, Dhanpat Rai & CO

ENGINEERING GRAPHICS
(Common to ECE, EEE & Mechanical)

Theory	: 2 Periods	Sessionals	: 30
Lab	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

COURSE OUTCOMES:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent sectional views of solids.
5. Develop the surfaces of regular solids.
6. Draw the isometric views of solids and combination of solids.

SYLLABUS**Introduction**

Lines, Lettering and Dimensioning. Geometrical Constructions.

Curves

Conic sections: General construction of ellipse, parabola and hyperbola. Construction of involutes. Normal and Tangent.

Projections of Points

Principal or Reference Planes, Projections of a point situated in any one of the four quadrants

Projections of Straight Lines

Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of straight line inclined to both the reference planes:

Projections of Planes

Projection of Perpendicular planes: Perpendicular to both reference planes, perpendicular to one reference plane and parallel to other reference plane, perpendicular to one reference plane and inclined to other reference plane. Projection of Oblique planes. Introduction to Auxiliary Planes.

Projections of Solids

Types of solids: Polyhedra and Solids of revolution. Projection of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Projections of Section of Solids

Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone .

Development of Surfaces

Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Isometric Views

Introduction to Isometric projection, Isometric scale and Isometric view. Isometric views of simple planes. Isometric view of Prisms, Pyramids, cylinder and cone. Isometric view of a combination of solids.

TEXT BOOK:

1. Elementary Engineering Drawing by N.D.Bhatt, Charotar Publishing House.

REFERENCE BOOK:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah, Tata Mc-Graw Hill.

PROFESSIONAL ETHICS AND MORAL VALUES

(Common to ECE, EEE & Mechanical)

Theory : 2 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

COURSE OUTCOME:

1. By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS

Ethics and Human Values

Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

Engineering Ethics

Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics, Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

Engineering as Social Experimentation

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders, Role of engineers in promoting ethical climate, balanced outlook on law.

Safety Social Responsibility and Rights

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

Global Issues

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

TEXT BOOKS:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

REFERENCE BOOK:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

PHYSICS LAB
(Common to ECE, EEE & Mechanical)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

LIST OF EXPERIMENTS

1. Sonometer – verification of laws of transverse vibrations in stretched strings.
2. Melde’s Experiment – Determination of the frequency of an electrically maintained tuning fork.
3. Newton’s Rings – Determination of radius of curvature of a convex lens.
4. Diffraction Grating – Determination of Wavelengths of lines of mercury spectrum using spectrometer by normal incidence method.
5. Determination of Cauchy’s constants of the material of the given prism using Spectrometer and mercury light.
6. Wedge Method – Determination of thickness of a paper by forming parallel interference fringes.
7. Variation of magnetic field along the axis of current carrying circular coil – Stewart and Gee’s apparatus.
8. Carey Foster’s bridge – Verification of laws of resistances.
9. Lee’s Method – Determination of coefficient of thermal conductivity of a bad conductor.
10. Calibration of voltmeter using potentiometer.
11. Calibration of low range Ammeter using potentiometer.
12. Laser – Diffraction.

REFERENCE BOOK:

1. Advanced Practical Physics Vol I & II by SP Singh and MS Chauhan, Pragathi Prakasam Publications

WORKSHOP
(Common to ECE, EEE & Mechanical)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

COURSE OBJECTIVE:

1. To impart hands-on training on basic engineering trades.

COURSE OUTCOMES:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry

Bench Work, tools used in carpentry.

Jobs for Class work – half lap joint, mortise and tenon joint, half – lap dovetail joint, corner dovetail joint, central bridle joint.

Sheet Metal

Tools used in sheet metal work, Laying development of the sheet metal jobs, soldering.

Jobs for class works – Square tray, taper tray (sides), funnel, elbow pipe joint, 60⁰ pipe joint.

Fitting

Tools used in fitting work, Different files, chisels, hammers and bech vice.

Jobs for class work – Square, hexagon, rectangular fit, circular fit and triangular fit.

REFERENCE BOOK:

1. Elements of workshop technology, Vol.1 by S. K. and A. K. Choudary, Media Promoters & Publishers Pvt. Ltd.

Code: B16 ENG 1114

NCC/NSS
(ECE, EEE & Mechanical)

Contact Hrs. : 3 Hrs.

This is an Audit Course. Every student should choose either NCC or NSS at the starting of the semester and pursue the same in that semester. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD -ON COURSE
TECHNICAL ENGLISH
(Common to All Branches)

Theory: 2 Periods

COURSE OBJECTIVES:

1. To improve the language proficiency of the students regarding the technical aspects.
2. To enhance the awareness about the formal speaking and writing skills in different contexts.
3. Motivating the students to improve their personal skills.

COURSE OUTCOMES:

1. Students improve their language skills in formal/ technical contexts.
2. They enhance their understanding of technical terms.
3. They improve their personal skills.

SYLLABUS

1. Defining Technical Writing.
2. Writing Skills
Business letters, E-mails
Report Writing
Writing Proposals & Business Quotations
Brochures, Newsletters
How to write an abstract & research paper
3. Speaking Skills
Technical Descriptions
PPT Presentations
Paper Presentations
4. Vocabulary
Technical Terms
5. Behavioral / Attitudinal Skills

TEXT BOOK:

1. Effective Technical Communication by Ashraf Rizvi: Tata McGraw Hill.

REFERENCE BOOKS:

1. Handbook for Technical Writing by David A. McMurrey and Joanne Buckley: Cengage Learning.
2. Technical Writing: Process and Product by Sharon J. Gerson and Steven M. Gerson: Pearson Publications

COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

GROUP – B (ECE, EEE & Mechanical) Branches

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1201	Mathematics-III *	4	3	1	--	4	30	70	100
B16 ENG 1203	Chemistry	4	3	1	--	4	30	70	100
B16 ENG 1205	Computer Programming Using C & Numerical Methods	4	3	1	--	4	30	70	100
B16 ENG 1207	History of Science and Technology	2	2	--	--	2	30	70	100
DS	Department Subject #	4	3	1	--	4	30	70	100
B16 ENG 1210	Chemistry Lab	2	--	--	3	3	50	50	100
B16 ENG 1212	Computer programming Using C & Numerical Methods Lab	2	--	--	3	3	50	50	100
B16 ENG 1213	English Language Lab *	2	--	--	3	3	50	50	100
B16 ENG 1214	Sports (Audit)	--	--	--	--	3	--	--	--
Total		24	14	4	9	30	300	500	800

Note: Add-On Course: Technology Course-I *

*** Common to both Group-A and Group-B**

ECE: B16 EC1208: Electronic Devices and Circuits

EEE: B16 EE1208: Circuit Theory

ME: B16 ME1208: Metallurgy and Materials Engineering

Code: B16 ENG 1201

MATHEMATICS – III
(Common to All Branches)

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

COURSE OBJECTIVES:

Students learn

1. Equations of a line, shortest distance between skew lines
2. About sphere, cone and cylinder
3. Double integrals, change of order of integration
4. Beta function, Gamma function, error function
5. Applications of double integrals to Areas, Volumes, mass, centre of gravity, Moment of inertia
6. The concept of Fourier Integral, Fourier Transform, properties.
7. Parseval's formulae

COURSE OUTCOMES:

Students will be able to

1. Utilize knowledge of line, sphere etc. in his engineering subjects
2. Utilize the knowledge of Beta and Gamma functions and multiple integrals to evaluate the integrals they come across in their applications
3. Utilize the knowledge of Fourier Transform in courses like Signals and Systems and in the solution of partial differential equations at a later stage

SYLLABUS

Solid Geometry

Equations of a plane, Normal form, Intercept form, Equations of Straight Line – Conditions for a line to lie in a Plane – Coplanar lines – Shortest distance between two skew lines - Intersection of three Planes – Equations of Sphere – Tangent Plane to a Sphere –Cone – Cylinder.

Multiple Integrals-1

Double Integrals - Change of Order of Integration - Double Integrals in Polar Coordinates - Triple Integrals - Change of Variables.

Multiple Integrals-2

Beta Function - Gamma Function - Relation between Beta and Gamma Functions-Error Function - Area enclosed by plane curves - Volumes of solids - Area of a curved surface - Calculation of mass - Center of gravity of a plane lamina- Moment of inertia.

Fourier Transforms

Introduction – definition - Fourier integral - Sine and Cosine integrals - Complex form of Fourier integral - Fourier transform - Fourier Sine and Cosine transforms -Finite Fourier Sine and Cosine transforms - properties of Fourier transforms, Convolution theorem for Fourier transforms - Parseval's identity for Fourier transforms - Fourier transforms of derivatives of a function.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

CHEMISTRY
(Common to ECE, EEE & Mechanical)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To develop full pledged laboratories for the analysis of engineering materials.
2. To develop full pledged lab for pollution, corrosion, polymers and electrochemical cells.
3. Extend the infrastructure facilities of the department for the project works of engineering students and also guide the project work under faculty supervision.
4. To develop full pledged lab for analysis of Water, Cement, Hydrogen Peroxide, pollutants and raw materials.

COURSE OUTCOMES:

1. Students learn in-depth about the topics of desalination of sea water, CNG, LPG Biogas, Semiconductors, Liquid crystals, Conducting polymers, fiber reinforced plastics, building materials.
2. Students understand the basic and advanced applied concepts.
3. Students learn to interrelate the theory and with the relevant experiment.
4. Students learn experimental techniques and understand the theory about experiments.

SYLLABUS**Water Chemistry**

Source of water- impurities- Hardness and its determination by EDTA method- Boiler troubles and their removal. Water softening methods- lime soda, zeolite and ion exchange. Municipal water treatment- Break point chlorination. Desalination of sea water – electro dialysis and reverse osmosis methods.

Building Materials

Portland cement: Manufacture-Chemistry involved in setting and hardening of cement –Cement concrete -RCC –Decay of concrete.

Refractories: Classification-Properties and Engineering applications.

Ceramics: Classification-Properties and uses.

Solid State Chemistry

Classification of solids-Types of crystals-properties-Imperfections in crystals. Band theory of solids. Chemistry of semi conductors –Intrinsic, Extrinsic, compound and defect. Organic semi conductors Purification of solids by Zone refining-Single crystal growth – epitaxial growth. Elementary ideas on Liquid crystals.

Corrosion Chemistry

Definition of corrosion- Types of corrosion-chemical & electrochemical corrosion –Pitting, stress corrosion, Galvanic corrosion, Water line corrosion Factors affecting corrosion – Prevention of corrosion- Cathodic protection. Corrosion inhibitors Protective coatings-Metallic coatings, electro plating, electroless plating, chemical conversion coatings- phosphate coatings, chromate coatings, anodizing. Organic coatings-Paints.

Polymers And Plastics

Definition-Types of polymerization – mechanism of free radical polymerization. plastics-Thermosetting and thermoplastic resins cellulose derivatives, vinyl resins, nylon 6,6, bakelite-Compounding of plastics-Fabrication of plastics. Fiber reinforced plastics – conducting polymers -Engineering applications of polymers.

Fuels And Lubricants

Solid fuels: Coal –Analysis of coal – Metallurgical coke- manufacture-Engineering applications.

Petroleum-refining-Knocking and Octane number of gasoline-Cetane number of diesel oil. Synthetic petrol – LPG-CNG–Applications. Rocket fuels – Propellants-classification.

LUBRICANTS: Principles of lubrications, classification of lubricants and properties of lubricants (any five)

TEXT BOOKS:

1. Engineering chemistry by Jain & Jain 15th edition Dhanpatrai publishing company.
2. Engineering chemistry by Dr. K.AnjiReddy and Dr. M.Sita rama reddy, silicon publications, 2015.

REFERENCE BOOKS:

1. A Text Book of Engineering Chemistry by S.S.Dara; S.Chand & Company Ltd.
2. Engineering Chemistry by B.Siva Sankar Mc.Graw Hill Education(India) Pvt.Ltd.
3. A Text Book of Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS
(Common to ECE, EEE & Mechanical)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To make the student familiar with programming in C and enable the student to implement the numerical methods described in this course using C as Programming Language.
2. To introduce the programming principles
3. To familiarize the variables construction, declarations, initializations etc.,
4. To introduce the Control statement
5. To introduce the arrays, structures, Unions, Files etc.,
6. To give the overall idea on Pointers
7. To give the idea about dynamic memory allocation
8. The nature of C language is emphasized in the wide variety of examples and applications.
9. To know about some popular programming languages.
10. To solve various numerical Methods
11. Student can write various programs by applying elementary concepts like arrays, structures, Unions, Files etc.,

COURSE OUTCOMES:

1. Student can understand basic terminology used in C programming.
2. Student can write programs by applying elementary algorithms to solve problems in C language.
3. Student can write, compile and debug programs in C language.
4. Student can Write programs to solve numerical methods
5. Student can be familiar with finite precision computation.

SYLLABUS

Introduction To C

Basic structure of C program, Constants, Variables and data types, Operators and Expressions, Arithmetic Precedence and associativity, Type Conversions. Managing Input and Output Operations, Formatted Input, Formatted Output.

Decision Making, Branching, Looping, Arrays & Strings: Decision making with if statement, Simple if statement, The if...else statement, Nesting of if...else statement, the else..if ladder, switch statement, the (?:) operator, the GOTO statement., The while statement, the do statement, The for statement, Jumps in Loops, One, Two-dimensional Arrays, Character Arrays. Declaration and initialization of Strings, reading and writing of strings, String handling functions, Table of strings.

Functions

Definition of Functions, Return Values and their Types, Function Calls, Function Declaration, Category of Functions: No Arguments and no Return Values, Arguments but no Return Values, Arguments with Return Values, No Argument but Returns a Value, Functions that Return Multiple Values. Nesting of functions, recursion, passing arrays to functions, passing strings to functions, The scope, visibility and lifetime of variables. .

Pointers

Accessing the address of a variable, declaring pointer variables, initializing of pointer variables, accessing variables using pointers, chain of pointers, pointer expressions, pointers and arrays, pointers and character strings, array of pointers, pointers as function arguments, functions returning pointers, pointers to functions, pointers to structures-Program Applications

Structure and Unions

Defining a structure, declaring structure variables, accessing structure members, structure initialization, copying and comparing structure variables, arrays of structures, arrays within structures, structures within structures, structures and functions and unions, size of structures and bit-fields- Program applications.

File handling

Defining and opening a file, closing a file, Input/ Output operations on files, Error handling during I/O operations, random access to files and Command Line Arguments- Program Applications.

Numerical Methods

Solutions of Algebraic and Transcendental Equations: Bisection Method, Newton Raphson Method.

Interpolation - Newton's forward and backward Interpolation, Lagrange's Interpolation in unequal intervals.

Solutions of Linear Equations: Gauss Elimination Method, Jacobi and Gauss Seidel Methods. Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule.

Solutions of Ordinary First Order Differential Equations: Euler's Method, Modified Euler's Method and Runge-Kutta Method.

TEXT BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt. Ltd.

REFERENCE BOOKS:

1. Let Us C, Yashwant Kanetkar, BPB Publications, 5th Edition.
2. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
3. The C –Programming Language' B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt. Ltd.
4. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific

HISTORY OF SCIENCE AND TECHNOLOGY

(Common to ECE, EEE & Mechanical)

Theory : 2 periods
Exam : 3 Hrs.

Sessionals :30
Ext. Marks :70
Credits : 2

COURSE OBJECTIVES:

1. To know the contributions of scientists for the development of society over a period of time.
2. To understand the Science and Technological developments that lead to human welfare.
3. To appreciate the Science and Technological contributions for the development of various sectors of the economy.
4. To identify the technological transfer versus economic progress of the countries.

COURSE OUTCOMES:

1. By the end of this course the students should be able to understand the contribution of Scientific and Technological developments for the benefit of society at large.

SYLLABUS**Historical Perspective of Science and Technology**

Nature and Definitions; Roots of Science – In Ancient Period and Modern Period (During the British Period); Science and Society; Role of Scientist in the Society.

Policies and Plans after Independence

Science and Technology Policy Resolutions; New Technology Fund; Technology Development (TIFAC); Programs aimed at Technological Self Reliance; Activities of Council of Scientific and Industrial Research.

Science and Technological Developments in Critical Areas

Space – The Indian Space Program: India's Geostationary Satellite Services – INSAT System And INSAT Services; **Defense Research and Technology** – Research Coordination, Research efforts and Development of technologies and Spin-off technologies for civilian use; **Nuclear Energy** –Effects of a nuclear explosion and India's safety measures.

Impact of Science and Technology in Major Areas

Ocean Development: Objectives of Ocean Development, Biological and Mineral resources, Marine Research and Capacity Building; **Biotechnology:** Meaning, Biotechnology techniques- Bioreactors, Cell fusion, Cell or Tissue Culture, DNA Fingerprinting, Cloning, Artificial Insemination and Embryo Transfer Technology and Stem Cell Technology; Application of Biotechnology – Medicine, Biocatalysts, Food Biotechnology, Fuel and Fodder and Development of Biosensors.

Technology Transfer and Development

Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques;
Appropriate Technology - Criteria and Selection of an Appropriate Technology; Barriers of Technological Change. .

TEXT BOOKS:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi-58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East –West Press (P) Ltd., New Delhi.

REFERENCE BOOKS:

1. Dr. G.R. Kohli, History of Science and Technology and Environmental Movements in India, Surjeet Publications, New Delhi

ELECTRONIC DEVICES AND CIRCUITS (For ECE)

Theory	: 3 periods	Sessionals	: 30
Tutorials	: 1 period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To give exposure to students on semiconductor physics of the intrinsic and extrinsic semiconductors.
2. To give exposure to students on the basics of semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED, Schottky barrier diode, PIN diode, varactor diode and tunnel diode.
3. To give exposure to students on rectifier circuits using diodes.
4. To give exposure to students on basics of BJT, JFET and MOSFET and biasing of BJT and FET.
5. To give exposure to students on the analysis of transistor at low and high frequencies.

COURSE OUTCOMES:

After completion of the course the students will be able to

1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJTs and FETs.
2. Use this knowledge to analyze and design basic electronic application circuits.
3. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

Transport Phenomena in SemiConductors

Mobility and conductivity, intrinsic and extrinsic semiconductors, mass action law, charge densities in a semiconductors, Hall Effect, generation and recombination of charges, drift and diffusion currents, continuity equation, injected minority carrier charge, potential variation in graded semiconductors.

PN junction diode

Open circuited PN junction , PN junction as a rectifier, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, Zener diode, Tunnel Diode, Photo diode, Point Contact diode, Schottky barrier diode, varactor diode, PIN diode, LED.

Diode Rectifiers

Half wave, Full wave and Bridge Rectifiers with and without filters, Ripple factor and regulation characteristics

Bipolar junction transistors

Introduction to BJT, operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α, β, γ relation between α and β, γ early effect or basewidth modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.

Field Effect transistors

JFET and its characteristics, pinch off voltage, FET small signal model, MOSFET and its characteristics

Transistor Biasing Circuits

The operating point, Bias stability, different types of biasing techniques, stabilization against variation in I_{co} , V_{BE} , & β . Bias compensation, thermal runaway, thermal stability, Biasing of FETs.

Transistors at low and High frequencies

Transistor hybrid model, h-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.

TEXT BOOKS:

1. Integrated electronics analog and digital circuits and systems by Jacob Millman, C Halkias, Chetan D Parikh, Tata McGrawhill publications.
2. Electronic Devices and Circuits Theory by Boylestad, Prentice Hall Publications.

REFERENCE BOOKS:

1. Electronic Principles by Albert Malvino, David J Bates, McGrawhill publications.
2. Electronic Devices and Circuits an Introduction by Allen Mottershead, Prenticehall India.
3. Electronic Devices and Circuits by Salivahanan, Tata McGrawhill publications.

CIRCUIT THEORY
(For EEE)

Theory	: 3 periods	Sessionals	: 30
Tutorials	: 1 period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To enlighten and enrich the students to acquire knowledge about the basics of circuit analysis, concepts of AC circuits, DC/AC network theorems, coupled & three phase circuits.
2. To inculcate problem solving skills and understanding of circuit theory through the application of techniques and principles of Electrical Circuits to common circuit problems.

COURSE OUTCOMES:

1. Able to develop an understanding of the basic fundamental electrical laws, elements of electric Networks and learn the techniques to measure voltage and current.
2. Develops the ability to apply circuit theorems to DC and AC circuits.
3. Able to analyse the coupled & three phase circuits.

SYLLABUS**Fundamentals of Electric Circuits**

Concepts of Electric circuit: EMF, Current, Potential difference, Power and Energy; Concepts of Network: Active and Passive elements, classification of Linear, Non-linear, Unilateral, Bilateral Lumped and Distributed elements; Reference directions for current and voltage; Voltage and Current sources; Voltage-Current Relations of R, L, C elements; Voltage and Current division; Series and Parallel combinations of Resistance, Inductance and Capacitance

D.C Circuits

Kirchhoff's Laws; Nodal Analysis, Mesh Analysis, Source Transformation, Linearity and Superposition Theorem, Thevenin's And Norton's Theorems, Reciprocity theorem, Maximum Power Transfer Theorem, Star-Delta Transformation.

AC Circuits

The Sinusoidal Forcing Function, Phasor Concept, Average and Effective Values of Voltage and Current, Instantaneous and Average Power, Complex Power, Steady State Analysis using Mesh and Nodal Analysis, Resonance.

Three Phase Circuits

Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits, Measurement of Power in Three Phase Power Circuits.

Magnetic Circuits

Magneto motive force(MMF), Reluctance, Magnetic flux; Analysis of magnetic circuit, Analogy between Electric & Magnetic circuits, Series Magnetic circuits, Magnetic leakage, B-H curve, Faraday's Laws of Electromagnetic Induction, Induced EMF, Dynamically Induced EMF, Statically Induced EMF, Self-Inductance, Mutual Inductance.(simple numerical problems)

TEXT BOOKS:

1. Engineering Circuit Analysis By W.H. Hayt Jr & J.E. Kemmerly, 5th Ed., Mc.Graw Hill.

REFERENCE BOOKS:

1. Basic Electrical Engineering By V.K Mehta & Rohit Mehta- S.Chand Pub.
2. Sudhakar & Syam Mohan “ Network Analysis”, Mcgraw Hill

METALLURGY AND MATERIALS ENGINEERING
(For Mechanical Engineering)

Theory	: 3 periods	Sessionals	: 30
Tutorials	: 1 period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

PREREQUISITES:

Basic knowledge in solid state chemistry, alloy systems.

COURSE OBJECTIVES:

The overall objective of the course is to impart knowledge about the engineering materials and their properties and predict their behavior under different working conditions and methods to enhance the mechanical properties of the material Specific objectives include:

1. This course will help in learning crystal structures, crystal defects, space lattices.
2. This course will be helps students to understand systems of solid solutions and their nature.
3. To acquaint the knowledge about the cooling curves and phase diagrams of different alloy systems, calculation of weight percent of phases in alloy system by applying Lever rule.
4. To acquaint knowledge about ferrous alloys, particularly Steel and Cast Irons.
5. To impart knowledge about Equilibrium and Non-Equilibrium diagrams.
6. To impart knowledge about different heat treatment and surface hardening methods in improving the mechanical properties of steels.
7. To impart knowledge about the plastic deformation of materials.
8. To impart knowledge about composite materials.

COURSE OUTCOMES:

Upon completion of this course, students should be able to:

1. Understand crystalline solids and their atomic structures.
2. Suggest and recommend necessary engineering materials for specific applications keeping in view of the cost, design, reliability, life, working conditions and properties of the products.
3. Understand different phase transformations in Iron-Iron Carbide diagram and distinguish between steels and cast irons.
4. Select different materials for tools and components based on functional requirements.
5. Use composite materials for different engineering applications like aerospace, automobile, ship building industry, sports item etc.

SYLLABUS

Structure of crystalline solids

Atomic structure & bonding in solids- Crystal structures-calculations of radius, Coordination Number and Atomic Packing Factor for different cubic structures - Imperfection in solids, point defects, Linear defects, Planar defects and Volume defects- Concept of Slip & twinning.

Phase diagrams

Basic terms- phase rule- Lever rule & free energy of phase mixtures cooling curves- Phase diagram & phase transformation - construction of phase diagrams- binary phase diagrams - Brass, Bronze, Al-Cu and AlSi phase diagrams- Invariant reactions, eutectic, peritectic, eutectoid, peritectoid, metatectic & monotectic reactions, Iron carbon phase diagram & microstructures of plain carbon steel & cast iron

Heat treatment

Heat treatment of steel- Annealing, and its types, normalizing, hardening, tempering, martempering, austempering - TTT diagrams, drawing of TTT diagram, TTT diagram for hypo- & hypereutectoid steels, effect of alloying elements, CCT diagram- Martensitic transformation, nature of martensitic transformation- Surface hardening processes like case hardening, carburizing, cyaniding, nitriding Induction hardening, hardenability, Jominy end-quench test, Age hardening of Al & Cu alloys Precipitation Hardening

Engineering Alloys

Properties, composition, microstructure and uses of low carbon, mild medium & high carbon steels. stainless steels, high speed steels, Hadfield steels, tool steels - Cast irons, gray CI, white CI, malleable CI, SC iron- The light alloys- Al & Mg & Titanium alloys- Copper & its alloys: brasses & bronzes- super alloys, Smart materials- Nano materials.

Composite Materials

Classification of composite materials, dispersion strengthened, particle reinforced and fiber reinforced composite laminates properties of matrix and reinforcement materials and structural applications of different types of composite materials.

TEXT BOOKS:

1. "Materials Science & Engineering- An Introduction", William D. Callister Jr. Wiley India Pvt. Ltd. 6th Edition, 2006, New Delhi.
2. Physical Metallurgy, Principles & Practices", V Raghavan. PHI 2nd Edition 2006, New Delhi.

REFERENCE BOOKS:

1. Introduction to Physical Metallurgy by Sidney H Avner, Tata McGraw-Hill Education 1997
2. Materials Science And Engineering: A First Course By V. Raghavan Phi 5th Edition 2011, New Delhi

CHEMISTRY LAB
(Common to ECE, EEE & Mechanical)

Lab : 3 periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

LIST OF PRACTICAL EXPERIMENTS:

1. Estimation of Sodium hydroxide with HCl (Na_2CO_3 primary standard).
2. Estimation of Iron as Ferrous iron in an Ore Sample.
3. Estimation of Oxalic acid by a redox method
4. Estimation of Calcium in a sample of Portland cement.
5. Estimation of Volume strength of Hydrogen Peroxide.
6. Estimation of Mohr's salt by Potassium dichromate
7. Determination of Hardness of an underground water sample.
8. Estimation of Zinc by EDTA method.
9. Determination of Alkalinity of water sample.

DEMONSTRATION EXPERIMENTS:

10. Determination of Viscosity and Viscosity index of a lubricant.
11. Printed Circuit Board
12. Determination of dissolved oxygen in given water sample.
13. Potentiometric titrations.
14. P^{H} Determination by using P^{H} meter.

REFERENCE BOOKS:

1. Essentials Experimental Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co Pvt. Ltd.
2. Laboratory Manual on Engineering Chemistry by Dr. Sudha Rani, Dhanpatrai Publishing Company.
3. Engineering Chemistry Laboratory Manual by Dr. K.Anji Reddy, Tulip Publications.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS LAB
(Common to ECE, EEE & Mechanical)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF PROGRAMMES:

1. Write a program to read x, y coordinates of 3 points and then calculate the area of a triangle formed by them and print the coordinates of the three points and the area of the triangle. What will be the output from your program if the three given points are in a straight line.
2. Write a program which generates 100 random numbers in the range of 1 to 100. Store them in an array and then print the array. Write 3 versions of the program using different loop constructs (eg. for, while and do-while).
3. Write a set of string manipulation functions eg. for getting a sub-string from a given position, copying one string to another, reversing a string and adding one string to another.
4. Write a program which determines the largest and the smallest number that can be stored in different data types like short, int, long, float and double. What happens when you add 1 to the largest possible integer number that can be stored?
5. Write a program which generates 100 random real numbers in the range of 10.0 to 20.0 and sort them in descending order.
6. Write a function for transporting a square matrix in place (in place means that you are not allowed to have full temporary matrix).
7. First use an editor to create a file with some integer numbers. Now write a program, which reads these numbers and determines their mean and standard deviation.
8. Implement bisection method to find the square root of a given number to a given accuracy.
9. Implement Newton Raphson Method to determine a root of polynomial equation.
10. Given a table of x and corresponding f(x) values, write a program which will determine f(x) value at an intermediate x value using Lagrange's Interpolation.
11. Write a function which will invert a matrix.
12. Implement Simpson's $1/3^{rd}$ rule for numerical integration.
13. Implement Trapezoidal rule for numerical integration.
14. Write a program to solve a set of linear algebraic equations.
15. Write a program to solve a differential equation using Runge-Kutta Method.

REFERENCE BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd

ENGLISH LANGUAGE LAB
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

The Language Lab focuses on the production and practices of sounds of language and familiarizes the students with use of English in everyday situations and contexts.

COURSE OBJECTIVES:

1. To make students recognize the sounds of English through Audio-Visual aids.
2. To help students build their confidence and help overcome their inhibitions and self consciousness while speaking in English. *The focus shall be on fluency.*
3. To familiarize the students with stress and intonation and enable them to speak English effectively.

COURSE OUTCOMES:

1. Students will be sensitized towards recognition of English sound pattern.
2. The fluency in speech will be enhanced.

SYLLABUS

1. English Sound Pattern – Letters
2. Sounds of English
3. Pronunciation
4. Stress and Intonation

Laboratory Practice Sessions:

1. Letters and Sounds, Worksheet-1
2. Interactions-1, Worksheet-2
3. The Sounds of English, Worksheet-3
4. Interactions-2, Worksheet-4
5. Pronouncing Words-Some Important Patterns, Worksheet-5
6. Interactions-3, Worksheet-2
7. Stress and Intonation, Worksheet-2

REFERENCE BOOKS:

1. Speak Well, Board of Editors, Orient Black Swan Publishers, Hyderabad, 2012
2. Cambridge English Pronouncing Dictionary, Cambridge University Press, India, 2012.
3. A Textbook of English phonetics for Indian students by T. Balasubramanian, Macmillan publisher, 1981.

Code: B16 ENG 1214

SPORTS
(Common to ECE, EEE & Mechanical)

Contact Hrs. 3 Hrs.

This is an Audit Course. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD ON COURSE
TECHNOLOGY COURSE – I
(Professional C Programming)
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. On successful completion of course students will be able to reinforce their learning of C, read code, and write good procedural code.

COURSE OUTCOMES:

1. Students will be able to solve a series of graduated problems
2. Students will be able to do projects in ‘C’ Language.

SYLLABUS

Introduction to Linux

Introduction to C with problem solving

Functions usage and application with solved problems

Introduction to pointers and applications with examples

Structures and Unions and file handling with program applications

Project using C

RESOURCES:

Talent Sprint Problem Bank, Project Euler



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

II/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 2101	Mathematics - IV	4	3	1	-	4	30	70	100
B16 EE 2104	Circuit Analysis & Synthesis	4	3	1	-	4	30	70	100
B16 EE 2105	Electrical Technology	4	3	1	-	4	30	70	100
B16 EC 2101	Analog Electronic Circuits	4	3	1	-	4	30	70	100
B16 CS 2104	Elementary Data Structures	4	3	1	-	4	30	70	100
B16 EC 2102	Probability Theory & Random Processes	4	3	1	-	4	30	70	100
B16 EE 2107	Network And Machines Lab	2	-	-	3	3	50	50	100
B16 EC 2105	Electronic Devices & Circuits Lab	2	-	-	3	3	50	50	100
B16 ENG 2104	English Proficiency	2	1	1	--	2	50	50	100
B16 ENG 2106	Industry Oriented Training	1	--	--	2	2	50	--	50
Total		31	19	7	8	34	380	570	950

Code: B16 ENG 2101

MATHEMATICS – IV
(Common to CIV,ECE,EEE & ME)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. The concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Green's, Stokes' and Divergence theorems
3. Classification of 2nd order Partial Differential Equations as well as solution of 1-Dimensional Wave equation and 1-Dimensional Heat equation
4. the concept of Analytic function, CR equations
5. Cauchy's Integral Theorem and Integral Formula
6. Taylor and Laurent series, Residues and Residue theorem

Course Outcomes:

Students will be able to

1. Apply the concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Determine scalar potential, circulation and work done
3. Evaluate integrals using Green's, Stokes' and Divergence theorems
4. Obtain the solution of 1-D wave equation and 1-D heat equation
5. Determine the zeroes and poles of functions and residues at poles
6. Evaluate certain real definite integrals that arise in applications by the use of Residue theorem

SYLLABUS

Vector Calculus-1

Definitions of Scalar and Vector point functions, Differentiation of vectors, Vector differential operator del, Del applied to scalar point function – gradient, Del applied to vector point function- divergence and curl, physical interpretation of gradient, divergence and curl(without proof), Del applied twice to a point function, Del applied to product of two functions, Irrotational and Solenoidal Fields, scalar potential

Vector Calculus-2

Integration of vectors, line integral, circulation, work done, surface integral, Flux, Green's, Stokes' and Gauss Divergence Theorems (Without proofs). Introduction to orthogonal curvilinear coordinates, cylindrical polar coordinates and spherical polar coordinates.

Applications Of Partial Differential Equations

Classification of second order partial differential equations, Method of separation of variables, One –dimensional wave equation- vibrations of a stretched string (no derivation)-, one-dimensional heat equation – Heat flow along a long horizontal bar (no derivation) (problems on heat equation involving homogeneous end conditions only), two dimensional Laplace equation in Cartesian coordinates.

Complex Variables-1

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n , e^z , $\sin z$, $\sinh z$ and $\log z$.

Limit and continuity of a function of the complex variable, derivative, analytic function, properties of Analytic functions, Cauchy- Riemann equations, Harmonic functions and Orthogonal system, application of analytic function to flow problems, geometric representation of $w=f(z)$, conformal mapping – Bilinear transformation only.

Complex Variables-2

Integration of complex functions, Cauchy's theorem, Cauchy's integral formula (statements only) . Taylor and Laurent series expansions of functions (statement of theorems only), zeros and singularities, Residue, calculation of residues, Cauchy's Residue theorem (without proof), Evaluation of real and definite integrals- integration around a unit circle

Text Book:

1. "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

CIRCUIT ANALYSIS & SYNTHESIS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To learn the concept of network theory and definitions of circuit elements for modeling practical electric circuits
2. To learn various theorems and techniques in electric circuit analysis and to know their significance and applications
3. To learn phasor concept and apply it to analysis of circuits in sinusoidal steady state.
4. To learn the concept of two-port network analysis and find models using different parameter sets.
5. To learn the concept of Laplace transform and its application to circuit analysis.

Course Outcomes

1. Students will learn circuit conventions and analyze DC circuits using various techniques like mesh analysis, nodal analysis and theorems.
2. Students will learn the significance of energy storing elements (Inductance & Capacitance) in circuits and analyze transient and steady state responses.
3. Students will learn the concepts of single and three-phase balanced circuits and analyze sinusoidal steady-state using phasor concept.
4. Student will learn the concept of network functions and analyze poles, zeros and time domain behavior from pole-zero plots.
5. Student will learn the concept of positive real functions and test whether the given network function is Hurwitz and positive real or not.

SYLLABUS**Analysis of DC Circuits:**

Active elements, Passive elements, Reference directions for current and voltage, Kirchhoff's Laws, Voltage and Current Division Nodal Analysis, Mesh analysis, Linearity and superposition, Thevenin's theorem and Norton's theorem, Reciprocity theorem, Z, Y, H, S-parameters.

DC transients:

Inductor, Capacitor, source free RL, RC and RLC response, Evaluation of Initial conditions, Application of unit-step function to RL, RC and RLC circuits, concepts of Natural, Forced and Complete response.

Sinusoidal Steady State Analysis:

The sinusoidal forcing function, Phasor Concept, Average and Effective value of Voltage and Current, Instantaneous and Average Power, Complex Power, Steady State Analysis using mesh and node analysis, Application of network theorems to AC circuits, resonance, Concept of Duality.

Network Functions:

Network functions for single port and two port, Calculation of Network functions for Ladder and General Networks, Poles and Zeroes, Restriction of Poles and Zeroes for Driving point and Transfer functions, Time Domain Behavior from Pole Zero plot, Transfer Functions in terms of Y and Z functions, Scaling Network Functions.

Positive Real Functions:

Positive real function and other properties, Herwitz polynomials, Computation of residues, even and Odd functions, Test for Positive Real Functions.

TextBooks:

1. Engineering Circuit Analysis, William H.Hayt Jr. and Jack E. Kemmerley, 5th Edition, McGraw Hill International Edition.
2. Modern Network Synthesis, M. E. Van Valkenburg, Wiley Eastern.

Reference Books:

1. Introduction to modern Network Synthesis : Van Valkenburg; John Wiley
2. Network Analysis: Van Valkenburg; PHI
3. Basic circuit theory:DasoerKuh; McGraw Hill.
4. A Course in Electrical Circuit Analysis by Soni& Gupta; DhanpatRai Publication.
5. Circuit Analysis : G.K. Mithal; Khanna Publication.
6. Networks and Systems :D.RoyChoudhury; New Age International.
7. Network Analysis, M. E. Van Valkenburg, 3rd Edition, PHI.
8. Network Analysis &Synthesis :UmeshSinha; SatyaPrakash Pub.
9. Network Analysis &Synthesis :F.F.Kuo; John Wiley & Sons Inc.

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ELECTRICAL TECHNOLOGY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To learn the basic principles of magnetic field theory.
2. To know the Construction and working of D.C Machines and A.C Machines.
3. To learn the characteristics of D.C Machines and A.C Machines.
4. To learn different tests of DC machines and A.C Machines to know its performance.

Course Outcomes

1. Classify the parts of DC Machines, Transformers, Three Phase Induction motors & Three Phase Synchronous machines.(k2)
2. Interpret the operation and working principle of DC Machines, Transformers, Three Phase Induction motors , Three Phase Synchronous machines.(k2)
3. Develop performance characteristics of various machines.(k3)
4. Construct experiments on various machines.(k3)
5. Analyze the application of electrical machines in various fields of engineering.(k4)

SYLLABUS**Magnetic Circuits:**

Definitions of magnetic circuit, Reluctance, MMF, Magnetic flux, Hysteresis loss. Faraday's laws of Electromagnetic induction, Induced E.M.F., Dynamically induced E.M.F., Statically induced E.M.F., Self-inductance, Mutual inductance, Lenz's law

DC Machines:

Principle of operation DC Generator - EMF equation - types - DC motor types - torque equation – speed control methods- applications - three point starter-Testing-Load test on D.C Shunt Motor, D.C Series Motor, Swinburne's test.

Transformers:

Principle of operation of single phase transformers - EMF equation - equivalent circuit – losses - efficiency and regulation-Testing- Open circuit and Short circuit tests.

Induction Motors:

Construction - Principle of operation of induction motor - slip - torque characteristics - Power flow diagram.

Synchronous Machines :

Construction-Principle of operation of alternators – EMF equation of alternator- regulation by synchronous impedance method, Principle of operation of synchronous motors, methods of starting, applications.

Text Books:

1. Electrical Machinery by Dr.P.S Bimbhra,Khanna publications.
2. Electrical machines by J.B Gupta, Published by S K Kataria and Sons.
3. A Textbook of Electrical Technology By R.K.Rajput,Laxmi Publications.
4. Principles of Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications.
5. A Textbook Of Electrical Technology B. L. THERAJA (Volume II)

Reference Books:

1. Basic Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications.
2. Basic Electrical Engineering, S.N. Singh, PHI.

ANALOG ELECTRONIC CIRCUITS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

The aim of this course is to

1. Understand the concepts of different type of amplifiers and analyze them.
2. Learn the classification of feedback amplifiers and analyze them.
3. Compare the voltage and power amplifiers and analyze them.
4. Understand the principle of oscillator and analyze different types of sinusoidal oscillators.
5. Learn the classification of tuned amplifiers and analyze them.
6. Understand the concept and analyze applications of op-amp.

Course Outcomes

After the completion of the course, students will be able to:

1. Know the equivalent circuit of multistage amplifier and its analysis. [K3]
2. Identify the different feedback topologies and analyze them. [K1]
3. Explain the principle of oscillator and design different types of sinusoidal oscillators.[K3]
4. Define the difference between voltage and power amplifiers and design different
5. classes. [K1, K3]
6. Know that Tuned amplifiers amplify a narrow band of frequencies and will also be able to analyze them.[K2, K3]
7. Identify that Op-amp not amplifies but also perform different operations and analyze some applications.[K1,K2]

SYLLABUS**Multistage Amplifiers**

Transistor at high frequencies, CE short circuit current gain and concept of GainBandwidth Product. BJT and FET RC Coupled Amplifiers at low and high frequencies. Frequency Response and calculation of Band Width of Multistage Amplifiers.

Feed Back Amplifiers

Concept of Feed Back Amplifiers - Effect of Negative Feedback on the amplifier characteristics. Four feedback topologies, Method of analysis of Voltage Series, Current Series, Voltage Shunt and Current Shunt feedback Amplifiers.

Sinusoidal Oscillators

Condition for oscillations –LC Oscillators – Hartley, Colpitts, Clapp and TunedCollector Oscillators – Frequency and amplitude Stability of Oscillators – CrystalOscillators – RC Oscillators -- RC Phase Shift and Weinbridge Oscillators.

Power Amplifiers

Classification of Power Amplifiers – Class A, Class B and Class AB power Amplifiers. Series Fed, Single Ended Transformer Coupled and Push Pull Class A and Class B Power Amplifiers. Cross-over Distortion in Pure Class B Power Amplifier, Class AB Power Amplifier – Complementary Push Pull Amplifier with trickle Bias, Derating Factor – Heat Sinks.

Tuned Voltage Amplifiers

Single Tuned and Stagger Tuned Amplifiers – Analysis – Double Tuned Amplifier – Bandwidth Calculation.

Operational Amplifiers

Concept of Direct Coupled Amplifiers. Ideal Characteristics of an operational Amplifier – Differential Amplifier - Calculation of common mode Rejection ratio – Differential Amplifier supplied with a constant current – Normalized Transfer Characteristics of a differential Amplifier – Applications of OP-Amp as an Inverting and Non-Inverting Amplifier, Integrator, Differentiator Summing and Subtracting Amplifier and Logarithmic Amplifier. Parameters of an Op-Amp, Measurement of OP-Amp Parameters.

Text Books:

1. Millman's Integrated Electronics by Jacob Millman, Christos Halkias, Chetan Parikh.
2. Electronic Devices and Circuits by Salivahanan

References:

1. Electronic Devices and Circuits by G.S.N.Raju.
2. Electronic devices and circuits-Mottershead
3. Op-amps and Linear Integrated Circuits – Gayakwad

ELEMENTARY DATA STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The aim of this course is to

1. Student will learn about data structures and the algorithms for manipulating them, and how to analyse the time and memory requirements of them.
2. Student will master some complex searching and sorting algorithms and their data structures, advanced types of trees, and graph representations and graph algorithms.
3. Student will learn when and how to use techniques for developing algorithms.
4. Student will also become skilled in algorithmic analysis and algorithm development using the latest techniques.

Course Outcomes:

After the completion of the course, students will be able to:

1. Be able to write programs and class libraries given a specification;
2. Implement various data structures.
3. Implement and analyse various sorting algorithms.
4. Understand abstract data types and how they are implemented in C.

SYLLABUS**Revision of C language:overview****Arrays and Functions:**

Organization and use of one Dimensional , Two dimensional and Multi dimensional Arrays, Handling of character strings, string operations, Concept of function, Parameter passing , Recursion.

Structures, Pointers and Files:

Definition of structure and Union, Programming examples, Pointers , Pointer Expressions, Programming examples, Dynamic Memory Allocation, Preprocessor Directives

Linear Data structures:

Stack -Representation, Operations, Queue- representation, Operations, Circular Queue, Linked List- Representation, Operations, Double Linked List and Circular List.

Non-linear Data Structures:

Trees, Binary Tree Representation, Tree Traversals, Conversion of a General Tree to Binary Tree.

Graphs

Representation of Graphs, Linked Representation of Graphs, Graph Traversals and Spanning Trees.

Searching & Sorting:

Basic search Techniques- Linear and Binary searching, Tree searching, Sorting-Insertion, Selection, Bubble, Quick and Merge Sorting.

Text Books:

1. Programming With C, by Schaum's Outlines, Byron Gottfried.
2. Fundamentals of Data Structures in C, by Sahni- Horowitz, Second Edition.

Reference Books:

1. An Introduction To Data Structures With Applications, Trembly and Sorenson.
2. The C-Programming Language, Kerningham and others.

PROBABILITY THEORY & RANDOM PROCESSES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the fundamental concepts and theorems of probability theory.
2. To introduce fundamental concepts of random variables and their statistical descriptions.
3. To discuss about various types of random processes, their properties, spectral representation and applications.
4. To understand the difference between time averages and statistical averages, Stationarity and Ergodicity.
5. To apply elements of stochastic processes for problems in real life.
6. To understand the concept of Noise as applicable to linear Systems.

Course Outcomes:

On completion of the course the student will be able to

1. Understand the axiomatic formulation of modern probability theory.
2. Characterize Probability Models and functions of Random variables based on single and multiple random variables.
3. Evaluate and apply moments and characteristic functions and understand the concept of Inequalities and probabilistic limits.
4. Understand the concept of Random process and determine covariance and spectral density of stationary random processes.
5. Demonstrate the specific applications to Poisson and Gaussian process and representation of low pass and band pass noise models.
6. Analyze the response of random inputs to linear time invariant systems.

SYLLABUS**Probability Theory**

Definitions of Probability, Axioms of Probability, Probability Spaces, Properties of Probabilities, Joint and Conditional Probabilities, Independent Events

Random Variables

Probability Distribution Functions, Probability Density Functions, Joint Distribution of Two Variables, Conditional Probability Distribution and Density, Independent Random Variables.

Statistical Averages

Functions of Random Variables and Random Vectors, Statistical Averages, Characteristic Function of Random Variables, Inequalities of Chebyshev and Schwartz, Convergence Concepts, Central Limit Theorem.

Random Processes

Stationarity, Ergodicity, Covariance Function and their Properties, Spectral Representation, Wiener-Kinchine Theorem.

Linear Systems and Random Noise Processes

Linear operations, Gaussian processes, Poisson Processes, Low-pass and Band-pass Noise Representation.

Text Books:

1. Probability Theory and Random Processes, S. P. Eugene Xavier, S. Chand and Co. New Delhi, 1998 (2nd Edition).
2. Probability Theory and Random Signal Principles, Peebles, Tata McGraw Hill Publishers.

Reference Books:

1. Signal Analysis, Papoulis, McGraw Hill N. Y., 1977.
2. Introduction to Random Signals and Noise, Davenport W. B. Jrs. and W. I. Root, McGraw Hill N.Y., 1954.
3. B.P. Lathi, "Signals, Systems & Communications", B.S. Publications, 2003.
4. Probability, Random Variables & Random Signal Principles, STARK et al, Pearson, 2002

NETWORKS AND MACHINES LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To learn to make simple electric circuits by using different sources, loads and components and verify basic laws.
2. To experimentally verify various theorems of circuit analysis.
3. To learn to find circuit models for two-terminal devices and two-port networks.
4. Conducting experiments on characteristics of generators & motors
5. Load tests on series, shunt, compound motors and compound generators-Swinburne's, Hopkinson's test.
6. OC & SC tests on single phase transformers, Sumpner's test.

Course Outcomes

1. Students will gain the skill to make and experiment with practical electric circuits.
2. Students will be able to measure voltage, current, power in practical electric circuits.
3. Students will know the significance of various theorems and their applications.
4. Students will be able to model devices for circuit analysis.
5. Students will be able to assess the behavior of different electrical machines.
6. Students will be able to predetermine the efficiency and regulation of different machines.

LIST OF EXPERIMENTS

1. Maximum Power Transfer Theorem
2. Superposition Theorem
3. Thevenin's Theorem
4. Series Resonance
5. Ohm's Law And Characteristics Of Filament Lamp
6. Parameters Of Iron Cored Inductor
7. Swinburne's Test
8. Load Test on Dc Shunt Motor
9. Load Test on Dc Series Motor
10. Load Test on 3 Phase Slip ring Induction Motor
11. OC and SC Test on Single Phase Transformer
12. Voltage Regulation of An Alternator by Synchronous Impedance Method
13. Speed Control of Dc Shunt Motor

Reference: Lab Manual

**ELECTRONIC DEVICES & CIRCUITS LAB
(Common to ECE & EEE)****Lab : 3 Periods**
Exam : 3 Hrs.**Sessionals : 50**
Ext. Marks : 50
Credits : 2**Course Objectives:**

1. To familiarize the student with test and measuring equipment like CROs, Multimeters, Ammeters, Voltmeters etc. and also to prepare the student to use signal generators, bread boards and to make the student identify the terminals of basic electronic devices like diodes, transistors and JFETs.
2. To familiarize the student with features of Multisim and to prepare the student to construct and simulate various electronic circuits using Multisim.
3. To make the student study experimentally the characteristics of basic electronic devices like ordinary pn diodes, LEDs, Zener diodes, BJTS, JFETs and rectifiers with and without filters.
4. To make the student to conduct experiments to analyze various parameters of BJT amplifiers and FET amplifiers.

Course Outcomes:

After the successful completion of the lab course, the students will be able

1. To understand the role of basic electronic devices like ordinary Pn diodes, Zener diodes, LEDs, BJTS and JFETs in achieving various functionalities like rectification, voltage regulation, amplification, switching action etc. in various electronic circuits.
2. To construct and simulate different electronic circuits using Multisim.
3. To have the hardware skills and software skills required in the design of electronic systems for various applications.

LIST OF EXPERIMENTS

1. V-I characteristics of semiconductor diode, LED and Zener diode.
2. Half wave and full wave rectifier with and without filters.
3. Input and output characteristics of transistor in CE configuration.
4. Transistor biasing circuits and transistor as a switch.
5. CE amplifier.
6. JFET common source amplifier.

LIST OF SIMULATION EXPERIMENTS

7. V-I characteristics of semiconductor diode, LED and Zener diode.
8. Regulation characteristics of Zener diode.
9. Input and output characteristics of transistor in CB configuration
10. JFET Characteristics.
11. CC amplifier
12. JFET common source amplifier

Reference: Lab Manual

ENGLISH PROFICIENCY
(Common to All Branches)

Theory	: 1 Period	Sessionals	: 50
Tutorial	: 1 Period	Ext. Marks	: 50
Exam	: 3 Hrs.	Credits	: 2

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

Students be able to

1. Understand the importance of professional communication.
2. Learn language skills and vocabulary in order to improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self-attributes which require improvement and motivation.
5. Build their confidence and overcome their inhibitions.
6. Improve their strategies in reading skills.

Course Outcomes:

1. Students enhance their vocabulary and use it in the relevant contexts .
2. They improve speaking skills.
3. They learn and practice the skills of composition writing.
4. They enhance their reading and understanding of different texts.
5. They enrich their communication both in formal and informal contexts.
6. They strengthen their confidence in presentation skills.

SYLLABUS

Speaking Skills

PPT

Describing event/place/thing

Picture Description

Extempore

Debate

Telephonic Skills

Analyzing Proverbs

Vocabulary

Affixes

Pairs of Words

Reading Skills

Reading Comprehension

Reading/Summarizing News Paper Article

Writing Skills

Designing Posters

Essay writing

Resume Writing

Reference Books:

1. Interchange (4th edition) Student's books 1&2 by Jack C. Richards, CUP.
2. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
3. English and Communication Skills for Students of Science and Engineering, by S.P.
4. Dhanavel, Orient Blackswan Ltd. 2009
5. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
6. The Oxford Guide to Writing and Speaking by John Seely OUP

(***Note: Sessional Marks will be evaluated based on Continuous Comprehensive Evaluation of the students' Performance - 40M, Attendance – 10M and External Marks will be evaluated based on Presentation Skills – 30M, Project 20M)

INDUSTRY ORIENTED TRAINING
(Common to ECE & EEE)

Lab : 2 Periods

Exam : 3 Hrs.

Sessionals : 50

Credits : 1

Course Objectives:

1. Be familiar with basic Data structures.
2. Master the implementation of linear data structures.
3. Master the implementation of non linear data structures.
4. Be familiar with Object Oriented Concepts.

Course Outcomes:

1. Application using implementation of Data structures.
2. Application using implementation of Linear and non linear Data structures in view of industry.
3. Applications using Object Oriented Concepts in view of industry.

Syllabus: Industry Oriented Applications on following topics.

BASIC CONCEPTS

System Life Cycle, Algorithm Specification, Recursive Algorithms, Data Abstraction, Performance Analysis, Space Complexity, Time Complexity, Asymptotic Notation, Comparing Time Complexities

IMPLEMENTATION (Using C)

Arrays
Stacks
Queues
Linked List
Double linked lists
Trees
Graphs

Applications of linear and nonlinear data structures and solving simple to complex problems in perspective of industry requirements.

Basic Concepts of OOP

Procedural Paradigms, Object Oriented Paradigm, OOP Principles and Terminology, OOP benefits, Procedure and Object Oriented programming languages, advantages and disadvantages, creating class, defining objects in C++ and JAVA.

Applications using OOP in solving simple to complex problems in perspective of industry requirements.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest- 25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 EC 2201	Switching Theory and Logic Design	4	3	1	--	4	30	70	100
B16 EC 2202	Electromagnetic Field theory & Transmission Lines	4	3	1	--	4	30	70	100
B16 EC 2203	Pulse and Digital Circuits	4	3	1	--	4	30	70	100
B16 EC 2204	Analog Communications	4	3	1	--	4	30	70	100
B16 EC 2205	Signals & Systems	4	3	1	--	4	30	70	100
B16 ENG 2201	Environmental Studies	2	3	1	--	4	30	70	100
B16 EC 2207	Analog Communications Lab	2	--	--	3	3	50	50	100
B16 EC 2208	Analog Electronic Circuits Lab with Simulation	2	--	--	3	3	50	50	100
B16 EC 2209	Industry oriented Technology Lab	1	--	--	2	2	50	--	50
B16 ENG 2204	Industry Oriented Training	1	--	--	2	2	50	--	50
Total		28	18	6	10	34	380	520	900

SWITCHING THEORY AND LOGIC DESIGN

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To provide insight of number systems and minimization of Boolean functions
2. To introduce the design of various combinational and sequential circuits
3. To introduce the design of Finite State Machines and Asynchronous Machines

Course Outcomes

By the end of the course the learners (students) will be able to

1. Able to understand various basic number system conversion and simplification of Boolean expressions
2. Design and analyze combinational and sequential circuits using logic gates, latches and flip-flops.
3. Analyze and design Finite State Machines
4. Analyze and design Asynchronous Machines

SYLLABUS**Number Systems And Codes**

Number Systems, Base Conversion Methods, Complements of Numbers, Codes, Error detecting and Error Correcting Codes.

Logic Gates And Minimization Of Boolean Functions

Symbols and Truth Tables of Gates – AND, OR, NOT, NAND, NOR and XOR. Proof of Boolean theorems and functions– Karnaugh Map (up to 6 variables) and QuineMcClusky methods.

Combinational Logic Circuits And Design

Logic Design of Combinational circuits – Binary addition, Subtraction, Multiplexers, Demultiplexers, Decoders, Encoders, Code Conversion, Priority Encoders, Seven – segment Displays, Comparators and PLDs.

Sequential Logic Circuits And Design

The Flip-flops – SR, RS, JK, MSJK, T and D-Flip-flops. Design of Clocked Flip-flops, Flip-Flop conversion from one type to another. Design of Shift Registers and Counters.

Traditional Approaches to Sequential Analysis and Design

Analysis and Design of Finite State Machines, State Reduction.

Asynchronous Finite State Machines

Analysis and Design of Asynchronous Machines, Cycles, Races and Hazards.

Text Books:

1. Switching and Finite Automata Theory, 2nd Edition, ZviKohavi, Tata McGraw-Hill, 1978.
2. Fundamentals of Digital Circuits-A.Anand Kumar-2nd Edition-PHI Learning Private Limited.

Reference Books:

1. Introduction to Switching Theory and Logical Design, 3rd Edition, Frederick J. Hill and Gerald R. Peterson, John Wiley and Sons, 1981.
2. An Engineering Approach to Digital Design, William I. Fletcher, PHI.

ELECTRO MAGNETIC FIELD THEORY & TRANSMISSION LINES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the concepts of static electric field , steady magnetic field and time varying electromagnetic fields in real time applications.
2. To introduce Maxwell's equations and their applications in practical situations.
3. To introduce the fundamental theory of electromagnetic wave propagation in bounded and unbounded media.
4. To study the propagation of energy in practical transmission lines and wave guides.

Course Outcomes:

After the completion of the course, Students should have the

1. Ability to apply the knowledge of mathematics, Science and engineering to the analysis and design of systems involving electric and magnetic fields as well as electromagnetic Waves.
2. Ability to identify, formulate and solve engineering problems in the area of electric and Magnetic fields and waves.
3. Ability to use Maxwell's equations to solve electromagnetic field problems.
4. Ability to apply the knowledge of electromagnetic fields in practical transmission lines and waveguides.

SYLLABUS**Electrostatics:**

Introduction, Coulomb's law and electric field intensity, electric field due to different types of charge distributions, electric flux density, gauss's law and applications. Energy and potential, electric field in terms of potential gradient, electric dipole, stored energy in static electric field and energy density, convection and conduction currents, continuity equation, conductors in electric field, relaxation time, dielectrics in electric field, Laplace's and Poisson's equations, uniqueness theorem, different capacitance configurations, Boundary conditions on $\vec{E}$ & $\vec{D}$ at the interface between two media, Related Problems.

Magneto statics:

Introduction, Biot-savart's law, Ampere's circuital law, applications of Ampere's circuital law, magnetic flux density, Gauss's law for magnetic fields, scalar and vector magnetic potentials, forces due to magnetic fields, magnetization in materials, inductance, boundary conditions on $\vec{H}$ & $\vec{B}$ at the interface between two media, energy stored in steady magnetic field, Related problems.

Time varying fields and Maxwell's equations:

Introduction, Faraday's law of electromagnetic induction, Transformer emf and motional emf, Maxwell's equations in integral and differential forms, word statements, Maxwell's equations using phasor notation, Boundary conditions on $\vec{E}$, $\vec{D}$, $\vec{H}$ & $\vec{B}$ at the interface between two media, Related problems.

Electromagnetic Waves:

Introduction, Wave equations for free space and for a conductive medium, uniform plane waves, properties of uniform plane waves, Relation between E and H in uniform plane wave, wave propagation in lossless and lossy media, Propagation in good conductors and good dielectrics, depth of penetration, polarization, Reflection of plane waves by a perfect conductor for normal and Oblique incidences, Reflection of plane waves by a perfect dielectric for normal and Oblique incidences, Brewster angle and critical angle, Poynting's theorem, Related Problems.

Transmission lines:

Introduction, types of transmission lines, equivalent circuit of transmission line, Primary and secondary constants of the line, Transmission line equations, characteristic impedance and expression for characteristic impedance, Reflection coefficient, standing wave ratio, lossless line, distortion less line, input impedance of transmission line, shorted and open circuited lines, impedance transformation with $\frac{\lambda}{8}$, $\frac{\lambda}{4}$ and $\frac{\lambda}{2}$ lines, Construction of smith chart, applications of smith chart, Single stub matching, Related problems.

Rectangular Waveguides:

Introduction, TM modes in rectangular waveguides, TE modes in rectangular waveguides, Impossibility of TEM mode in waveguides, Characteristics of TE and TM modes, cutoff frequency, cutoff wavelength, phase and group velocities, characteristic wave impedance, dominant mode, related problems.

Text Books:

1. Principles of Electromagnetics - N.O.Sadiku, Oxford University Press, 4th edition.
2. Electromagnetic field theory and Transmission Lines – G.SasibhusanaRao, Wiley India Pvt.Ltd.

Reference Books:

1. EM Waves and Radiating Systems – E.C.Jordan, Printice Hall India.
2. Electromagnetics with applications – Kraus and Fleisch, McGraw Hill.
3. Electromagnetic field theory and Transmission lines – G.S.N.Raju, Pearson Education Pvt.Ltd.
4. Engineering Electromagnetics – W.A.Hayt and JABuck, Tata McGraw Hill.

PULSE AND DIGITAL CIRCUITS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students should learn

1. Applications of integrator, differentiator, clippers and clamper circuits.
2. Design of various multivibrators for various applications.
3. Design various Time base generators.
4. Synchronization techniques for sweep circuits.
5. Different logic families, realize logic gates using diodes and transistors.

Course Outcomes:

Students will be able to

1. Understand the applications of integrator, differentiator, clippers and clamper circuits.
2. Design different multivibrators for various applications.
3. Design different time base generators.
4. Analyze synchronization techniques for sweep circuits.
5. Understand different logic families & realize logic gates using diodes and transistors.

SYLLABUS**Linear Wave Shaping:**

High pass and Low pass RC circuits, Response of High pass and Low pass RC circuits to sinusoidal, step, pulse, square, exponential and Ramp inputs, High pass RC circuit as a differentiator, Low pass RC circuit as an integrator. Attenuators and its application as CRO probe, RL and RLC Circuits and their response for step input, Ringing Circuit.

Nonlinear Wave Shaping:

Diode clippers, Transistor Clippers, Clipping at two independent levels, Comparator, Applications of voltage Comparators, Diode Comparator, Clamping Operation, Clamping Circuits using Diode with Different Inputs, Clamping Circuit Theorem, Practical Clamping circuits, Effect of diode Characteristics on Clamping Voltage.

Bistable Multivibrators:

Transistor as a switch, Switching times of a transistor, Design and Analysis of Fixed-bias and self-bias transistor binary, Commutating capacitors, Triggering schemes of Binary, Transistor Schmitt trigger and its applications.

Monostable And AstableMultivibrators:

Design and analysis of Collector coupled Monostable Multivibrator, Expression for the gate width and its waveforms. Design and analysis of Collector coupled Astable Multivibrator, expression for the Time period and its waveforms, The Astable Multivibrator as a voltage to frequency convertor.

Time Base Generators:

General features of a time-base signal, Methods of Generating time base waveform, Exponential voltage sweep circuit, Basic principles of Miller and Bootstrap time base generators, transistor Miller sweep generator, transistor Bootstrap sweep generator, Current Sweep circuit, Linearity correction through adjustment of driving Waveform.

Synchronization And Frequency Division:

Principles of Synchronization, Frequency division in sweep circuit, Synchronization of AstableMultivibrators, Synchronization of a sweep circuit with symmetrical signals, Sine wave frequency division with a sweep circuit.

Logic Families:

Realization of gates using diodes and Transistors, RTL, DTL.

TEXT BOOKS:

1. Pulse Digital and Switching Waveforms, J. Millman ,H. Taub, andM.S.PrakashRao McGraw-Hill, Second Edition .
2. Pulse and Digital Circuits, A. Anand Kumar, PHI, Second Edition, 2005.

Reference Books:

1. Pulse and Digital Circuits, K.VenkatRao, Pearson Education India, 2nd Edition, 2010.
2. Pulse Digital and Switching Waveforms, J. Millman and H. Taub, McGraw-Hill, 2nd Edition 1991.
3. Pulse Switching and Digital Circuits – David A.Bell, PHI,5thEdn., Oxford University Press.

ANALOG COMMUNICATIONS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand the fundamental concepts of communication systems.
2. To familiarize with the concepts of linear or amplitude modulation (AM, DSB-SC, SSB and VSB) and demodulation techniques.
3. To familiarize with the concepts of angular or non-linear modulation (FM and PM) and demodulation techniques.
4. To understand and compare different analog modulation schemes.
5. To familiarize with techniques for generating and demodulating narrow-band and wide-band frequency and phase modulated signals.
6. To provide a good understanding of the behavior of analog communications in the presence of noise.
7. To develop a clear insight into the relations between the input and output signals in various stages of a transmitter and a receiver of AM & FM systems.
8. To classify and discuss different types of transmitters and receivers as applicable to analog communication systems.

Course Outcomes:

Upon successful completion of this course the student will be able to

1. Understand the need for modulation and learn about the basic elements of communication system.
2. Understand the concepts of Analog Modulation and Demodulation techniques.
3. Evaluate various parameters of analog modulated waveform in Time and Frequency domain.
4. Analyze and compare the performance of various analog modulation techniques in the presence of noise.
5. Analyze different characteristics of transmitters.
6. Analyze different characteristics of receivers.

SYLLABUS

Linear Modulation Systems:

Need for Modulation, Frequency Translation, Method of Frequency Translation, Amplitude Modulation, Modulation Index, Spectrum of AM Signal, Modulators and Demodulators (Diode detector), DSB-SC Signal and its Spectrum, Balanced Modulator, Synchronous Detectors, SSB Signal, SSB Generation Methods, Power Calculations in AM Systems, Application of AM Systems.

Angle Modulation Systems:

Angle Modulation, Phase and Frequency Modulation and their Relationship, Phase and Frequency Deviation, Spectrum of an FM Signal, Bandwidth of Sinusoidally Modulated FM Signal, Effect of the Modulation Index on Bandwidth, Spectrum of Constant Bandwidth FM, Phasor Diagram for FM Signals. FM Generation: Parameter variation method, Indirect method of Frequency Modulation (Armstrong Method), Frequency Multiplication, PLL FM Demodulator, Pre – emphasis and De – emphasis, Comparison of FM and AM.

Noise In AM and FM Systems:

Sources of Noise, Resistor Noise, Shot Noise, Noise in AM Systems, Noise in Angle Modulation Systems, Comparison between AM and FM with respect to Noise, Threshold in Frequency Modulation

Radio Transmitters:

Classification of Radio Transmitters, AM and FM Transmitters, Radio Telegraph and Telephone Transmitters, SSB Transmitters

Radio Receivers:

Radio receiver Types, AM Receivers – RF Section, Frequency Changing and Tracking, Intermediate Frequency and IF Amplifiers, Automatic Gain Control (AGC); FM Receivers – Amplitude Limiting, FM Demodulators, Ratio Detectors, ISB Receiver, Comparison with AM Receivers. Communication Receivers: Extensions of the Super-heterodyne Principles, Additional Circuits.

Text Books:

1. Principles of Communication Systems, H. Taub and D. L. Schilling, McGraw Hill, 1971.
2. Electronic Communications System : Fundamentals Through Advanced by Wayne Tomasi.

Reference Books:

1. Modern Digital and Analog Communication Systems, B. P. Lathi (2nd Edition).
2. Electronic Communications Modulation and Transmission, Robert J. Schoenbeck, PHI N. Delhi, 1999.
3. Communication systems, R.P.Singh and S.D.Sapre 2nd edition TMH 2008.
4. Communication Systems, Simon Haykins (2nd Edition).
5. Electronic Communication Systems, G. Kennedy, McGraw Hill, 1977 (2nd Edition)

SIGNALS AND SYSTEMS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To introduce the fundamental concepts and techniques associated with the understanding of signals and systems.
2. To familiarize with techniques suitable for analyzing both continuous-time and discrete time LTI systems using transforms.
3. To familiarize with development of the mathematical skills to solve problems involving convolution, filtering, and sampling.

Course Outcomes:

After completion of the course the student will be able to

1. Understand the basic concepts of signals and systems.
2. Analyze the spectral characteristics of Continuous Time and Discrete Time periodic and aperiodic signals using Fourier analysis.
3. Analyze system properties based on impulse response and Fourier analysis
4. Classify systems based on their properties and determine the response of LTI systems using convolution and also understand the concept of correlation between signals.
5. Apply Z- transforms for analyzing discrete-time signals and systems
6. Understand the process of sampling and the effects of under sampling.

SYLLABUS**Introduction to signals and linear time Invariant systems**

Continuous –Time and Discrete –Time signals, Signal Energy and Power, Periodic Signals, Even and odd Signals, Continuous- Time complex Exponential and Sinusoidal Signals, Discrete –Time complex Exponential and Sinusoidal Signals, Periodicity of Continuous –Time and Discrete –Time Complex Exponentials, Continuous-Time Unit impulse and Unit step Signals, Discrete- Time Unit Impulse and Unit Step Sequences, Continuous –Time and Discrete –Time Systems, Interconnections of Systems, Basic System Properties of Continuous –Time and Discrete –Time Systems, Introduction of continuous–Time LTI Systems and Discrete –Time LTI Systems, Casual LTI Systems Described by Differential and Difference Equations, Singularity Functions.

Fourier Series Representation of Periodic Signals

Introduction, Fourier Series Representation of continuous time Periodic Signals, convergence of the Fourier Series, Properties of continuous time Fourier Series, Fourier Series representation of discrete time periodic signals, Properties of discrete time Fourier Series.

Continuous and Discrete time Fourier Transform

Introduction, Representation of Aperiodic signals, the continuous time Fourier Transform, The Fourier Transform for periodic signals, Properties of the continuous time Fourier Transform, Systems characterized by linear constant-coefficient Differential equations. Discrete time Fourier Transform, Fourier Transform for periodic signals, Properties of the Discrete time Fourier Transform, Systems characterized by linear constant co-efficient Difference equations.

Convolution of signals

Introduction of convolution integral and convolution sum, Graphical interpretation of Convolution, System analysis by Convolution, Convolution as a superposition of impulse response, Convolution of a function with a unit impulse, Convolution relationships, Signal comparison.

Correlation of signals

Introduction of Correlation of signals, properties of correlation functions, Introduction of Energy Density Spectrum(ESD) and Power Density Spectrum(PSD), Relation between Autocorrelation function and ESD/PSD, Relation between Convolution and Correlation. Correlation functions for nonfinite energy signals.(8 Periods)

Sampling Theorem and Z-transform

Introduction to signal reconstruction from its samples using interpolation, the effect of under sampling: aliasing, Discrete time processing of continuous time signals, sampling of Discrete time signals. The Z-Transform, The Inverse Z-Transform, Properties of Z-Transform, The initial and final value theorems, some common Z-transform pairs, Analysis and characterization of LTI systems using the Z-Transforms, System function algebra and block diagram representation.(10 Periods)

Text Books:

1. Signals and Systems- Alan V. Oppenheim, Alan S. Willsky and Ian T. Young, PHI, 2ndEdn.
2. Signals Systems and Communication-B. P. Lathi, BS Publication.

Reference Books:

1. Signals and Systems – P.RamakrishnaRao, TMH.
2. Signals and Systems- A.AnandaKumar,PHI.
3. Signals and Systems – P.RameshBabu,SciTech Publication.
4. Signals and Systems-K. Raja Rajeswari and B. V. Rao, Prentice Hall of India

ENVIRONMENTAL STUDIES
(Common to ECE,EEE & ME)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

Students learn

1. To develop an awareness and sensitivity to the total environment and its related problems.
2. To participate actively participation in environmental protection and improvement.
3. To develop skills for active identification and development of solutions to environmental problems
4. To evaluate environment programmes in terms of social, economic, ecological and aesthetic factors.
5. To Create a “CONCERN AND RESPECT FOR THE ENVIRONMENT”

Course Outcomes:

Students will be able to

1. Get awareness among the students about the nature and natural ecosystems.
2. Learn sustainable utilization of natural resources like water, land, minerals, air.
3. Learn resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation.
4. Safe guard against industrial accidents particularly nuclear accidents.
5. Learn Constitutional provisions for the protection of natural resources.

SYLLABUS

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness.
Global Environmental Crisis

Ecosystems:

Basic concepts, Forest Ecosystems, Grassland Ecosystems and Desert Ecosystems, Aquatic Ecosystems

Biodiversity:

Introduction to Biodiversity, Value of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of bio-diversity.

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people, **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems

Energy Resources: Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and global warming, acid rain and Ozone layer depletion.

Environmental Problems in India:

Drinking water, Sanitation and Public health, population growth and environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

Text Books:

1. Environmental Studies (From Crisis to Cure) by R. Rajagopalan, Oxford university Press, 2008
2. Environmental Studies by Anubha Kaushik & C.P. Kauskik, New Age International (P) Ltd, New Delhi, 2006

Reference Books:

1. Environmental Sciences by G.Tyler Miller, JR,10th ed, Thomson publishers, 2004

ANALOG COMMUNICATION LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. The purpose of this course is to provide the student with a practical perspective of various Analog communication modules.
2. This course also helps the student to implement various analog modulation and demodulation schemes using discrete components.
3. To be familiar with the design of various parameters of LPF , BPF and HPF
4. To design IF and RF amplifiers and plot their frequency response.
5. To be familiar with different types of experiments like pre-emphasis, de-emphasis and DSB-SC waveform generators.

Course Out Comes:

1. Design and implement modulation and demodulation circuits for amplitude modulation technique.
2. Design and implement modulation and demodulation circuits for frequency modulation technique.
3. Design second order passive and active filters for various frequency bands.
4. Construct the circuit and study the characteristics of different transmitter and receiver circuits such as Harmonic generator, RF Amplifier, IF Amplifier, pre-emphasis and -emphasis.

SYLLABUS

1. Generation of AM Signal and measurement of Modulation Index.
2. Diode Detector for AM Signals.
3. Generation of FM Signal.
4. FM Detector.
5. Receiver Measurements.
6. Balanced Modulator.
7. Passive Filters (LPF, HPF, BPF).
8. Active Filters.
9. Attenuator.
10. Equalizer and Twin-T-Network.
11. Frequency Multiplier/Limiter.
12. SSB Generation and Detection.
13. Pre-emphasis and De-emphasis.
14. PLL.
15. IF Amplifier.

Reference : Lab Manuals.

ANALOG ELECTRONIC CIRCUITS LAB WITH SIMULATION
(Common to ECE & EEE)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. This laboratory course enables students to get practical experience in design, assembly and evaluation of analog electronic circuits. They will use Multisim to test their electronic designs.

Course Outcomes:

Students will be able to:

1. Acquire a basic knowledge on simple applications of operational amplifier.
2. Observe the amplitude and frequency responses of negative feedback amplifier and two stage RC coupled amplifier.
3. Design and test sinusoidal oscillators.
4. Design and test a power amplifier.
5. Design, construct, and take measurement of the analog electronic circuits to compare experimental results in the laboratory with theoretical analysis.
6. Use Multisim to test their electronic design.

LIST OF EXPERIMENTS

1. Design of LC Oscillators (Hartley Oscillator, Colpitts Oscillator)
2. Design of RC Oscillators (Wien Bridge Oscillator, RC phase Shift Oscillator)
3. Design of Basic Applications of Operational Amplifier.
4. Frequency response of Two Stage RC Coupled Amplifier.
5. Frequency response of Current Series Feedback Amplifier (with and without feedback)
6. Measurement of resonant frequency, bandwidth and quality factor of single Tuned Voltage Amplifier.
7. Calculation of Collector Circuit efficiency of Class B Push Pull Power Amplifier.

LIST OF EXPERIMENTS
(Simulation)

8. Design of LC Oscillators (Hartley Oscillator, Colpitts Oscillator)
9. Design of RC Oscillators (Wien Bridge Oscillator, RC phase Shift Oscillator)
10. Design of Basic Applications of Operational Amplifier.
11. Frequency response of Two Stage RC Coupled Amplifier.
12. Frequency response of Current Series Feedback Amplifier (with and without feedback)
13. Measurement of resonant frequency, bandwidth and quality factor of single Tuned Voltage Amplifier.
14. Calculation of Collector Circuit efficiency of Class B Push Pull Power Amplifier.

Reference : Lab Manuals

INDUSTRY ORIENTED TECHNOLOGY LAB

Lab	: 2 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Credits	: 1

Course Objectives:

This laboratory course enables students to get practical experience in design, interfacing of different sensors and motors with the microcontrollers like Aurdino and Single Board Computer like Raspberry Pi.

Experiments:

1. Familiarization with Aurdino microcontroller and raspberry pi, different sensors, drivers and motors.
2. Control of conducting and non-conducting periods of LED using Aurdino microcontroller.
3. Interfacing humidity and temperature sensor (DHT11) with Aurdino microcontroller.
4. Interfacing ultrasonic sensor, PIR sensor with Aurdino microcontroller.
5. Interfacing DC motor with Aurdino microcontroller using L298 motor driver.
6. Interfacing and control of servo motor with Aurdino microcontroller.
7. Introduction to Raspberry Pi and interfacing different sensors and motors to it.
8. Interfacing camera with Raspberry Pi and performing different operations using open CV (Computer Vision).
9. Controlling DC motor based on DHT11 sensor output using Raspberry Pi.
10. Controlling of DC motor based on ultrasonic sensor output using Raspberry Pi.
11. Image capturing based on PIR sensor output using Raspberry Pi.
12. Introduction to GSM/GPS module and interfacing them with Aurdino and Raspberry Pi.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Record/Report-10 Marks, Exam-10 Marks and Attedndance-5 Marks)

INDUSTRY ORIENTED TRAINING
(Common to ECE & EEE)

Lab : 2 Periods
Exam : 3 Hrs.

Sessionals : 50
Credits : 1

Course Objectives:

1. Be familiar with core JAVA.
2. Master the implementation of Applet programming.
3. Master the implementation of Networking concepts in core JAVA.
4. Be familiar with CORBA , J2EE, RMI concepts..

Course Outcomes:

1. Application using implementation of core JAVA concepts.
2. Application using implementation of AWT, Applets
3. Applications using Networking concepts in view of industry.

Syllabus: Industry Oriented Applications on following topics.

BASIC CONCEPTS

Fundamentals: HTML, OOP Concepts, Comparing JAVA with C & C++, JAVA Programming language Syntax, Variables, Data types, statements and expressions.

Control Statements: If else, for, while, and do while loops, Switch statements.

Arrays & Structures: One Dimensional & Two Dimensional Arrays, Named Structures.

Functions: Parameter Passing, Static Modifier.

IMPLEMENTATION (Using JAVA)

Classes and Interfaces
Threads and multithreaded programming
packages.

Applications of AWT, Applets and Networking concepts and solving simple to complex problems in perspective of industry requirements.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest- 25 Marks)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

III/IV B.TECH

(With effect from 2016-2017 Admitted Batch onwards)

Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 EC 3101	Linear ICs and Applications	4	3	1	-	4	30	70	100
B16 ENG 3101	Principles of Economics and Management	4	3	1	-	4	30	70	100
B16 EC 3102	Computer Architecture and Organization	4	3	1	-	4	30	70	100
B16 EC 3103	Antennas and Propagation	4	3	1	-	4	30	70	100
B16 EC 3104	Electronic Measurements and Instrumentation	4	3	1	-	4	30	70	100
B16 EC 3106	Linear Integrated Circuits & Pulse Circuits Lab with Simulation	4	3	1	-	4	30	70	100
B16 EC 3107	Digital Integrated Circuits & Hardware Descriptive Language	2	-	-	3	3	50	50	100
B16 ENG 3102	Verbal & Quantitative Aptitude-I	2	5	--	--	5	100	--	100
#M-I	MOOCS-I	2	4	--	--	4	100	--	100
#M-II	MOOCS – II	2	4	--	--	4	100	--	100
Total		34	31	6	6	43	580	520	1100

#M-I	B16 ENG 3103	Basic Coding
	B16 EC 3109A	Microwave Theory and Techniques
	B16 EC 3109B	Analysis and Design Principles of Microwave Antennas
	B16 EC 3109C	Principles of Digital Communications
#M-II	B16 EC 3110A	Introduction to Wireless and Cellular Communications
	B16 EC 3110B	Fabrication Techniques for MEMS-based Sensors: Clinical Perspective
	B16 EC 3110C	Introduction to information theory, Coding and cryptography
	B16 EC 3110D	Digital Image Processing

LINEAR ICS AND APPLICATIONS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To understand the internal diagram and characteristics of operational amplifier.
2. To learn about the linear and non-linear applications of operational amplifier.
3. To know the concepts of Active filters and waveform generator
4. To understand the concepts of Analog to Digital Converters and Digital to Analog Converters
5. To understand the industrial applications using 555 timer, PLL.

Course Outcomes:

Upon completion of the course, students will be able to

1. Understand the terminal characteristics of op-amps and design/analyze fundamental circuits based on op-amps.
2. Analyze the effect of feedback on the performance of op-amp.
3. Design and analyze of non-linear circuits and active filters.
4. Design and Analyze of various applications using IC 565 and IC 555.
5. Understand the operation of Analog to Digital and Digital to Analog Converters

SYLLABUS

Applications of Operational Amplifiers : Basics of Op-Amp, Block Diagram, open loop and closed loop op-amp configurations, Frequency compensation Techniques, Logarithmic Amplifier, Instrumentation Amplifiers, Voltage to Current and Current to Voltage Converters. Op-amp As a Comparators, Schmitt trigger, Wave form Generators, Sample and Hold Circuits, Rectifiers, Peak Detection

Active Filters: Butterworth type LPF, HPF, BPF, BEF, All-pass Filters, Higher Order Filters and their Comparison, Switched Capacitance Filters.

Oscillators: Op-Amp Phase Shift, Wein-bridge and Quadrature Oscillator, Voltage Controlled Oscillators, Analog Multiplexers.

Special ICs: 555 Timers, 556 Function Generator ICs and their Applications, Three Terminal IC Regulators, IC 565 PLL and its Applications, Voltage to Frequency and Frequency to Voltage Converters.

Digital to Analog and Analog to Digital Converters: DAC techniques, Weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, Different types of ADCs-parallel Comparator type ADC, Counter type ADC, Successive approximation ADC and ADC specifications.

Text Books:

1. Op-Amps and Linear ICs- Ramakanth Gayakwad, PHI, 1987.
2. Integrated Circuits- Botkar, Khanna Publications.

Reference Books:

1. Microelectronics- Jacob Millman.
2. Applications of Linear ICs- Clayton.

PRINCIPLES OF ECONOMICS AND MANAGEMENT

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives

1. To familiarise Engineering students with the real world of business.
2. To provide capacity to analyse business decisions by understanding the origin & concepts of economics.
3. To provide capacity to analyse business decisions by understanding the concepts of management.
4. To help engineering students know the need and importance of entrepreneurship.

Course outcomes

1. Students will be able to gain empirical knowledge and understand the complete frame work of business.
2. To analyse the concepts pertaining to economic decision making.
3. To analyse the concepts of Managerial decision making.
4. To inculcate the spirit of Entrepreneurship and gain knowledge for setting up an enterprise.

SYLLABUS

Introduction to Managerial Economics:

Wealth, Welfare and Scarce Definitions of Economics, Micro and Macro Economics, Demand –Law of Demand, Elasticity of Demand, Types of Elasticity and Factors Determining price Elasticity; Demand :Utility-Law of Diminishing Marginal Utility and its limitations.

Conditions of Different Market Structures:

Perfect Competition, Monopolistic Competition, Monopoly, Oligopoly and Duopoly.

Forms of Business Organisation:

Sole Proprietorship, Partnership, Joint Stock Company-Private Limited and public limited Companies. Public Enterprise and their types.

Introduction to Management:

Functions of Management – Taylor's Scientific Management; Henry Fayol's Principles of Management;

Human Resource Management:

Basic functions of HR Manager; Man Power Planning, Recruitment, Selection, Training, Development, Placement, Compensation and Performance Appraisal (In Brief).

Production Management:

Production Planning and Control, Plant Location, Break-Even Analysis, Assumptions and Applications.

Financial Management:

Types of capital; Fixed and Working Capital and Methods of Raising Finance; Depreciation; Straight Line and Diminishing Balance Methods.

Marketing Management:

Functions of Marketing and Distribution channels

Entrepreneurship-

Entrepreneurial Functions, Entrepreneurial Development; Objectives, Training, Benefits; Phase of Installing a Project.

Text Books:

1. K KDewtt, Modern Economic Theory ,S.Chand& Company New Delhi-55
2. S.C.Sharma and Banga T.R, Industrial Organisation and engineering EconomicsKhanna Publications Delhi-6

Reference Books:

1. A.R.Araryasri Management Science,Tata McGraw-Hill,New Delhi-20
2. A.R.Araryasri, Managerial Economics and financial analysis, Tata McGraw-Hill,New Delhi-20

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Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand Basic building blocks of a Digital Computer.
2. To understand the designing of a CPU.
3. Analyze the concept of interfacing of peripheral devices & Memory.
4. To understand the concept of Memories and design of Main memory.

Course Outcomes:

After successfully completing the course students will be able to:

1. Understand how computers represent and manipulate data.
2. Develop the general architecture design of a digital computer.
3. Learn the art of Microprogramming.
4. Develop independent learning skills to interface main memory & I/O.

SYLLABUS

Register Transfer and Micro operations:

Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.

Basic Computer Organization:

Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference Instructions, Input - Output and Interrupt, Complete Computer Description.

Micro programmed Control:

Control Memory, Address Sequencing, Microinstruction Formats, Micro program Example, Design of Control Unit.

CPU Organization:

Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer (RISC).

Input – Output Organization:

Peripheral Devices, Input - Output Interface, Asynchronous Data Transfer, Modes of Transfer, Priority Interrupt, Direct Memory Access (DMA), Input- Output processor, CPU-IOP communication.

Memory Organization:

Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.

Text Books:

1. Computer System Architecture, M. Morris Mano, PHI Publications, (3rd Edition May 1996).
2. Computer Architecture and Organization. John P.Hayes. McGraw Hill International, (3rd Edition).

Reference Books:

1. Computer Organization, V. Carl Hamacher, Zvonko G. Vranesic and Safwat G. Zaky, McGraw Hill International, (4th Edition).
2. Digital Computer Fundamentals, Thomas C. Bartee.

ANTENNAS AND PROPAGATION

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Understand the radiation mechanism of antennas and to learn about basic parameters like impedance, gain, directivity, bandwidth, effective length, beam width and radiation pattern etc.
2. Derive fields and power radiated by elemental antenna, Half wave dipole, quarter wave monopole and values of their radiation resistance.
3. Understand the necessity of antenna arrays and to learn about theory of uniform linear arrays, broad side and end fire arrays, non uniform linear arrays like binomial arrays and pattern multiplication.
4. Learn about antenna synthesis and different methods of synthesis and design antenna arrays for different applications.
5. Have knowledge about practical LF, HF, VHF, UHF and Microwave antennas and be able to design practical antennas.
6. Have knowledge about various antenna measurements and be able to conduct different types of antenna measurements.
7. Have knowledge about various types of radio wave propagation like Ground wave, Ionospheric, space wave and Duct propagation and be able to design different types of communication links

Course Outcomes:

After completion of the course the student will be able to

1. Understand Radiation mechanism and functions of antennas, identify antenna Parameters derive expressions for antenna parameters.
2. Analyze and design wire and aperture antennas for different applications.
3. Analyze and design (or synthesize) Antenna arrays.
4. Capable of performing various antenna measurements and come up with conclusions about antenna parameters and performance.
5. Identify characteristics of radio wave propagation and be able to design different types of communication links for different frequency bands

SYLLABUS

Fundamentals of Antennas & Radiation from Antennas:

Definition of antennas, functions of Antennas, properties of antennas, antenna parameters, polarization, basic antenna elements, radiation mechanism, radiating fields of alternating current element, radiated power and radiation resistance of current element, different types of current distribution on linear antennas, radiated fields, radiated power and radiation resistance of half-wave dipole and quarter – wave monopole, directional characteristics of dipole antennas.

Linear Arrays:

Uniform linear arrays, field strength of a uniform linear arrays, locations of principal maximum, null and secondary maxima, first side lobe level, analysis of broad side and endfire , Pattern multiplication, binomial arrays, effect of earth on vertical patterns, methods of excitation of antennas, impedance matching techniques, transmission loss between transmitting and receiving antennas – Friis formula, antenna noise temperature and signal-to-noise ratio.

Antenna array synthesis:

Introduction, synthesis methods, fourier transform method, Woodward Lawson method, Dolph-chebychev linear method, Taylor method, Laplace transform method.

Practical Antennas – LF,MF,HF,VHF & UHF antennas

Classification of antennas according to type of radiation and type of current distribution of antennas – Isotropic, Omnidirectional & directional antennas, standing wave and travelling wave antennas, Classification according to frequency of operation – LF, MF, HF, VHF & UHF, brief introduction to LF & MF antennas, earth mat, counterpoise earth, top capacitance hat.

HF,VHF& UHF Antennas - V Antennas, Inverted V Antennas, Rhombic antennas, folded dipole, Yagi-Uda antenna, Log periodic antenna, Loop And Helical Antennas.

Microwave antennas:

Introduction, types of reflector antennas, corner reflector, parabolic reflector, feed systems for parabolic reflector, horn antennas, slot antennas and impedance of slot antennas, Babinet's principle, lens antennas and microstrip antennas.

Antenna measurements

Introduction, measurement ranges, antenna impedance measurements, antenna gain and directivity measurement, measurement of radiation pattern, beamwidth and SLL, measurement of polarization, measurement of phase, measurement of radiation resistance.

Wave Propagation

Types of radio wave propagation, groundwave propagation and sommerfeld's analysis of ground wave propagation, wave tilt of ground wave, structure of ionosphere, refractive index of ionosphere, mechanism of wave bending by ionosphere, critical frequency, MUF, Skip distance,fading and remedial measures, effect of earth's magnetic field on ionosphere propagation,faraday rotation, tropospheric (space wave) propagation, range of space wave propagation, effective earth radius, field strength of space wave, atmospheric effects on spacewave propagation, duct propagation and scatter propagation.

Text Books:

1. EM waves and Radiating systems – by E. C. JORDAN and K. G. Balmain – PHI , New Delhi.
2. Antenna theory- by C. A. Balanis, john wiley.

Reference Books:

1. Antennas – By J.D. Kraus, McGrawhill.
2. Antenna and wave propagation – by G.S.N Raju, Pearson education.

CONTROL SYSTEMS
(Common to ECE & EEE)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To learn the modeling of linear systems using transfer functions and obtain transfer functions for physical electrical and mechanical systems.
2. To learn to represent systems using block diagrams and signal flow graphs and derive their transfer functions..
3. To learn the significance of time response and find it for system analysis in transient and steady state.
4. To learn the concept of stability and know different techniques of stability analysis.
5. To learn the concept of frequency response and its application for control system analysis.

Course Outcomes:

1. Students will be able to model electrical and mechanical physical systems by applying laws of physics
2. Students will be able to represent mathematical models of systems using block diagrams & Signal Flow Graphs and derive their transfer functions
3. Students will be able to analyze systems in time domain for transient and steady-state behaviour
4. Students will learn the concept of stability and use RH criterion and Root locus methods for stability analysis.
5. Students will learn to obtain frequency response plots of systems and use them for system analysis and stability assessment.

SYLLABUS

Introduction to control systems : Open loop and closed loop systems- Transfer Functions of Linear Systems– Impulse Response of Linear Systems – Mathematical Modeling of Physical Systems – Equations of Electrical Networks – Modeling of Mechanical Systems – Equations of Mechanical Systems, Analogous Systems.

Block Diagrams of Control Systems: Signal Flow Graphs (Simple Problems) – Reduction Techniques for Complex Block Diagrams and Signal Flow Graphs (Simple Examples)- Feedback Characteristics of Control Systems

Time Domain Analysis of Control Systems: Time Response of First and Second Order Systems with Standard Input Signals – Steady State Error Constants – Effect of Derivative and Integral Control on Transient and Steady State Performance of Feedback Control Systems.

Concept of Stability:Routh-Hurwitz Criterion, Relative Stability Analysis, the Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus (Simple Problems to understand theory).

Frequency Domain Analysis of control systems: Bode Plots- Log Magnitude versus Phase Plots- Polar Plots -Correlation between Time and Frequency Responses -Nyquist Stability Criterion -Assessment of Relative Stability -All Pass and Minimum Phase Systems - Constant M and N Circles.

Text Books:

1. I.J.Nagrath and M. Gopal, 'Control Systems Engineering',New Age International Publishers(6th Edition).
2. Benjamin C. Kuo, ' Automatic Control Systems', PHI (5th Edition).

Reference Books:

1. Katsuhiko Ogata, 'Modern Control Engineering', , PHI (4th Edition).
2. Richard C.Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers(8th Edition)

ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The student will

1. Select the instrument to be used based on the requirements.
2. Understand and analyze different signal generators and analyzers
3. Understand the design of oscilloscopes for different applications.
4. Understand the principle of operation and working of various types of bridges for measurement of parameters

Course Outcomes:

The student will be able to

1. Evaluate basics of measurement systems, principle of basic meter
2. Evaluate how a signal can be generated using different types of meters.
3. Investigate a signal / waveform with different oscillators.
4. Use bridges of many types and measure appropriate parameters.
5. Design different transducers for measurement of different parameters.

SYLLABUS

Performance characteristics of instruments, Static characteristics, Accuracy, Resolution, Precision, Expected value, Error, Sensitivity. Errors in Measurement, Dynamic Characteristics-speed of response, Fidelity, Lag and Dynamic error. DC Voltmeters- Multi-range, Range extension/Solid state and differential voltmeters, AC voltmeters, True RMS responding voltmeter, Multi-meter for Voltage, Current and resistance measurements.

Signal Generator- fixed and variable, AF oscillators, Standard and AF sine and square wave signal generators, Function Generators, Square pulse, Random noise, sweep, Arbitrary waveform. Wave Analyzers, Harmonic Distortion Analyzers, Spectrum Analyzers, Digital Fourier Analyzers.

Oscilloscopes CRT features, vertical amplifiers, horizontal deflection system, sweep, trigger pulse, delay line, sync selector circuits, Dual beam CRO, .Dual trace oscilloscope, sampling oscilloscope, digital storage oscilloscope, Lissajous method of frequency measurement, standard specifications of CRO, CRO probes.

AC Bridges Measurement of inductance- Maxwell's bridge, Anderson bridge. Measurement of capacitance - Schering Bridge. Wheat stone bridge. Wien Bridge, Errors and precautions in using bridges.

Transducers- active & passive transducers : Resistance, Capacitance, inductance; Strain gauges, LVDT, Piezo Electric transducers, Resistance Thermometers, Thermocouples, Thermistors, Sensistors.

Text Books:

1. Electronic instrumentation, second edition - H.S.Kalsi, Tata McGraw Hill, 2004.
2. Modern Electronic Instrumentation and Measurement Techniques – A.D. Helfrick and W.D. Cooper, PHI, 5th Edition, 2002.

Reference Books:

1. Electronic Instrumentation & Measurements - David A. Bell, PHI, 2nd Edition, 2003.
2. Electronic Test Instruments, Analog and Digital Measurements - Robert A.Witte, Pearson Education, 2nd Ed., 2004.

LINEAR INTEGRATED CIRCUITS & PULSE CIRCUITS LAB WITH SIMULATION**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2**

Course Objectives:

This laboratory course enables students to get practical experience in design, assembly and evaluation of Linear integrated circuits & Pulse Circuits. They will use Multisim to test their electronic designs.

Course Outcomes :

Students will be able to:

1. Design and conduct experiments on RC low pass and high pass circuits.
2. Observe operation of UJT Sweep Generator.
3. Design and test different types of Multi vibrators
4. Acquire a basic knowledge on simple applications of operational amplifier.
5. Design, construct Schmitt trigger using operational amplifier.
6. Use Multisim to test their electronic designs.

LIST OF EXPERIMENTS

1. Linear Wave Shaping
 - a) Passive RC Differentiator
 - b) Passive RC Integrator
2. Non Linear Wave shaping
 - a) Clipping Circuits
 - b) Clamping Circuits
3. Self biasbistableMultivibrator
4. Schmitt Trigger Using μA 741
5. UJT Sweep Generator
6. AstableMultivibrator using 555 timer

LIST OF EXPERIMENTS (Simulation)

1. Linear Wave Shaping
 - a) Passive RC Differentiator
 - b) Passive RC Integrator
2. Non Linear Wave shaping
 - a) Clipping Circuits
 - b) Clamping Circuits
3. Self biasbistableMultivibrator
4. Schmitt Trigger Using μA 741
5. UJT Sweep Generator
6. AstableMultivibrator using 555 timer

Reference Books:

- 1.Op-Amps Linear ICs- RamakanthGayakwad, PHI, 1987.

DIGITAL INTEGRATED CIRCUITS & HARDWARE DESCRIPTIVE LANGUAGE

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. Learn and understand the basics of digital electronics, Boolean algebra, and able to design the simple logic circuits and test/verify the functionality of the logic circuits
2. Design combinational and sequential logic circuits using digital ICs.
3. This laboratory course enables students to get practical experience in design, assembly and evaluation of digital integrated circuits and HDL lab. Students use digital trainer kit and Xilinx ISE simulator to test their electronic designs.

Course Outcomes:

Upon completion of the course, students will be able to

1. Synthesize, simulate and implement a digital design in a configurable digital circuit with computer supported aid tools and digital trainer kit.
2. Acquire Knowledge of analysis and synthesis of combinational and sequential circuits with simulators and digital trainer kits.
3. Build high level programming (HDL programming) skills for digital circuits.
4. Adapt digital circuits to electronics and telecommunication field.

LIST OF EXPERIMENTS**A. HARDWARE**

1. Verify the operation of following digital components using Digital Trainer Kit
 - a. Logic gates
 - b. Full adder using gates
 - c. Full subtractor using gates
2. Design and verify the logic function of multiplexer and de-multiplexers using digital trainer kit
3. Design code convertors using digital trainer kit
 - a. BCD TO SEVEN segment display
 - b. Priority encoder
 - c. Decoder
4. Verify the operation of following flip-flops using Digital Trainer Kit
 - a. RS flip flop
 - b. JK flip flop
 - c. D flip flop
 - d. T flip flop
5. Design a following synchronous counters using Digital Trainer Kit
 - a. mod 16 counter
 - b. mod 8 counter
6. Verify the following logical functions of shift registers using Digital Trainer Kit
 - a. SIPO
 - b. PISO

B. SOFTWARE

7. Verify the operation of following digital components using ISE Simulator
 - a. logic gates
 - b. full adder using gates
 - c. full subtractor using gates
8. Design and verify the logic function of multiplexer and demultiplexers using ISE Simulator
9. Design code convertors using ISE Simulator
 - a. BCD TO SEVEN segment display
 - b. Priority encoder
 - c. Decoder
10. Verify the operation of following flip-flops using ISE Simulator
 - a. RS flip flop
 - b. JK flip flop
 - c. D flip flop
 - d. T flip flop
11. Design a following synchronous counters using ISE Simulator
 - a. mod 16 counter
 - b. mod 8 counter
12. Verify the following logical functions of shift registers using ISE Simulator
 - a. SIPO
 - b. PISO

Reference Books:

1. Digital design by john f. wakerly 3rd edition pearson publications

VERBAL & QUANTITATIVE APTITUDE – I
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal and Soft Skills-I

Course objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion

Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance

Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions

Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms

Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

BASIC CODING
(Common to ECE & EEE)
(MOOCS-I)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:**At the end of the course students will be able to**

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS**UNIT I Review of Programming constructs**

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations vs pointed locations, Referencing a 2D-Matrix

UNIT IV User-defined datatypes, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by RobotLaphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

MICROWAVE THEORY AND TECHNIQUES (MOOCS-I)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Instructor Name: Prof. Girish Kumar

Institute: IIT Bombay

Course Intro: The course will be broadly focusing on analysis, design and development of microwave circuits and systems. The detailed topics to be covered are given below: 1. Introduction to Microwaves-History and applications of Microwaves, Mathematical Model of Microwave Transmission-Microwave transmission modes, waveguides and transmission lines, Impedance Matching, Microwave Network Analysis- ABCD and Scattering parameters. 2. Passive and Active Microwave Devices- Directional Coupler, Power Divider, Magic Tee, Attenuator, Resonator. Microwave active components: Diodes, Transistors, Microwave Tubes 3. Microwave Design Principles- Microwave Filter Design, Microwave Amplifier Design, Microwave Power Amplifier Design, Low Noise Amplifier Design, Microwave Mixer Design, Microwave Oscillator Design. Microwave Antennas- Antenna parameters, Antenna for ground based, airborne and satellite borne systems, Planar Antennas. 4. Microwave Measurements, Microwave Systems, Effect of Microwaves on human body, Monolithic Microwave ICs, RFMEMS for microwave components, Microwave Imaging.

Pre Requisites: Electromagnetic Waves

Core/Elective: Core

UG/PG: Both

Industry Support: Telecom industry, defense industry and space organization

Reference: (1) R.E. Collins, Microwave Circuits, McGraw Hill (2) D.M. Pozar- Microwave Engineering (3) B. Bhat and S.K. Koul -Stripline Like Transmission Structures (4) I. J. Bahl and P. Bhartia- Microwave Solid State Circuit Design

About Instructor: Girish Kumar received the Ph.D. degree in electrical engineering from Indian Institute of Technology Kanpur, India, in 1983. From 1983 to 1985, he was a Research Associate with the Electrical Engineering Department, University of Manitoba, Winnipeg, Canada. From 1985 to 1991, he was an Assistant Professor with the Electrical Engineering Department, University of North Dakota, Grand Forks, ND, USA. Since 1991, he is with Indian Institute of Technology Bombay (IIT-Bombay), Mumbai, India, where he is currently a Professor in the Electrical Engineering Department. He has authored more than 290 papers in the international and national journals and conference proceedings. He is an author of two books and filed for five patents. His research interests include microstrip antennas and arrays, broadband antennas, microwave integrated circuits and systems.

Course plan:

S. No	week	Module Name
1	1	Introduction to Microwaves: History and Applications, Effect of Microwaves on human body
2	2	Microwave Transmission Modes, Waveguides, Transmission Lines
3	3	Impedance Matching, ABCD and S-parameters
4	4	Power dividers, Combiners, Couplers
5	5	Microwave Filters
6	6	Active Components-Diodes, Transistors, Tubes, Microwave Attenuator Design
7	7	RFMEMS, RF Switches, Phase Shifter,Mixer Design
8	8	Microwave Amplifiers and LNA Design
9	9	Power Amplifier Design, Oscillator Design
10	10	Antennas
11	11	Microwave Measurements, Microwave Systems and Imaging
12	12	Software session and Lab demonstration

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>

**ANALYSIS AND DESIGN PRINCIPLES OF MICROWAVE ANTENNAS
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Instructor Name: Amitabha Bhattacharya
Institute: IIT Kharagpur

Course Intro: The course Analysis and Design principles of Microwave Antennas covers a broad spectrum of the antenna design and analysis, starting with the basic concepts of microwave radiation to numerical computation of antenna currents, Impulse Radiating Antennas (IRAs), antenna pattern synthesis, etc.

This course will not only help the undergraduate and graduate students by providing fundamental concepts of antenna theory but also help the design engineer to get familiar with different state of the art techniques of antenna analysis and design. In a nutshell, this course would lay the foundation for further exploring the vast area of microwave antenna analysis and design.

Pre Requisites: Basic Knowledge of Electromagnetic Theory, Transmission line, Electronic network theory is required. Following are the NPTEL courses done earlier covering the above concepts (a) Electromagnetic Theory (b) Basic Tools in Microwave engineering (c) Basic Building Blocks of Microwave Engineering

Core/Elective: Core

UG/PG: UG

Industry Support: Radar Industry, Space Industry, Avionics industry, Defense Industry, Internal security Industry, Mining industry, Geo-exploration Industry.

Reference: Balanis, C. A., "Antenna Theory Analysis and Design", Third Edition, John Wiley and Sons, 2005

About Instructor: Amitabha Bhattacharya was born in Kolkata, West Bengal in the year 1964. He received his B.Tech, (E&ECE) Degree from IIT Kharagpur in 1986, M.E. (E&TCE) from Jadavpur University in 1994 and Ph.D. (E. & ECE) from IIT Kharagpur in 1998. He started his professional career in 1986 by joining as Junior Research Engineer in an ISRO- sponsored research project at IIT Kharagpur and continued thereafter as a Senior Research Assistant in a DRDO sponsored Research Project till 1991. In 1997, he joined SAMEER, Mumbai and then Defence Lab, Jodhpur as a Research Scientist. From 2000 onwards he joined teaching profession, first as an Assistant Professor in the Electronics and Instrument Department of Indian School of Mines, Dhanbad and then in 2007, in the faculty of Electronics and Electrical Communication Engineering Department of IIT Kharagpur in 2007.

Course plan:

S. No	week	Module Name
1	1	Microwave Radiation Fundamentals
2	2	Basic Antenna Parameters
3	3	Wire Antennas
4	4	Aperture Antennas
5	5	Array of Radiating Elements
6	6	Reflector Antenna
7	7	Generalized Antenna Analysis by
8	8	Potential Concept

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**PRINCIPLES OF DIGITAL COMMUNICATIONS
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Instructor Name: S N Merchant

Institute: IIT Bombay

Course Intro: The course is an introduction to modern digital communications at a senior undergraduate level. The primary goal of the course is to provide the student an understanding of the principles, techniques, trade-offs, and fundamental limits in modern digital communication systems. This course introduces the fundamentals of digital signaling, information theory and coding, digital transmission and reception

Pre Requisites: Signals and Systems, Basic Probability and Random variables

Core/Elective: Core

UG/PG: UG

Industry Support: Qualcomm, Samsung, BEL, ISRO, DRDO labs, communication companies

Reference: Digital Communications, 5th Ed, John G Proakis and MasoudSalehi, McGraw Hill, 2007/2008

About Instructor: Dr.Shabbir N. Merchant received his B. Tech, M. Tech, and PhD degrees all from Department of Electrical Engineering, Indian Institute of Technology Bombay, Mumbai. Currently he is a Professor in Department of Electrical Engineering at IIT Bombay. He has more than 35 years of experience in teaching and research. He has made significant contributions in the field of signal processing and its applications. His noteworthy contributions have been in solving state of the art signal and image processing problems faced by Indian defence. His broad area of research interests are wireless communications, wireless sensor networks, signal processing, multimedia communication and image processing and has published extensively in these areas. He is a co-author with his students who have won Best Paper Awards. He has served on Technical Program Committees of many IEEE premier conferences.

Course plan:

S. No	week	Module Name
1	1	Introduction I
2	2	Source Coding Theorem II
3	3	Mutual Information I
4	4	Signal Space Representations II
5	5	Optimum Receiver I
6	6	Quantizer Design I
7	7	Quantizer Design VI
8	8	Pulse Shaping I
9	9	Equalization II
10	10	Digital Modulation Methods V
11	11	Digital Modulation Methods X
12	12	Error Control Coding I

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**INTRODUCTION TO WIRELESS AND CELLULAR COMMUNICATIONS
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Layout:

Week 1: Overview of Cellular Systems and evolution 2g/3G/4G/5G

Week 2: Cellular Concepts – Frequency reuse, Cochannel and Adjacent channel Interference, C/I, Handoff, Blocking, Week 3: Wireless propagation Part 1 - Link budget, Free-space path loss, Noise figure of receiver

Week 4: Wireless propagation Part II - Multipath fading, Shadowing, Fading margin, Shadowing margin,

Week 5: Antenna Diversity

Week 6: Wireless Channel Capacity

Week 7: MIMO

Week 8: CDMA Part I

Week 9: CDMA Part II

Week 10: OFDM and LTE Part I

Week 11: OFDM and LTE Part II

Week 12: Large Scale Propagation effects and Channel Models

Suggested Reading

1. T. S. Rappaport, "Wireless Communications – Principles and Practice" (2nd edition) Pearson, 2010, ISBN 9788131731864.
2. A. Molisch, "Wireless Communications," Wiley, 2005 Haykin & Moher, "Modern Wireless Communications" Pearson .
3. J. G. Proakis, "Digital Communications," McGraw Hill.
4. A. Goldsmith, "Wireless Communications," Cambridge Univ Press, 2005.
5. D. Tse and P. Viswanath, "Fundamentals of Wireless Communications," Cambridge Univ Press, 2005

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>

**FABRICATION TECHNIQUES FOR MEMS-BASED SENSORS: CLINICAL
PERSPECTIVE
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Instructor Name: HardikJeetendraPandya

Institute: IISc Bangalore

Course Intro: This course is designed with an aim of educating students in the area of microtechnology and its use to fabricate sensors and systems. The students will have an exposure to sensors and its importance in the real world. The students will also be able to understand how to fabricate some of those sensors. Several examples of engineering devices used in clinical research will be also covered. Class 10000 non-conventional clean room and some equipment within it will also be shown. Below are some of the course outcomes. *if*~ Ability to understand microfabrication process

if~ Understand sensors used in electronics and biomedical areas *if*~ Understand Clean Room (Class 1 to Class 10000)

if~ Understand Microengineering Technology *if*~ Design the process flow for fabricating microheater required in gas sensors.

if~ Design the process flow for fabricating force sensors for biomedical application. *if*~ Design microheater for gas sensors as per specifications. *if*~ Design force sensors as per specifications. *if*~ Understand fabrication of microfluidic platforms, micro-cantilevers, flexible force sensors, inter-digitated electrodes, polymer-glass bonding etc. for clinical research

Pre Requisites: Basic Electronics

Core/Elective: Elective

UG/PG: Both

Industry Support: BEL, TI, ON Semiconductor, TSMC, Intel

Reference: J.D. Plummer, M.D. Deal, P.G. Griffin, Silicon VLSI Technology, Pearson Education, 2001.

S.A.Campbell, The Science and Engineering of Microelectronic Fabrication, Oxford University Press, 2001.

S.M. Sze (Ed), VLSI Technology, 2nd Edition, McGraw Hill, 1988

Senturia S. D., Microsystem Design, Kluwer Academic Publisher, 2001

Madou, M Fundamentals of Microfabrication, CRC Press, 1997.

Gad-el-Hak, M., Ed.; The MEMS Handbook; CRC Press: New York, NY, 2002.

About Instructor:

Dr.Hardik J. Pandya is a core faculty member in the Department of Electronic Systems Engineering, Division of Electrical Sciences, IISc Bangalore where he is developing Advanced Microsystems and Biomedical Devices Facility for Clinical Research and Biomedical and Electronic (10-6-10-9) Engineering Systems Laboratory to carry out cutting-edge research on novel devices to solve unmet problems in biology and medicine. He is recipient of prestigious Early Career Research Award from Science and Engineering Research Board, Government of India as well as a start-up grant of 228 Lacs from IISc. He has taught Design for Analog Circuits, Analog Integrated Circuits, VLSI technology, and Semiconductor Devices to undergraduate and graduate students from Electronic Engineering, Instrumentation Engineering, and Applied Physics. He seek to understand and exploit novel ways of fabricating microengineering devices using glass, silicon, polymers and integrate with unusual classes of micro/nanomaterials.

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>

INTRODUCTION TO INFORMATION THEORY, CODING AND CRYPTOGRAPHY (MOOCS-II)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Instructor Name: Prof. Ranjan Bose

Institute: IIT Delhi

Course Intro: Information theory, coding and cryptography are the three load-bearing pillars of any digital communication system. In this introductory course, we will start with the basics of information theory and source coding. Subsequently, we will discuss the theory of linear block codes (including cyclic codes, BCH codes, RS codes and LDPC codes), convolutional codes, Turbo codes, TCM and space time codes. Finally, we will introduce the basics of secure communications by focusing on cryptography and physical layer security. Wherever possible, applications of the theory in real world scenarios have been provided. The underlying aim of this course is to arouse the curiosity of the students.

Pre Requisites: Basic exposure to linear algebra and probability theory, as well as, a course in digital communications.

Core/Elective: Core

UG/PG: Both

Industry Support: Telecommunication companies, Internet companies, Information security companies

Reference: Basic text book R. Bose, Information theory, coding and cryptography, McGraw-Hill, 3rd Edition, 2016.

About Instructor: Ranjan Bose received his B.Tech. degree in electrical engineering from the Indian Institute of Technology (IIT), Kanpur, India in 1992 and the M.S. and Ph.D. degrees in electrical engineering from the University of Pennsylvania, Philadelphia, USA in 1993 and 1995, respectively. He worked at Alliance Semiconductor Inc., San Jose, CA, as a Senior Design Engineer from 1996 to 1997. Since November 1997 he has been with the Department of Electrical Engineering at Indian Institute of Technology, Delhi, where currently he is the Microsoft Chair Professor. His research interests lie in the areas of secure communications, coding theory, ultra-wideband (UWB) communications, broadband wireless access and wireless security. He currently heads the Wireless Research Lab in IIT Delhi. His lectures on wireless communications form a part of the video courses offered by the National Program on Technology Enhanced Learning (NPTEL).

Course plan

S. No	week	Module Name
1	1	Lecture 1: Introduction to Information Theory Lecture 2: Entropy, Mutual Information, Conditional and Joint Entropy Lecture 3: Measures for Continuous Random Variable, Relative Entropy
2	2	Lecture 4: Variable Length Codes, Prefix Codes Lecture 5: Source Coding Theorem Lecture 6: Various source coding techniques: Huffman, Arithmetic, Lempel Ziv, Run Length
3	3	Lecture 7: Optimum Quantizer, Practical Application of Source Coding: JPEG Compression Lecture 8: Introduction to Super Information Lecture 9: Channel Models and Channel Capacity
4	4	Lecture 10: Noisy Channel Coding Theorem Lecture 11: Gaussian Channel and Information Capacity Theorem Lecture 12: Capacity of MIMO channels
5	5	Lecture 13: Introduction to Error Control Coding Lecture 14: Introduction to Galois Field Lecture 15: Equivalent Codes, Generator Matrix and Parity Check Matrix
6	6	Lecture 16: Systematic Codes, Error Detections and Correction Lecture 17: Erasure and Errors, Standard Array and Syndrome Decoding Lecture 18: Probability of Error, Coding Gain and Hamming Bound
7	7	Lecture 19: Hamming Codes, LDPC Codes and MDS Codes Lecture 20: Introduction to Cyclic Codes Lecture 21: Generator Polynomial, Syndrome Polynomial and Matrix Representation
8	8	Lecture 22: Fire Code, Golay Code, CRC Codes and Circuit Implementation of Cyclic Codes Lecture 23: Introduction to BCH Codes: Generator Polynomials Lecture 24: Multiple Error Correcting BCH Codes, Decoding of BCH Codes
9	9	Lecture 25: Introduction to Reed Solomon (RS) Codes Lecture 26: Introduction to Convolutional Codes Lecture 27: Trellis Codes: Generator Polynomial Matrix and Encoding using Trellis
10	10	Lecture 28: Viterbi Decoding and Known good convolutional Codes Lecture 29: Introduction to Turbo Codes Lecture 30: Introduction to Trellis Coded Modulation (TCM)
11	11	Lecture 31: Ungerboeck's design rules and Performance Evaluation of TCM schemes Lecture 32: TCM for fading channels and Space Time Trellis Codes (STTC) Lecture 33: Introduction to Space Time Block Codes (STBC)
12	12	Lecture 34: Real Orthogonal Design and Complex Orthogonal Design Lecture 35: Generalized Real Orthogonal Design and Generalized Complex Orthogonal Design Lecture 36: Introduction to Cryptography: Symmetric Key and Asymmetric Key Cryptography Lecture 37: Some well-known Algorithms: DES, IDEA, PGP, RSA, DH Protocol Lecture 38: Introduction to Physical Layer Security: Notion of Secrecy Capacity Lecture 39: Secrecy Outage capacity, Secrecy Outage probability, Cooperative jamming

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>

DIGITAL IMAGE PROCESSING (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Outline:

Digital image processing deals with processing of images which are digital in nature. Study of the subject is motivated by three major applications. The first application is in improvement of pictorial information for human perception i.e. enhancing the quality of the image so that the image will have a better look. The second is for autonomous machine applications which have wider applications in industries, particularly for quality control in assembly automation and many similar applications. This course will introduce various image processing techniques, algorithms and their applications

Course plan:

Week	Topics
1	Introduction and signal digitization
2	Pixel relationship
3	Camera models & imaging geometry.
4	Image interpolation
5	Image transformation
6	Image enhancement I.
7	Image enhancement II.
8	Image enhancement III.
9	Image restoration I
10	Image restoration II & Image registration
11	Colour image processing
12	Image segmentation.
13	Morphological image processing
14	Object representation, description and recognition

References:

1. Digital Image Processing by Rafael C Gonzalez & Richard E Woods, 3rd Edition
2. Fundamentals of Digital Image Processing by Anil K Jain
3. Digital Image Processing by William K Pratt

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R16)
III/IV B.TECH
 (With effect from **2016-2017** Admitted Batch onwards)
 Under Choice Based Credit System
ELECTRONICS AND COMMUNICATION ENGINEERING
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 EC 3201	Microwave Engineering	4	3	1	--	4	30	70	100
B16 EC 3202	Microprocessors and its Applications	4	3	1	--	4	30	70	100
B16 EC 3203	Digital Communication	4	3	1	--	4	30	70	100
#ELE-I	ELECTIVE-I	4	3	1	--	4	30	70	100
B16 EC 3206	Digital Signal Processing	4	3	1	--	4	30	70	100
#ELE-II	ELECTIVE-II	4	3	1	--	4	30	70	100
B16 EC 3211	DSP Laboratory	2	--	--	3	3	50	50	100
B16 EC 3212	Microprocessors and Microcontrollers Lab	2	--	--	3	3	50	50	100
B16 ENG 3202	Verbal & Quantative Aptitude-II	2	5	--	--	5	100	--	100
B16 EC 3213	Mini Project	2	--	--	3	3	50	--	50
#M-III	MOOCS-III	2	4	--	--	4	100	--	100
Total		34	27	6	9	42	530	520	1050

# ELE-I	B16 EC 3204	Radar & Navigation
	B16 EC 3205	Information Theory And Coding
	B16 CS 3210	Object oriented programming
	B16 CS 3211	Web technologies
	B16 CS 3212	Software Engineering
#ELE-II	B16 EC 3207	Embedded Systems & Microcontrollers
	B16 EC 3208	Micro Electronics
	B16 EC 3209	Telecommunication Switching Systems
	B16 EC 3210	Digital Signal Processors And Architectures
#M-III	B16 ENG 3204	Advanced Coding
	B16 EC 3214A	Principles of Signals and Systems
	B16 EC 3214B	Analog Circuits
	B16 EC 3214C	Microprocessors and Microcontrollers

MICROWAVE ENGINEERING

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. The purpose of this course is to provide the operational characteristics and conceptual understanding of active and passive components at microwave frequencies.
2. This course also emphasizes formulation and application of scattering matrix for the analysis of different microwave passive components.
3. Further, this course also provides the understanding of measurement techniques of different parameters and the fabrication techniques of MMIC's at microwave frequencies.

Course Outcomes:

By the end of the course the learners (students) will be able to

1. Explain the working principle of different passive waveguide components used at microwave frequencies
2. Understand the conceptual and operational characteristics of different microwave signal generators and amplifiers
3. Apply the properties of scattering matrix for solving the scattering matrix of different passive microwave components for both ideal and practical considerations and analyze their operation
4. Understand different fabrication techniques involving Microwave integrated circuits
5. Understand and implement different experimental procedures involving measurement of microwave parameters.

SYLLABUS**Microwave Components:**

Introduction to Microwave Engineering- microwave spectrum bands, advantages and applications of microwaves, Wave-guide Components, coupling mechanisms, Directional Couplers, Magic Tee, Attenuators, Ferrite Devices, Isolators, Circulators, Cavity Resonators, Re-entrant Cavities.

Microwave Tubes:

Limitations of conventional tubes at microwave frequencies, Resonant Cavities, Linear beam tubes- Reflex Klystron, applegate diagram and principle of working, Two – Cavity Klystron, Multi – Cavity Klystron, Traveling Wave Tube, Crossed Field Device- Magnetron, Hull cut-off voltage Equation.

Microwave Solid state Devices

Negative resistance phenomenon, Gunn Diode, domain formation, RWH theory, Tunnel Diode- principle of operation, IMPATT- principle of operation, TRAPATT, BARITT, PIN Diodes.

Scattering Matrices of Microwave Components:

Scattering Matrix and its Properties, Scattering Matrix of Isolator, circulator, directional coupler, E Plane Tee, H plane Tee and Magic Tee.

Microwave Measurements:

VSWR & Impedance measurements, Frequency, Guided Wavelength, measurements of S parameters of reciprocal/non reciprocal devices.

Microwave Integrated Circuits:

Introduction, advantages and disadvantages of MMICs, comparison of MMICs with HMICs, Applications of MICs, materials used for MMICs, Substrate, Conductor, Dielectric and Resistive Materials, Growth of MMIC, Fabrication Techniques, MOSFET Fabrication, - MOSFET formation, NMOS fabrication process.

Text Books:

1. Foundations For Microwave Engineering, R. R. Collin, McGraw Hill.
2. Microwave Engineering, G.S.N. Raju, IK International Publishers,

Reference Books:

1. Microwave Engineering, Microwave Engineering, 4th Edition, David M. Pozar, November 2011.
2. Microwave Engineering, Annapurnadas, McGraw Hill.

MICROPROCESSORS AND ITS APPLICATIONS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To identify the Basic building blocks of a digital computer
2. To understand the architectures of Microprocessors 8085 and 8086
3. To be familiar with Assembly Language Programming
4. To understand the concept of interfacing peripheral devices and memory to 8085 Microprocessor

Course Outcomes:

By the end of the course the learners (students) will

1. Understand and analyze architecture of the 8085 and 8086 microprocessors
2. Be familiar with the 8085 and 8086 Assembly Language Programming
3. Learn about Hardware and software requirements in interfacing and designing microprocessor based products for practical applications

SYLLABUS

Internal Architecture and Functional Description of INTEL 8085 Microprocessor Pin out & Signals, Flag Register, Memory Read/Write and I/O Read /Write Cycles, Stack Memory, Interrupt Structure of 8085, Instruction Set and Timing Diagrams, Addressing Modes.

Programming The 8085:

Introduction to 8085 Assembly Language Programming, Programming model of 8085, Addressing modes of 8085 with examples, Instruction Set, Sample Programs, Subroutines and Interrupt Service Routines.

Interfacing of Semiconductor Memory and I/O Devices to 8085:

Classification of Read Write and Read only Memories, Interfacing of SRAMs, DRAMs and EPROMs. Interfacing of Parallel I/O (8255), Timer/Counter (8253/8254), Serial I/O (8251A) with 8085 Microprocessor.

Internal Architecture and Functional description of Intel 8086/8088 microprocessor, Memory Segmentation and physical address generation and physical address generation ,Register organization, Status flags and machine control flags of 8086,pinout and signals in detail, Memory read/write and I/O read/Write Bus cycles, 8086 memory Banks,8086 minimum and maximum modes of operation.

Introduction to 8086 Assembly language programming, programmable register array of 8086,Data Addressing modes of 8086 with examples, classification of 8086 instructions, sample 8086 assembly language programs using data transfer, Arithmetic and logical instructions.

Text Books:

1. Architecture Programming and Applications, Ramesh S. Goankar, New Age International Pvt. Ltd., (3rd Edition).
2. Microprocessors and interfacing , Douglas V.Hall ,Tata McGraw-Hill Revised 2nd Edition

Reference Book:

1. Microprocessors: The 8086/8088, 80186/80286, 80386/80486 And The Pentium Family,Nilesh B. Bahadure, Phi Learning Pvt. Ltd., 2010.

DIGITAL COMMUNICATION

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand the basic principles of digital communication system.
2. Analyze noise & the error performance of digital modulation techniques.
3. Design optimum receivers for digital modulation techniques.
4. To understand the concept of spread spectrum communication system.

Course Outcomes:

By the end of the course the learners (students) will be able to

1. Understand concept of different modulation techniques.
2. Understand the effect of noise in various digital communication systems and learn about optimum detection.
3. Compare performance of two basic digital modulation techniques.
4. Analyze performance of spread spectrum communication system.

SYLLABUS

Pulse Modulation and Digital Representation of Analog Signal: Sampling, Pulse Amplitude Modulation and Concept of Time Division Multiplexing, Pulse Width Modulation, Pulse Position Modulation, Digital representation of analog signal :Quantization of signals, Quantization error, Pulse Code Modulation, Companding, T1 Digital system, Differential Pulse Code Modulation, Delta Modulation, Adaptive Delta Modulation, Continuously Variable Slope Delta Modulation.

Digital Modulation and Transmission: Binary Phase-Shift Keying, Differential Phase-Shift Keying, Differentially-Encoded PSK (DEPSK), Quadrature Phase-Shift Keying (QPSK), M-ary PSK, Binary Frequency Shift-Keying, Comparison of BFSK and BPSK, M-ary FSK, Minimum Shift Keying (MSK), Duo-binary Encoding.

Mathematical Representation of Noise: Some Sources of Noise, Frequency-domain representation of Noise, Spectral Components of Noise, Response of a Narrowband Filter to Noise, Effect of a Filter on the Power Spectral Density of Noise, Superposition of Noises, Linear Filtering, Noise Bandwidth, Quadrature Components of Noise, Noise in Pulse Code Modulation and Delta Modulation Systems:PCM Transmission, Calculation of Signal-to-Noise Ratio in PCM, Delta Modulation(DM) Transmission, Calculation of Signal-to-Noise Ratio in DM, Comparison of PCM and DM .

Optimal Reception of Digital Signal: A Base-band Signal Receiver, Probability of Error, Optimum Receiver for both Baseband and Passband - Calculation of optimum filter Transfer function, Optimum filter realization using Matched filter, Probability of Error of the Matched Filter, Optimum filter realization using Correlator, Optimal of Coherent Reception: PSK, FSK, QPSK, Comparison of Modulation Systems.

Spread Spectrum Modulation: Direct Sequence (DS) Spread Spectrum, Use of Spread Spectrum with Code Division, Multiple Access (CDMA), Ranging using DS Spread Spectrum, Frequency Hopping (FH) Spread Spectrum, Generation and Characteristics of PN Sequences, Acquisition (Coarse Synchronization) of a FH Signal, Tracking (Fine Synchronization) of a FH Signal, Acquisition (Coarse Synchronization) of a DS Signal, Tracking of a DS Signal.

Text Books:

1. Principles of Communication Systems by Herbert Taub, Donald L Schilling and Goutam Saha, 3rd edition, Tata McGraw-Hill Publications, 2008 New Delhi.
2. Advanced Digital Communication By Wayne Thomas 3rd Edition Pearson Publications

Reference Books:

1. Digital Communications by Simon Haykins John Wiley, 2005
2. Modern Analog and Digital Communications by B.P. Lathi, Oxford reprint, 3rd Edition, 2004.

RADAR & NAVIGATION (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To provide insight of basic working principle of Radar
2. To apply different methods to measurement the Range, angle information etc. of the target from the radar,
3. To introduce different types of Radar systems and other types of tracking Radars,
4. To provide insight of advantages, limitations and applications of various Radar.
5. To provide insight of basics of various navigational aids and their working principles, applications, limitations and different methods to overcome their limitations

Course Outcomes:

By the end of the course the learners (students) will be able to

1. Able to understand the basic working principles of various Radars
2. Apply various mathematical equations to measure the Range and angle information of the targets from the radar
3. Analyze and design of radar signals, MTI, Pulse Doppler radar and various tracking Radars
4. Analyze various Radar systems, advantages, limitations and their applications.
5. Analyze various Navigational Aids like LORAN, DECCA, OMEGA, TACAN, VOR.

SYLLABUS

Introduction to Radar:

Origin of Radar, Basic Principle of Radar, Range to a target, Pulse Repetition Frequency and Range Ambiguities, Radar Block Diagram and Operation, Radar Equation, Integration of Radar Pulses, Probability of Detection and Probability of False Alarm, Radar Antenna Parameters, System Losses and Propagation Effects, Applications of Radar.

Radar Receivers:

Noise Figure and Noise Temperature, Types of Duplexers, Types of Mixers, Radar Displays, Receiver Protectors.

MTI and Pulse Doppler Radar:

Introduction to Doppler Effect, Simple CW Doppler Radar, Pulse Doppler Radar, Butterfly effect, Coherent and Non Coherent Moving Target Indication Radar, Delay line Cancellers, Limitation to MTI performance, Moving target Detector, MTI from moving platform.

Tracking Radar:

Types of Tracking Radars, Sequential Lobing, Conical Scan, Monopulse tracking Radar, Low angle tracking, Synthetic Aperture Radar (SAR), Active and Passive Aperture Phased array Radars, MST Radar, ECM, ECCM.

Fundamentals of Navigational Aids:

Principles of Direction Finders, Sense Finders, VOR, TACCAN, Aircraft Homing and ILS, Radio Altimeter, LORAN, DECCA, OMEGA

Text Books:

1. Introduction to Radar Systems – Merrill I. Skolnik, SECOND EDITION, Tata McGraw-Hill, 2001.
2. Radar Engineering and Fundamentals of Navigational Aids, G S N Raju, IK International Publishers, 2008.

Reference Books:

1. Introduction to Radar Systems – Merrill I. Skolnik, SECOND EDITION, McGraw-Hill, 1981.
2. Fundamentals of RADAR, SONAR and Navigation Engineering – K.K.Sharma

INFORMATION THEORY AND CODING (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To introduce students to the basic concepts of information theory & and applications of coding for information compression & channel protection
2. To study entropy, source coding and the fundamental limits of data compression
3. To study mutual entropy & channel capacity to understand the fundamental limits of information transmission, and means to achieve it
4. To study channel coding & decoding techniques for error control
5. To give a foundation for studying modern wireless communications & information security.

Course Outcomes:

After taking this course, the student will be able to:

1. **Appreciate** the mathematical concept of information (uncertainty) via probability & **compute** the entropy of a source
2. **Understand** the need of source coding & variable length codes
3. **Device** source codes using Shannon-Fano & Huffman algorithms, **calculate** the efficiency of a code
4. **Compute** mutual entropy of a channel, understand the **concept** of channel capacity, **State** Shannon's noisy channel coding theorem which creates the field of channel coding, **compute** channel capacity of BSC & AWGN channels, **define** characteristics of an ideal communication system
5. **Realize** the need & benefits of channel coding, **Understand** Linear block codes structure, theory & **use** syndrome technique for **decoding** for linear block codes, **Study** cyclic codes (BCH, RS and CRC) structure, theory, implementation & decoding of cyclic codes
6. **Study** Convolutional codes representation, generation & decoding of convolutional codes using Viterbi algorithm, get **acquainted** with concatenated codes to increase coding gain & Trellis Coded Modulation (TCM), Ungerboeck trellis codes for bandwidth efficiency
7. **Differentiate** source coding and channel coding & **learn** applications of coding
8. **Know modern codes & pursue** modern wireless communications & information security courses

SYLLABUS

PART-I: INFORMATION THEORY (CARLSON & HAYKIN)

Information Theory and Source Coding: Discrete memory less sources, Information measure, Entropy and Information rate, Shannon's source coding theorem, Coding for a discrete memory less source, Shannon-Fano & Huffman algorithms, Predictive coding for sources with memory.

Information Transmission: Mutual information, Mutual entropy, Discrete channel capacity, Shannon's channel coding theorem, Coding for the Binary Symmetric Channel, Continuous channels: Continuous information, Entropy, Entropy maximization, AWGN channel capacity, Ideal communication system.

PART-II:ERROR CONTROL CODING (HAYKIN& K. DEERGHARAO)

Channel Coding: Block Codes & Cyclic Codes: Rationale for coding, Types of codes, Discrete memoryless channels, Linear block codes, Syndrome decoding, Cyclic codes, Properties of BCH, RS and CRC codes.

Convolutional Codes: Representation & generation, Decoding Convolutional codes, Exhaustive search method, Maximum Likelihood decoding of Convolutional codes, Viterbi Algorithm, Sequential decoding, Trellis codes, Burst error correction, Interleaving, Concatenated coding, Automatic Repeat Request (ARQ) schemes, Applications of coding, Comparison of error rates in coded and uncoded transmission.

Modern Codes: Turbo codes, Non-recursive and Recursive Systematic Convolutional(RSC) Encoders, Turbo Encoder, Low Density Parity Check(LDPC) Codes, Properties, Construction of Parity Check Matrix H, Tanner Graphs, LDPC Encoding, MIMO System, Space–Time-Coded MIMO System, Space-Time(ST) codes, Space–Time Block Codes (STBC), Alamouti2-transmit Code - (1Q at least)

Text Books:

1. Communication Systems, 3/e, by A.B. Carlson, Mcgraw Hill Publishers (Part-I)
2. Digital Communications by Simon Haykin, John Wiley & Sons (Part-I & II)

Reference Books:

1. Information Theory, Coding and Cryptography – Ranjan Bose, 2nd Ed, 2009, TMH.
2. Digital Communications-Fundamentals and Applications - Bernard Sklar, PE.

OBJECT ORIENTED PROGRAMMING (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives: On completing this course student will be able to

1. Understand the syntax and principles of Object oriented programming language, and to programs using control statements, classes and interfaces.
2. Design and development of secure and extendable C++ applications.
3. Understanding the concepts of oops, different predefined classes and packages.
4. Understand the concepts of polymorphism

Course Outcomes:

1. Students will be able to handle I/O streams and Run time errors.
2. Students will be able to construct applications and Identify where data structures are appearing in them

SYLLABUS

Basics of Object Oriented Programming:

Object Oriented Paradigm, Principles of OOP, benefits of OOP, datatypes, declarations, expressions and operator precedence, functions, scope of variables.

Introduction to C++:

Classes and objects, Constructors & Destructors, constructor with dynamic allocation, explicit constructor, Operator Overloading through Unary, Binary, Assignment and Stream operators & type conversions.

Inheritance and Manipulating Strings:

Derived classes, syntax of derived classes, making private members inheritable, single, multilevel, multiple, hierarchical, hybrid inheritance, Virtual base Class, abstract classes, Creating String Objects, Manipulating String Objects, Relational Operations, Accessing String Characteristics.

Polymorphism:

Pointers, virtual functions and polymorphism- pointers to objects, this pointer, pointers to derived classes, virtual and pure virtual functions, Dynamic polymorphism, Virtual destructor, Virtual Base Class, Dynamic Casting, Cross Casting, Down Casting.

Templates, Exception handling, Streams and Files in C++:

Class templates, Function templates, member function templates, exception handling, managing console I/O operations, Stream Classes, Formatted and Unformatted i/o operations, managing output with manipulators, working with files.

Text Books:

1. Object oriented Programming using C++: E. Balagurusamy, PHI

Reference Books:

1. Object Oriented Programming in C++: N. Barkakati, PHI
2. Object Oriented Programming through C++ by Robot Laphore.
3. Object Oriented Analysis and Design by Andrew Haigh– (Tata Mcgrah Hill.)

WEB TECHNOLOGIES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives

1. Understand the various steps in designing a creative and dynamic website.
2. Understand, analyze and create XML document and XML Schema.
3. Understand the fundamentals of PHP.
4. Understand the fundamentals of MySQL Database and installation process.
5. Understand the fundamentals of Web Hosting.

Course outcomes:

1. They will be able to write html, JavaScript, CSS codes.
2. They will have clear understanding of hierarchy of objects in HTML and XML.
3. Finally they can create good, effective and customized websites.

SYLLABUS

Introduction to HTML , Core Elements , Links and Addressing, Images , Text , Colors and Background, Lists, Tables and Layouts , Frames, Forms , Cascading Style Sheets.

Introduction to Java Scripts, Elements of Objects in Java Script, Dynamic HTML with Java Script.

Document type definition, XML Syntax, XML Schemas, Document Object model, Presenting XML, Using XML Processors.

Introduction to PHP, Language Basics, Functions, Strings, Arrays.

MYSQL Installation, Accessing MySQL Using PHP, Form Handling, Inserting Data into Tables , Selecting Data from a Table, Updating Table , Deleting data from Table. Cookies & Session Tracking.

Text Books:

1. Web Programming, building internet applications, 2nd Ed., Chris Bates, Wiley Dreamtech
2. The complete Reference HTML and DHTML, Thomas A. Powey
3. Learning Php, Mysql, Robin Nixon
4. Programming Php, Kevin Tatroe, Peter MacIntyre & Rasmus Lerdorf foreword by Michael Bourque.

Reference Books:

1. Internet , World Wide Web , How to program, Dietel , Nieto, PHI/PEA
2. Web Technologies, Godbole, kahate, 2nd Ed., TMH

SOFTWARE ENGINEERING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Students can learn basic concepts of Software Engineering
2. To Know the distinction between various types of requirements
3. To learn various Process development models
4. To know how to estimate the cost and schedule of Software Projects
5. To know how to test software Systems

Course Outcomes :

At the end of the course, the students can able

1. To Remember the basic concepts of software Engineering
2. To use various process development models
3. To apply various techniques for gathering and analyzing requirements
4. To gain the knowledge of Software Architecture views
5. To estimate the cost and Schedule of Projects by using various estimation models.
6. To apply various testing strategies to test the software systems.

SYLLABUS

Introduction : Problem domain, Software Engineering challenges, the software engineering approach

Software Processes : Software Processes, desired characteristics of Software Processes,

Software Process development models : Waterfall model, Prototyping, Iterative development, Time Boxing model

Software Requirement analysis and specification : Need for SRS, Requirement process, Problem analysis, Requirement specification: Characteristics of SRS, Components of SRS.

Software Architecture: Role of Software Architecture, Architecture views.

Planning Software Project : Process planning, Effort estimation, Project scheduling and staffing

Software design : Introduction to Function oriented Design and Object Oriented Design.

Testing : Testing fundamentals, Black box testing, White box testing, Testing Process.

Text Book :

1. An integrated approach to Software Engineering, PankajJalote, Third edition, Springer

Reference Book :

1. Software Engineering – A Practitioners approach, Roger S Pressman, 7th Edition, Mcgrawhill International Edition.

DIGITAL SIGNAL PROCESSING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. This course introduces students to the fundamental principles of Digital Signal Processing and develops essential analysis and design tools required for signal processing systems & implementations. Also this subject is an introduction to the graduate-level courses in a broad range of disciplines spanning communications, speech processing & image processing.
2. The topics include SS basics, sampling theorem, Z-transform, analysis of Discrete-time Linear Time-Invariant Systems, Realization structures, Frequency domain representation of signals and systems, DTFT, DFS, Discrete Fourier Transform (DFT), linear/circular convolutions, Fast Fourier Transform (FFT) algorithms, FIR & IIR digital filter design, Multi-rate DSP and a few DSP applications.

Course Outcomes:

At the end of this course, the students will be able to:

1. Describe the DSP fundamental theory and components, Develop an understanding of DSP advantages, limitations and fundamental tradeoffs
2. Carry-out LTI system analysis using convolution & Z-transform
3. Carryout data analysis & spectrum analysis using FFT
4. Design IIR & FIR digital filters to meet specifications
5. Knows multi-rate SP aspects, filter banks & applications
6. Tackle numerical & practical issues in DSP implementation
7. Apply DSP techniques to real world problems in information processing, filtering, communications, detection & estimation, Relate & translate DSP theory to applications, Ready to take advanced DSP courses & pursue research
8. Illustrate and implement real-time DSP principles using MATLAB & DSP Processors, Ready to work in DSP industry.

SYLLABUS**Discrete-Time Signals and Systems (Oppenheim & Proakis)**

Introduction to Digital Signal Processing, Basic elements of a DSP system, Advantages of Digital SP over Analog SP, Discrete-time signals and systems, DT-LTI systems described by Linear constant-coefficient difference equations, Properties & Analysis of DT-LTI systems, Discrete linear convolution, Frequency domain representation of DT Signals and Systems, DTFT, Review of the Z-transform, Properties, Inverse Z-transform, Analysis of DT-LTI systems in Z-Domain, System function, One-sided Z-transform, Solution of difference equations, Structures and Realization of Digital Filters, Direct-I, II, series and parallel forms.

Discrete Fourier Transform (DFT) and Fast Fourier Transform Algorithms (FFT): (Oppenheim & Proakis)

Frequency analysis of discrete time signals, DFS, Properties of DFS, Sampling of DTFT, DFT, Properties of DFT, Circular and linear convolution of sequences using DFT, Efficient computation of DFT, Radix-2 Decimation-in-Time(DIT) & Decimation-in-Frequency(DIF) FFT Algorithms, Inverse FFT.

Design of IIR & FIR Digital Filters: (Oppenheim & Proakis)

General considerations in Filter design, Analog filter approximations– Butterworth and Chebyshev, Frequency response specifications, Design of IIR digital filters from analog filters, Bilinear Transformation Method, Impulse Invariance Technique, Low-pass filter Design examples, Characteristics of FIR Digital Filters, Design of Linear Phase FIR digital Filters using Windows, Effect of Window selection & filter length on filter frequency response, Design examples, Comparison of IIR and FIR Filters.

DSP Applications: (Mitra & Proakis)

Overview of DSP applications, Spectral analysis of sinusoidal signals using FFT, Subband coding of speech signals, Signal compression, Transmultiplexers, Practical aspects of DSP system implementation, DSP hardware & DSP Processors, Finite precision arithmetic effects.

Multirate Digital Signal Processing Fundamentals (SK Mitra)

Introduction to Multirate DSP, Basic sampling rate alteration devices: upsampler, downsampler, Time and Frequency domain characterization of up/down samplers, Multirate structures for sampling rate conversion, Filters used in sampling rate alteration systems, Interpolator and decimator, Brief introduction to Digital Filter Banks.

Text Books:

1. Alan V. Oppenheim, Ronald W. Schaffer, “**Digital Signal Processing**” – PHI Ed., 2006
2. John G. Proakis, D.G. Manolakis, “**Digital Signal Processing: Principles, Algorithms and Applications**”, 3rd Ed., PHI, 1996.

Reference Books:

1. Sanjit K. Mitra, “**Digital Signal Processing: A Computer Based Approach**”, Tata McGraw Hill.
2. Sen M. Kuo, WoonSeng Gan, “**Digital Signal Processors: Architectures, Implementations and Applications**”, Prentice Hall, 2005.

EMBEDDED SYSTEMS & MICROCONTROLLERS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study the basics of embedded systems and its examples.
2. To discuss various software architectures in embedded systems.
3. To discuss Inter Task Communication procedures in RTOS and design issues of RTOS.
4. To study various embedded software development tools and debugging techniques.
5. To study the 8051 Microcontroller architecture and its instruction set.

Course Outcomes:

By the end of the course the learners (students) will

1. Ability to understand the concepts related to RTOS and its Inter Task communication Methods.
2. Ability to understand various design issues of RTOS.
3. Understand about embedded software development tools.
4. Understand the basic architecture of 8051 micro controller and instruction set.

SYLLABUS

Introduction to Embedded Systems: Examples, Typical Hardware, Memory, Architecture, Instruction set, Programming. Interrupt Basics, Shared-Data problem, Interrupt Latency.

Software Architectures: Round-Robin Architecture, Round-Robin with Interrupts Architecture, Function-Queue Scheduling Architecture, Real-Time Operating Systems Architecture, Selection of Architecture.. Real Time Operating System: Tasks and Task States, Tasks and Data, Semaphores and Shared Data, Semaphore Problems, Semaphore variants.

Inter Task Communication & Design issues of RTOS: Message Queues, Mailboxes, Pipes, Interrupt Routines in RTOS Environment, Principles of RTOS, Encapsulation Semaphores and Queues, Hard, Real-Time Scheduling Considerations.

Embedded Software development Tools & Debugging techniques: Host and Target Machines, Linker/Locator for Embedded Software, Getting Embedded Software into the Target System, Testing on your Host Machine, Instruction Set Simulators, Laboratory Tools used for Debugging.

Microcontroller: Introduction to Microcontroller, Architecture of 8051, Pin diagram of 8051, Memory organization, External Memory interfacing, stacks. Instruction set: Instruction timings, 8051 instructions: Data transfer instructions, Arithmetic instructions, Logical instructions, Branch instructions, Subroutine instructions and Bit manipulation instruction.

Text Books:

1. An Embedded Software Primer, David E. Simon, Pearson Education, 2005.
2. The 8051 Microcontroller Architecture, Programming & Applications, Kenneth J.Ayala, Penram International.

Reference Books:

1. Embedded Systems: Architecture, Programming and Design, Raj Kamal, Tata McGraw-Hill Education, 2008

MICRO ELECTRONICS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the technology, design concepts and fabrication process of Very Large Scale Integrated circuits
2. To bring both Circuits and System views on design together.
3. To Understand how to analyse and design of digital logic gates using MOSFETs and ECL circuits

Course Outcomes:

By the end of the course the learners (students) will

1. Be familiar with MOSFET basics and Fabrication process
2. Understand and analyze Digital CMOS circuits and other digital logic families

SYLLABUS

Integrated- Circuit Fabrication: Monolithic Integrated - Circuit (microelectronics) technology- The planar processes - Bipolar Transistor Fabrication - Fabrication of FETs - CMOS Technology - Monolithic Diodes - The Metal - Semiconductor Contact - IC Resistor - IC Capacitors - IC Packaging - Characteristics of IC Components - Microelectronic circuit layout.

Basic Digital circuits: MOS Technology - NMOS, CMOS, Inverters, Logic gates - ECL circuits.

Combinational Circuits: Arithmetic functions - Comparators - Multiplexers - Demultiplexers - Memory - Memory applications - PAL - PLAs.

Sequential Circuits: A1 - Bit memory - The circuits properties of bistable latch - The clocked SR Flip-Flop - J-K, T, and D-type Flip-flops. Shift-registers - Ripple Counters - synchronous counters - Applications of counters.

Text Book:

1. Microelectronics by Jacob Milliman, Arbin Grabel second edition, TMH.

Reference Books:

1. Part 2 of Integrated Circuits, Design Principles and Fabrications by editors, Warner and Fordemwalt, 1965, Motorola Series, McGraw Hill.
2. MOS LSI Design and Applications by Dr. William N. Carr and Dr. Jack P. Mize, McGraw Hill, 1972.
3. Micro electronic circuits and devices second edition Horenstien, PHI.

TELECOMMUNICATION SWITCHING SYSTEMS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The student will

1. Understand different types of switching
2. Understand both time and space based switches
3. Understand the means of measuring traffic.
4. Understand the implication of the traffic level on system design.
5. Understand about ISDN functions and applications.

Course Outcomes:

The student will be able to

1. Evaluate the time and space parameters of a switched signal
2. Establish the digital signal path in time and space, between two terminals
3. Evaluate the inherent facilities within the system to test some digital switch functions.
4. Investigate the traffic capacity of the system.
5. Able to understand different data rate and applications of ISDN

SYLLABUS

Introduction:

Evolution of Telecommunications, Simple Telephone Communication, Basics of Switching System, Manual Switching System, Major Telecommunication.

Crossbar Switching: Principles of Common Control, Touch Tone Dial Telephone, Principles of Crossbar Switching, Crossbar Switch Configurations, Cross point Technology, Crossbar Exchange Organization.

Electronic Space Division Switching:

Stored Program Control, Centralized SPC: Standby mode, Synchronous duplex mode, Distributed SPC, Software Architecture, Application Software, Enhanced Services, Need for networks, Two stage networks, Three stage networks and n-stage networks.

Time Division Switching:

Basic Time Division Space Switching, Basic Time Division Time Switching, Time multiplexed space switching, Combination Switching: Time Space (TS) Switching, Space-time (ST) Switching, Three-Stage Combination Switching, n- Stage Combination Switching.

Telephone Networks:

Signaling Techniques, In-channel Signaling, Common Channel Signaling, CCITT Signaling System no.6, CCITT Signaling System no.7.

Traffic engineering: grade of service and blocking probability, modeling switching system, blocked models and loss estimates, delay systems.

Integrated Services Digital Network:

Motivation for ISDN, New Services, Network and Protocol Architecture, Transmission Channels, User- Network Interfaces, Signaling, Numbering and Addressing, Service Characterization, Interworking, ISDN Standards, Expert Systems in ISDN, Broadband ISDN.

Text Books:

1. Telecommunication Switching Systems and Networks- ThiagarajanViswanathan, 2000, PHI.
2. Telecommunications Switching, Traffic and Networks- J. E. Flood, 2006, Pearson Education.

Reference Books:

1. Digital Telephony- J. Bellamy, 2nd Edition, 2001, John Wiley.
2. Data Communications and Networks- Achyut S. Godbole, 2004, TMH.
3. Principles of Communication Ststems- H. Taub& D. Schilling, 2nd Edition, 2003, TMH.
4. Data Communication & Networking- B. A. Forouzan, 3rd Edition, 2004, TMH.
5. Telecommunication System Engineering – Roger L. Freeman, 4th Ed., Wiley-Inter Science, John Wiley & Sons,2004.

DIGITAL SIGNAL PROCESSORS AND ARCHITECTURES
(Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study the basic DFT, FFT and rate conversion algorithms.
2. To study the number format, dynamic range and sources of errors in DSP systems
3. To learn about TMS programmable DSPs and their programming capabilities.
4. To understand basic DSP algorithms on TMS processors
5. To study the FFT algorithms on TMS320C54XX DSP device.

Course Outcomes:

After undergoing the course, students will be able to

1. Apply DFT and FFT algorithms for DSP application
2. Apply the number format, dynamic range and various sources of errors in DSP system
3. Implement application programs on a DSP processor
4. Implement various DSP algorithms on TMS processors
5. Implement FFT algorithms on TMS320C54XX DSP algorithm.

SYLLABUS

Introduction to Digital Signal Processing:

Introduction, A Digital signal-processing system, The sampling process, Discrete time sequences. Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT), Linear time-invariant systems, Digital filters, Decimation and interpolation.

Computational Accuracy in DSP Implementations:

Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.

Architectures for Programmable DSP Devices:

Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation UNIT, Programmability and Program Execution, Speed Issues, Features for External interfacing.

Programmable Digital Signal Processors:

Commercial Digital signal-processing Devices, Data Addressing modes of TMS320C54XX DSPs, Data Addressing modes of TMS320C54XX Processors, Memory space of TMS320C54XX Processors, Program Control, TMS320C54XX instructions and Programming, On-Chip Peripherals, Interrupts of TMS320C54XX processors, Pipeline operation of TMS320C54XX Processors.

Analog Devices Family of DSP Devices:

Analog Devices Family of DSP Devices – ALU and MAC block diagram, Shifter Instruction, Base Architecture of ADSP 2100, ADSP-2181 high performance Processor.

Introduction to Blackfin Processor - The Blackfin Processor, Introduction to Micro Signal Architecture, Overview of Hardware Processing Units and Register files, Address Arithmetic Unit, Control Unit, Bus Architecture and Memory, Basic Peripherals.

Interfacing Memory and I/O Peripherals to Programmable DSP Devices:

Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA).

Text Books:

1. Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.
2. A Practical Approach to Digital Signal Processing - K Padmanabhan, R. Vijaya rajeswaran, Ananthi. S, New Age International, 2006/2009
3. Embedded Signal Processing with the Micro Signal Architecture Publisher: Woon-SengGan, Sen M. Kuo, Wiley-IEEE Press, 2007

Reference Books:

1. Digital Signal Processors, Architecture, Programming and Applications – B. Venkataramani and M. Bhaskar, 2002, TMH.
2. Digital Signal Processing –Jonatham Stein, 2005, John Wiley.
3. DSP Processor Fundamentals, Architectures & Features – Lapsley et al. 2000, S. Chand & Co.
4. Digital Signal Processing Applications Using the ADSP-2100 Family by The Applications Engineering Staff of Analog Devices, DSP Division, Edited by Amy Mar, PHI
5. The Scientist and Engineer's Guide to Digital Signal Processing by Steven W. Smith, Ph.D.,California Technical Publishing, ISBN 0-9660176-3-3, 1997

DSP LABORATORY**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2**

Course Objective:

1. To implement Linear and Circular convolution
2. To analyze frequency response of Discrete Time Sequence and Systems.
3. To design Digital FIR and IIR filters.
4. To compute Fast Fourier Transform for given sequence.
5. To implement basic Image Processing techniques using MATLAB.

Course Outcomes:

1. Able to write the MATLAB coding for basic mathematical operations to complex operations like FFT.
2. Able to Design and Analyze LTI systems& Digital Filters.
3. Understand the image processing techniques

LIST OF EXPERIMENTS

1. Generation of discrete –time sequences and signals.
2. Program for Linear Convolution of two sequences.
3. Program for Circular Convolution of two sequences.
4. Frequency analysis of discrete time sequences.
5. Frequency analysis of discrete time Linear and Non Linear systems.
6. Frequency analysis of discrete time variant and Time invariant systems.
7. Frequency analysis of discrete time LTI system.
8. Design of IIR digital filter using Butterworth filter.
9. Design of IIR digital filter using Chebyshev filter.
10. Design of FIR digital filter using Hamming window and Rectangular window.
11. Program for sum of sinusoidal signals.
12. Program for Fast Fourier Transform.
13. Program for adjusting intensity values of a image using histogram equalization.
14. Program for Deblurring Images Using a Wiener Filter.

ReferenceBook:

1. Alan V. Oppenheim, Ronald W. Schafer, “**Digital Signal Processing**” – PHI Ed., 2006

MICROPROCESSORS AND MICROCONTROLLERS LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To understand the basics of Microprocessors 8085, 8086 and Microcontroller 8051.
2. To understand the internal organization of INTEL 8085, 8086 Microprocessors and Microcontroller 8051.
3. Developing Assembly Language Programs using the instruction sets of microprocessors and microcontroller and to study the interfacing of the processor with various peripheral devices.

Course Outcomes:

1. The objective of this course is to become familiar with the instruction set of Intel microprocessors and microcontroller and also to familiarize with Assembly language programming. The accompanying lab is designed to provide practical hands-on experience with microprocessor software applications and interfacing techniques.

SYLLABUS**Experiments Based On ALP (8085)**

1.
 - a. Assume that byte of data is stored at memory location 'X'. Write an ALP which tests bit 5 of this data. Write 'FF' in the location 'X+1' if the bit 5 is '1' and '00' if bit 5 is '0'.
 - b. Check the zero condition of this number and write '00' at location 'Y' if it is '0' and 'FF' at 'Y' if non zero.
 - c. For data value in the location 'X' compute the number of logic 1's and store the result in the location 'Y+1'.
2.
 - a. Write an ALP to swap the contents of location 'X' and 'X+1' using BC & HL Register pairs.
 - b. By using above logic, write an ALP to transfer a block of data into another block.
3.
 - a. Write an ALP to add and subtract two eight bit Number stored in the location 'X' and 'X+1' by assuming that content of 'X' is greater than content of 'X+1'
 - b. Modify this program to add two 16 bit numbers without using DAD instruction.
4. Two 8 bit numbers 34H and 43H are stored in locations 'X' and 'X+1' compute the product of these two numbers using
 - a. Repetitive addition method
 - b. Shift and add method
5. The number of the bytes of a block of data is in location 'X' and data starts from location 'X+1' onwards defining a stack pointers. Write an ALP to arrange this sequence of data in reverse order. Keep the reverse sequence from 'Y' onwards.
6. The number of bytes of a block of data is location 'X' and data starts from location 'X+1' onwards. Arrange this block of data in ascending order by using bubble sorting technique

7. Using 8279 write an ALP to generate the message of 4 characters. Activate the LED's individually and make the display ON &OFF for every 0.5 seconds

Experiments Based On ALP (8086)

8. Programs on Data Transfer Instructions
9. Programs on Arithmetic and Logical Instructions
10. Programs on Branch Instructions
11. Programs on Subroutines
12. Sorting of an array

Experiments based on Interfacing and Microcontroller (8051)

13. Programs on Data transfer instructions using 8051 Microcontroller
14. Programs on Arithmetic and Logical instructions using 8051 Microcontroller

Reference Book

1. Architecture Programming and Applications, Ramesh S. Goankar, New Age International Pvt. Ltd., (3rd Edition).

VERBAL & QUANTITATIVE APTITUDE – II
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing .

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by RashmiBansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and menstruation.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation

Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II :Puzzle test, blood Relations, permutations, Combinations and probability

Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III :Periods, Clocks, Calendars, Cubes and cuboids

Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles

Selective puzzles from previous year placement papers, sitting arrangement, problems-circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Menstruation

Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of menstruation-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

MINI PROJECT**Lab: 3 Hrs.****Sessionals : 50**
Credits : 2

Course Objectives:

1. To introduce the students to electronics & communication engineering projects.
2. To help the students in applying the core knowledge and programming skills learnt upto the end of 3rd year 1st semester, to find solutions to challenges facing industry and society.
3. To help the students develop practical skills and knowledge about practical tools.
4. To provide the students and opportunity to exercise their creative and innovative qualities in a group environment.

Course Outcomes:

After doing a mini project, students will be able to

1. Achieve practical knowledge within his chosen area of technology for project development.
2. Identify, analyze, formulate and handle electronics & communications projects with a systematic and comprehensive approach.
3. Contribute as an individual or as member of team in development of technical projects.
4. Develop effective communication skills for presentation of project related activities.

ADVANCED CODING
(Common to ECE & EEE)
(MOOCS-III)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates.
3. To investigate different methods to find time complexities.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS

UNIT I Review Coding essentials and modular programming

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific .
4. Object OrientedProgrammingin C++: N. Barkakati, PHI.
5. ObjectOriented Programming through C++ byRobatLaphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

**PRINCIPLES OF SIGNALS AND SYSTEMS
(MOOCS-III)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course outline

Week 1 : Introduction to Signals, Signal Classification, Continuous/ Discrete Time Signals

Week 2 : Definition and Classification of Systems, Linear Time Invariant (LTI) Systems

Week 3 : Properties of LTI Systems, Impulse Response, Convolution, Causality, Stability

Week 4 : Impulse Response of Discrete Time Systems, Discrete Time Convolution, Difference Equations and Analysis

Week 5 : Laplace Transform, Properties of Laplace Transform, Inverse Laplace Transform

Week 6 : Introduction to z-Transform, Properties of z-Transform, Region of Convergence, Inverse z-Transform

Week 7 : Introduction to Fourier Analysis, Fourier Series for Periodic Signals, Properties of Fourier Series

Week 8 : Introduction to Fourier Transform, Properties of Fourier Transform, Frequency Response of Continuous Time Systems, Examples of Frequency Response

Week 9 : Fourier Analysis of Discrete Signals, Discrete Time Fourier Transform (DTFT), Properties of DTFT, Examples of DTFT

Week 10 : Frequency Response of Discrete Time Systems, Discrete Fourier Transform (DFT), Properties of DFT, Examples of DFT

Week 11 : IIR/ FIR Filters, Direct Form Realization, Cascade and Parallel Form Realization, Problem Solving

Week 12 : Concept of State, State Space Analysis, State Space Representation of Continuous Time Systems, Solution of State Equations for Continuous Systems

Suggested Reading Materials:

- Signals and Systems, Oppenheim and Willsky, Prentice Hall

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>

ANALOG CIRCUITS (MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

UNIT I Introduction to Analog circuits

Course introduction; Need for nonlinear circuits; Incremental analysis of nonlinear circuits; Diode and its models; Incremental picture of a two port nonlinear circuit; Constraints on y-parameters and large signal characteristics to obtain a high gain; MOS transistor and its characteristics; AC coupling network to add signal to bias; AC coupling at input and output; Common source amplifier

UNIT II MOS transistor

Output conductance of a MOS transistor; Inherent gain limitation of a transistor; Sensitivity of g_m to transistor parameters; Biasing a transistor at a constant current; Drain feedback configuration; Current mirror;

UNIT III Common source Amplifier

Common source amplifier using drain feedback; Common source amplifier using current mirror bias; Common source amplifier using source feedback bias; Using a resistor instead of a current source for biasing; Further biasing techniques;

UNIT VI Voltage/Current Controlled Voltage Sources

VCVS using a transistor; Source follower biasing; CCVS using a transistor; CCVS using an opamp; Biasing a CCVS; Emitter degenerated amplifier; Common gate amplifier and its biasing; VCCS using a transistor and its biasing; pMOS transistor and its small signal model; Biasing a pMOS transistor; Converting nMOS circuits to pMOS; Amplifiers using a pMOS transistor

UNIT V Bipolar Junction Transistor

Bipolar junction transistor-large and small signal models; BJT circuits- Biasing; Common source amplifier; Emitter follower; BJT Common base amplifier, Transimpedance amplifier; Swing limits of amplifiers; Two transistors in feedback; Two transistors in feedback

WEBLINK:

<https://onlinecoursesnp. tel.ac.in>

Code: B16 EC 3214C

MICROPROCESSORS AND MICROCONTROLLERS
(MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Week 1: Introduction: General processor architecture, Microprocessors, Microcontrollers, Basic Computer Organization,

Week 2: 8085 Microprocessors,

Week 3: 8085 – Part II

Week 4: 8085 – Part III

Week 5: 8085 – Part IV

Week 6: 8051 – Part I

Week 7: 8051 – Part II

Week 8: PIC, AVR

Week 9: ARM – Part I

Week 10: ARM – Part II

Week 11: Interfacing examples – Part I

Week 12: Interfacing examples – Part II

REFERENCES:

1. R. Gaonkar, “Microprocessor Architecture, Programming and Applications with the 8085”, Prentice Hall, 2014.
2. M.A. Mazidi, R.D. McKinlay, J.G. Mazidi, “The 8051 Microcontroller: A Systems Approach”, Pearson, 2013.
3. M.Bates, “PIC Microcontrollers”, Newnes, 2011.
4. M.A. Mazidi, S. Naimi, S. Naimi, “The AVR Microcontroller and Embedded Systems: Using Assembly and C”, Prentice Hall, 2011.
5. W.A. Smith, “ARM Microcontroller Interfacing: Hardware and Software, Eketor, 2010

WEBLINK:

<https://onlinecoursesnp.tel.ac.in>



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 EC 4101	Digital Image Processing	4	3	1	-	4	30	70	100
B16 EC 4102	VLSI Design	4	3	1	-	4	30	70	100
B16 EC 4103	Fiber Optic Communications	4	3	1	-	4	30	70	100
B16 EC 4104	Microwave Engineering & Optical communications Lab	2	-	-	3	3	50	50	100
B16 EC 4105	Digital Communication Lab	2	-	-	3	3	50	50	100
B16 EC 4106	Project Phase-I	2	-	-	3	3	50	--	50
Total		18	9	3	9	21	240	310	550

DIGITAL IMAGE PROCESSING

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives: The student should be made to

- 1.Learn digital image fundamentals.
- 2.Be exposed to simple image processing techniques.
- 3.Be familiar with image compression techniques.
- 4.Learn to represent color image in form of features.
- 5.Be exposed to segmentation techniques.

Course Outcomes:

Upon successful completion of this course, students will be able to:

- 1.Discuss digital image fundamentals.
- 2.Apply image enhancement and restoration techniques.
- 3.Use image compression techniques.
- 4.Represent features of color images.
- 5.Use image segmentation techniques.

SYLLABUS

DIGITAL IMAGE FUNDAMENTALS Introduction – Origin – Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels.

IMAGE ENHANCEMENT

Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening of Spatial Filtering – Frequency Domain: Introduction to Fourier Transform – Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters.

IMAGE RESTORATION

Noise models – Mean Filters – Order Statistics – Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters – Inverse Filtering, Weiner Filtering .

IMAGE COMPRESSION Compression: Fundamentals – Image Compression models – Error Free Compression – Variable Length Coding – Bit-Plane Coding – Lossless Predictive Coding – Lossy Compression – Lossy Predictive Coding – Compression Standards.

COLOR IMAGE PROCESSING AND SEGMENTATION

Color fundamentals, colormodels,theRGB,the CMY and CMYK, the HSI colormodels,color transformations, color slicing, tone and color corrections, histogram processing, segmentation fundamentals, thresholding.

Text book:

1. Rafael C. Gonzales, Richard E. Woods, “Digital Image Processing”, Third Edition, Pearson Education, 2010.

Reference Books:

1. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, “Digital Image Processing Using MATLAB”, Third Edition Tata McGraw Hill Pvt. Ltd., 2011.
2. Anil Jain K. “Fundamentals of Digital Image Processing”, PHI Learning Pvt. Ltd., 2011.
3. William K Pratt, “Digital Image Processing”, John Willey, 2002.
4. Malay K. Pakhira, “Digital Image Processing and Pattern Recognition”, First Edition, PHI Learning Pvt. Ltd., 2011.
5. <http://eeweb.poly.edu/~onur/lectures/lectures.html>.
6. <http://www.caen.uiowa.edu/~dip/LECTURE/lecture.ht>

VLSI DESIGN

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives: Student will be introduced to

1. Use mathematical methods and circuit analysis models in analysis of CMOS digital electronics circuits, including logic components and their interconnections. Learn the various fabrication steps of NMOS, CMOS technologies
2. Apply CMOS technology-specific layout rules in the placement and routing of transistors and interconnect and to verify the functionality, timing, power and parasitic effects.
3. Learn some basic electrical properties of MOSFET and scaling models and limitations of scaling of MOS circuits.
4. Learn architectural issues and design some structured systems of MOS circuits.
5. The concepts and techniques of modern integrated circuit design and testing (CMOS VLSI). Design static CMOS combinational and sequential logic at the transistor level, including mask layout.

Course Outcomes:

By the end of the course the learners (students) will be able to

1. Apply the Concept of design rules during the layout of a circuit. Model and simulate digital
2. VLSI systems using hardware design language.
3. Synthesize digital VLSI systems from register-transfer or higher level descriptions
4. Understand current trends in semiconductor technology, and how it impacts scaling and performance.

SYLLABUS

Review of microelectronics and an introduction to MOS technology: Introduction to IC technology, MOS and related VLSI technology, NMOS, CMOS, Bi-CMOS Technologies, Production of E beam marks, I_{ds} versus V_{ds} Relationships, , Pull-up to Pull-down Ratio for NMOS inverter, Alternative forms of pull-up.

MOS and Bi-CMOS circuit design processes:

MOS layers, Stick diagrams, Design rules, and layout, 2 & 1.2 micro meter Double Metal, Double Poly. CMOS/Bi-CMOS rules, Layout diagrams, Symbolic diagrams.

Basic Circuit concepts:

Sheet resistance, Area capacitances of layers, Delay unit, Wiring Capacitances, Choice of layers.

Scaling of MOS Circuits: Scaling Models and Scaling Factors, Scaling Factors for device parameters, Limitations of scaling.

Sub system design and Layout:

Architectural issues, Switch logic, Gate Logic, Examples of Structural design (Combinational logic).

Sub system design process:

Design of ALU subsystem, Some commonly used Storage/Memory Elements, Aspects of design tools, Design for testability, Practical design for test guidelines, Built in self-test, CMOS project-an incrementer / decrementer, a comparator for two n-bit numbers. Ultra-fast systems, Technology development, MOSFET based design.

Text books:

1. Basic VLSI Design by Douglas A, Pucknell, Kamran Eshraghian, PrenticeHall, 1996, 3rd Edition.

Reference Books:

1. Mead, C.A and Conway, LA, “Introduction to VLSI Systems”, AddisonWesley, Reading, Massachusetts, 1980.
2. CMOS Digital Integrated Circuits Analysis and Design-Sung-MO kang, YusufLeblebici, Tata
3. McGraw Hill Education, 2003.

FIBER OPTIC COMMUNICATIONS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course objectives: The student will be introduced to

1. Functionality of each of the components that comprise a fiber- optic communication system
2. Properties of optical fiber that affect the performance of a communication link and types of fiber materials with their properties and the losses occur in fibers.
3. Principles of single and multi-mode optical fibers and their characteristics
4. Analyze the operation of LEDs, laser diodes, and PIN photo detectors (spectral properties, Band width, and circuits) and apply in optical systems.
5. Analyze and design optical communication and fiber optic sensor systems and analog and digital receivers.

Course outcomes: After going through this course the student will be able to

1. Choose necessary components required in modern optical communications systems.
2. Design and build optical fiber experiments in the laboratory, and learn how to calculate electromagnetic modes in waveguides, the amount of light lost going through an optical system, dispersion of optical fibers.
3. Use different types of photo detectors and optical test equipment to analyze optical fiber and light wave systems.
4. Choose the optical cables for better communication with minimum losses.
5. Design, build, and demonstrate optical fiber experiments in the laboratory.

SYLLABUS**Overview of optical fiber communication**

The general system, advantages of optical fiber communications. Optical fiber wave guides- Introduction, Ray theory transmission, Total Internal Reflection, Acceptance angle, Numerical Aperture, Skew rays, Cylindrical fibers- Modes, V-number, Mode coupling, Optical propagation through fiber modes. Step Index fibers, Graded Index fibers, Single mode fibers- Cut off wavelength, Mode Field Diameter, Effective Refractive Index, Related problems.

Transmission characteristics of optical fiber

Glass, Active glass, Plastic optical fibers. Signal distortion in optical fibers-Attenuation, Absorption, Scattering and Bending losses, Core and Cladding losses, Information capacity determination, Group delay, Types of Dispersion:- Material dispersion, Wave-guide dispersion, Polarization-Mode dispersion, Intermodal dispersion, Pulse broadening in Graded index fiber, Related problems.

Optical Sources & Detectors

LEDs, Structures, Materials, Quantum efficiency, Power, Modulation, Power bandwidth product. Injection Laser Diodes- Modes, Threshold conditions, External quantum efficiency, Laser diode rate equations, Resonant frequencies, Reliability of LED&ILD, Optical detectors- Physical principles of PIN and APD, Detector response time, Temperature effect on Avalanche gain, Comparison of Photo detectors, Related problems.

Fabrication, Cabling, Installation & Fiber connectors

Fabrication – Deposition methods. Fiber optic cables – Basic structure, Loose buffer cable, tight buffer cables, Cable classification. Installation- Classification, Procedure. Fiber connectors- Connector types, Single mode fiber connectors, Connector return loss, Fiber Splicing- Splicing techniques, Splicing single mode fibers, Fiber alignment and joint loss- Multimode fiber joints, single mode fiber joints.

Source to fiber Power launching

Output patterns, Power coupling, Power launching vs Wavelength, Equilibrium Numerical Aperture, Laser diode to fiber coupling, Optical receiver operation- Fundamental receiver operation, Digital signal transmission, error sources, Receiver configuration, Digital receiver performance, Probability of Error, Quantum limit, Analog receivers.

Optical system design – Point-to- point links- Link power budget, Rise time budget with examples, WDM, Necessity, Principles.

Text books:

1. Optical Fiber Communications – Gerd Keiser, McGraw-Hill International edition, 3rd Edition, 2000.
2. Optical Fiber Communications – John M. Senior, PHI, 2nd Edition, 2002.

Reference Books:

1. Fiber Optic Communications – D.K. Mynbaev , S.C. Gupta and Lowell L. Scheiner, Pearson Education, 2005.
2. Text Book on Optical Fiber Communication and its Applications – S.C. Gupta, PHI, 2005.
3. Fiber Optic Communications – Joseph C. Palais, 4th Edition, Pearson Education, 2004.

MICROWAVE ENGINEERING AND OPTICAL COMMUNICATION LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs	Ext. Marks	: 50
		Credits	: 2

Course Objectives: The aim of this course is to

1. Know about behaviour of Microwave Components.
2. To study the characteristics of microwave oscillators.
3. To analyze the characteristics and parameters of various microwave components.
4. To Study the radiation pattern of dipole and Yagi-uda antennas.
5. To study the performance parameters of optical source and detector and also plot the loss characteristics.

Course Outcomes: After the completion of the course, students will be able to:

1. Make use of microwave equipment.
2. Able to understand microwave measurements.
3. Measure performance of simple microwave circuits and devices.
4. Analyze the radiation patterns of antennas.
5. Assess the performance of optical devices.

LIST OF EXPERIMENTS

1. Measurement of Frequency and Guide Wavelength
2. Volt-Ampere characteristics of Gunn Diode
3. Measurement of Low VSWR and Unknown Load Impedance
4. Mode Characteristics of Reflex Klystron
5. Study of Directional Coupler Parameters
6. Measurement of losses in Optical Fiber
7. Measurement of Numerical Aperture
8. Study of Analog fiber Optic link
9. Study of Radiation pattern of Dipole Antenna in E-plane
10. Study of Radiation pattern of Dipole Antenna in H-plane
11. Study of Radiation pattern of Yagi-Uda Antenna in E-plane
12. Study of Radiation pattern of Yagi-Uda Antenna in H-plane

References:

1. Lab Manual

DIGITAL COMMUNICATION LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To know the steps involved in the analysis of digital communication systems.
2. To know how to synthesize a digital communication module with the given specifications.
3. This course gives students deep knowledge in digital communication systems at the Practical level.
4. To learn the fundamental concepts of Pulse modulation and digital modulation techniques.
5. To understand the building blocks of digital communication system.

Course Outcomes:

At the end of the course the student will

1. Be able to understand basic theories of Digital communication system in practical.
2. Be able to design and implement different modulation and demodulation techniques.
3. Be able to Perform the time and frequency domain analysis of the signals in a digital communication system.
4. Develop the skill to analyze and implement analogue to digital converters like PCM, DM.
5. Have the ability to design pass band digital modulation systems and techniques with desired specifications

LIST OF EXPERIMENTS

1. Sampling theorem Verification.
2. Pulse Amplitude Modulation (PAM) and Demodulation.
3. Pulse Width Modulation (PWM) and Demodulation.
4. Pulse Position Modulation (PPM) and Demodulation.
5. Pulse Code Modulation (PCM) and Demodulation.
6. Differential Pulse Code Modulation (DPCM) and Demodulation.
7. Delta Modulation (DM) and Demodulation.
8. Phase Shift Keying (PSK).
9. Frequency Shift Keying (FSK).
10. Analog to Digital and Digital To Analog Conversion.

References:

1. Lab Manual

PROJECT PHASE-I**Lab : 3 Periods****Sessionals : 50****Credits : 2**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Write report and present it effectively.

The phase-I of the project shall comprise of

- Problem identification in close collaboration with industry.
- Literature survey.
- Deriving work content and carry out of project requirement analysis.
- Submission of interim report.
- Presentation to an expert committee.

(Note: Sessionals 50 marks will be awarded based on
Continuous evaluation - 25 Marks
Seminar and Viva voce - 25 Marks.)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

ELECTRONICS AND COMMUNICATION ENGINEERING**II-SEMESTER**

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/ Week	Sessional Marks	Exam Marks	Total Marks
B16 EC 4201	Cellular and Mobile Communications	4	3	1	-	4	30	70	100
B16 EC 4202	Computer Networks	4	3	1	-	4	30	70	100
# ELE-III	ELECTIVE-III	4	3	1	-	4	30	70	100
B16 EC 4208	Project Phase-II	12	-	-	9	9	50	100	150
	Total	24	9	3	9	21	140	310	450

# ELE-III	B16 EC 4203	Internet of things.
	B16 EC 4204	Digital System design Through HDL
	B16 EC 4205	Bio Medical Signal Processing.
	B16 EC 4206	Satellite Communication
	B16 EC 4207	Digital TV

CELLULAR AND MOBILE COMMUNICATIONS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To make students familiar with fundamentals of mobile communication systems.
2. To understand and identify the problems and services of mobile communication systems.
3. To understand multiple access techniques (TDMA/FDMA/CDMA etc.) to reduce the interference effect in mobile communications.
4. To understand the basic implementations of GSM system.
5. To have an insight into the various propagation models and different path loss models.

Course Outcomes:

1. Students are able to understand the fundamentals of mobile communication systems.
2. Students are able to identify the problems and there remedies in wireless mobile communications.
3. Students are able to analyze multiuser systems with the help of different multiplexing techniques.
4. Students are able to understand the basics of GSM mobile communication standard, its architecture.
5. Students are able to understand the various mobile propagation channel models and path loss models.

SYLLABUS

Introduction to Mobile and Cellular Communication Systems: Introduction to wireless communications, examples of wireless communication systems, the cellular concept and system design fundamentals.

Elements of Cellular Radio Systems and Handoff Technologies:

Frequency reuse, Channel assignment strategies, Handoff strategies, Interference and system capacity, Trunk and grade services, Methods for improving coverage and capacity in cellular system.

Multiple Access Techniques:

Multiple access techniques for wireless communications FDMA, TDMA, Spread Spectrum techniques, SDMA, Packet Radio, CSMA, capacity of Cellular CDMA with multiple cells and capacity of SDMA.

GSM:

Wireless systems and standards, AMPS, IS-94, GSM traffic, Examples of GSM cell, Frame structure of GSM cell, details of forward and reverse CDMA channels.

Mobile Radio Propagation:

Introduction to mobile radio propagation, free space propagation models, Large scale path loss, Reflection, Diffraction, Scattering, Outdoor and Indoor propagation models.

Text Books:

1. Wireless Communications Principles and Practice, Second Edition, THEODORE S.RAPPAPORT.
2. Wireless and Cellular Communications by WILLIAM.C.Y.LEE

Reference Books:

1. Wireless digital Communications, DR. KAMILO FEHER.
2. Electronic Communication system, WAYNE TOMASI.
3. Wireless Communications, SANJAY SHARMA.

COMPUTER NETWORKS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To familiarize with the fundamental concepts of computer networking and network
2. Reference models.
3. To introduce basic concepts of analog and digital transmission techniques, switching techniques.
4. To understand error control and flow control mechanisms.
5. To familiarize with different multiple access protocols such as ALOHA, CSMA.
6. To familiarize with different networking devices and congestion control algorithms.
7. To familiarize with TCP and UDP header formats.

Course Outcomes:

Upon completion of the course, students will be able to

1. Explain basic computer network principles and layers of the OSI model and TCP/IP.
2. Explain the concepts of transmission media, switching and multiplexing techniques.
3. Explain and analyse the error control and flow control methods.
4. Explain different multiple access control protocols and IEEE standards for LANs and MANs.
5. Identify the different types of connecting devices and explain the basic concepts of congestion control algorithms and internetworking.
6. Explain TCP and UDP header formats.

SYLLABUS

Uses of Computer Networks, Line Configuration, Topology, Transmission mode, Categories of Networks-LAN, MAN, WAN; Network Software- Protocol Hierarchies, Design issues of layers, Connection Oriented and Connectionless services; Reference Models- The OSI Reference Model, The TCP/IP Reference Model, The B-ISDN ATM Reference Model.

Theoretical basis for Data communication, Transmission media- Guided and Unguided Transmission media; The Telephone System-Structure of Telephone system, Trunks and Multiplexing, Frequency Division Multiplexing, Time Division Multiplexing, Switching- Circuit

Switching, The Switch Hierarchy, Crossbar switches, Space Division Switches, Time Division Switches; Narrow band ISDN, Broadband ISDN and ATM- Virtual Circuits versus Circuit Switching.

DATA LINK LAYER

Design issues, Error Detection and Correction, Elementary Data link protocols, Sliding window protocols, HDLC, **Medium access sub layer**-The Channel allocation problem, Multiple Access Protocols-ALOHA, Carrier Sense Multiple Access protocols; IEEE standard for 802 LANs, Satellite Networks

NETWORK LAYER

Design considerations, Difference between Gateways, Ethernet switch, Router, Hub, Repeater, Congestion Control algorithms- General principles of Congestion Control, Congestion prevention policies. The Leaky bucket algorithm and Token bucket algorithm, The Network Layer in the Internet- The IP Protocol, IP Addresses.

TRANSPORT LAYER

The Transport layer Service, Elements of Transport protocols, The Internet Transport Protocols- UDP, TCP.

APPLICATION LAYER

The Domain Name System, Electronic mail, The World Wide Web.

Text Books:

1. Data Communications and Networking by Behrouz A. Forouzan, 2nd edition, Tata McGraw Hill.
2. Computer Networks — Andrew S Tanenbaum, 3rd Edition, Pearson Education/PHI.

Reference Books:

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education
2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson

INTERNET OF THINGS (IOT) (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To assess the vision and introduction about IoT
2. To understand IoT architecture and application perspective
4. To understand how the objects interact with people, with information system and with
2. other objects
3. To implement hands-on project development to find innovative solutions
4. To implement data and management to use of devices in IoT

Course Outcomes

1. Interpret the vision of IoT from a global context
2. Determine the Iot Architecture and application perspective
3. Identifying and describing different kinds of Internet-connected product concepts.
4. Analyzing, designing, and developing prototypes models of Internet-connected products using various tools.
5. Understanding the challenges and applying right techniques for user-interaction with connected-objects.

SYLLABUS

Introduction

IoT overview, The IoT paradigm, Smart objects, IoT Platforms (like Aurdino, ARM Cortex, Raspberry Pi / Intel Galileo), Bits and atoms, Convergence of Technologies.

Introduction to Internet and web networking basics: HTTP, Rest, JSON, XML, Interfacing to Cloud Harnessing mobile computing for IoT

Introduction to Technologies behind IoT:

RFID, NFC, Mobil Data Technologies (GPRS, 3G, 4G), Wifi, Powering the IoT using low power wireless technologies like Bluetooth smart technology, Zigbee, WSN. RTLS + GPS, Agents and Multi agent systems.

IoT Architecture:

Machine to Machine, Web of Things, IoT protocols (The Layering concepts ,IoT Communication Pattern, IoT protocol Architecture, The 6LoWPAN - IPv6 over Low power Wireless Personal Area Networks)

IoT Applications and issues:

Combination scenarios, Breaking assumptions, IoT in retail, IoT in healthcare, IoT in manufacturing.

Prototyping Connected Objects: Open source prototype platforms, Arduino based internet communication. Integrating and accessing Internet services, Raspberry PI / Beagle board based Gateways, Data Analysis Techniques.

Case Studies:

Case studies from Industry for different verticals like Retail, Healthcare, Home automation etc

Text Books :

1. 6LoWPAN: The Wireless Embedded Internet, Zach Shelby, Carsten Bormann, Wiley
2. Internet of Things: Converging Technologies for Smart Environments and Integrated
3. Ecosystems, Dr.OvidiuVermesan, Dr. Peter Friess, River Publishers
4. Building the Internet of Things. Sara Cordoba, WimerHazenber, Menno Huisman. BIS Publishers. 2011.

Reference Books:

1. Internet of Things: A Hands-On Approach By Arshdeep Bahga, Vijay Madisetti
2. Fundamentals of Wireless Sensor Networks: Theory and Practice By Waltenegus Dargie, Christian Poellabaue

DIGITAL SYSTEM DESIGN THROUGH HDL
(Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Learn basic digital design paradigms and the necessary Verilog HDL constructs that would help them to build combinational & sequential logic circuits and run simulations using CAD tools.
2. Design functional units including adders, multipliers, ROMs, SRAMs, and PLAs
3. Testing of logic circuits using different design techniques.

Course Outcomes:

Upon completion of the course, students will be able to

1. To understand and design complex digital systems at several level of abstractions.
2. To create circuits that realizes specified digital functions.
3. To identify logic and technology-specific parameters to control the functionality
4. To design and model complex digital system
5. To verify several digital circuits using different techniques.

SYLLABUS

Digital Logic Design using VHDL

Introduction, designing with VHDL, design entry methods, logic synthesis , entities , architecture, packages and configurations, types of models: dataflow , behavioural , structural, signals vs. variables, generics, data types, concurrent vs. sequential statements , loops and program controls.

Combinational Logic Circuit Design using VHDL

Combinational circuits building blocks: Multiplexers, Decoders, Encoders, Code converters, Arithmetic comparison circuits , VHDL for combinational circuits , Adders-Half Adder, Full Adder, Ripple-Carry Adder, Carry Look-Ahead Adder, Subtraction, Multiplication.Sequential Logic Circuit: Design using VHDLFlip-flops, registers & counters, synchronous sequential circuits: Basic design steps, Mealy Statemodel, Design of FSM using CAD tools, Serial Adder Example.

Digital Logic Design using Verilog HDL

Introduction, Verilog Data types and Operators, Binary data manipulation, Combinational andSequential logic design, Structural Models of Combinational Logic, Logic Simulation,DesignVerification and Test Methodology, Propagation Delay, Truth Table models using Verilog.

Digital Logic Circuit Design Examples using Verilog HDL

Behavioral modeling , Data types, Boolean-Equation-Based behavioral models of combinational logics , Propagation delay and continuous assignments , latches and level-sensitive circuits in Verilog, Cyclic behavioral models of flip-flops and latches and Edge detection, comparison of styles for behavioral model; Behavioral model, Multiplexers, Encoders and Decoders, Counters, Shift Registers, Register files, Dataflow models of a linear feedback shift register.

Testing of Digital Logic Circuit Design

Machines with multi cycle operations, ASM and ASMD charts for behavioural modeling. Testing of logic circuits, fault model, complexity of a test set, path-sensitization, circuits with tree structure, random tests, Testing of sequential circuits, Built in self test.

Text books:

1. Stephen Brown & Zvonko Vranesic, "Fundamentals of Digital logic design with VHDL", Tata McGraw Hill, 2nd edition.
2. Michael D. Ciletti, "Advanced digital design with the Verilog HDL", Eastern economy edition, PHI.

Reference books:

1. Stephen Brown & Zvonko Vranesic, "Fundamentals of Digital logic with Verilog design", Tata McGraw Hill, 2nd edition.
2. J. Bhaskar, "VHDL Primer", 3rd Edition, PHI Publications.
3. Ian Grout, "Digital systems design with FPGAs and CPLDs", Elsevier Publications.

BIO MEDICAL SIGNAL PROCESSING (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives: The objectives of this course are to:

1. Describe the origin, properties and suitable models of important biological signal such as ECG and EEG.
2. Introduce students to basic signal processing techniques in analyzing biological signals.
3. Develop the mathematical and computational skills relevant to the field of biomedical signal processing.
4. Develop thorough understanding on basics of ECG signal compression algorithms.
5. Increase the student's awareness of the complexity of various biological phenomena and cultivate an understanding of the promises, challenges of the biomedical engineering.

Course outcomes: At the end of the course, students will be able to:

1. Possess the basic mathematical skills necessary to analyse ecg and eeg signals.
2. Possess the basic scientific skills necessary to analyse ecg and eeg signals.
3. Possess the basic computational skills necessary to analyse ecg and eeg signals.
4. Apply classical and modern filtering and compression techniques for ecg and eeg signals.
5. Develop a thorough understanding on basics of ecg and eeg feature extraction.

SYLLABUS

Introduction to Biomedical Signals

The nature of Biomedical Signals, Examples of Biomedical Signals, Objectives and difficulties in biomedical analysis, Electrocardiography: Basic electrocardiography, ECG lead systems, ECG signal characteristics, Simple signal conversion systems, Conversion requirements for biomedical signals, Signal conversion circuits

Signal Averaging and Adaptive Noise cancelling

Basics of signal averaging, signal averaging as a digital filter, a typical averager, software for signal averaging, limitations of signal averaging. Principal noise canceller model, 60-Hz adaptive cancelling using a sine wave model, other applications of adaptive filtering

Data Compression Techniques

Turning point algorithm, AZTEC algorithm, Fan algorithm, Huffman coding, data reduction algorithms The Fourier transform, Correlation, Convolution, Power spectrum estimation, Frequency domain analysis of the ECG

Cardiological signal processing

Basic Electrocardiography, ECG data acquisition, ECG lead system, ECG signal characteristics (parameters and their estimation), Analog filters, ECG amplifier, and QRS detector, Power spectrum of the ECG, Band pass filtering techniques, Differentiation techniques, Template matching techniques, A QRS detection algorithm, Real-time ECG processing algorithm, ECG interpretation, ST segment analyzer, Portable arrhythmia monitor

Neurological signal processing and Analysis of EEG signals

The brain and its potentials, the electrophysiological origin of brain waves, The EEG signal and its characteristics (EEG rhythms, waves, and transients), Correlation, Detection of EEG rhythms, Template matching for EEG, spike and wave detection

Text Books:

1. Biomedical Digital Signal Processing- Willis J. Tompkins, PHI2001.
2. Biomedical Signal Processing Principles and Techniques- D C Reddy, McGraw- Hill publications2005

Reference Books:

1. Biomedical Signal Analysis- Rangaraj M. Rangayyan, John Wiley & Sons 2002.

SATELLITE COMMUNICATION
(Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives: The student will be introduced to

1. Functionality of KEPLAR'S laws planetary motion.
2. Be aware of space segment equipment.
3. To know the Principles of deploying earth station. Understand various parameters of link design.
4. Analyse the various multiple access techniques

Course outcomes: After going through this course the student will be able to

1. Choose necessary components required in modern satellite communications systems.
2. Design and build space segment, depending upon the requirement.
3. Design link margin for various applications.
4. Choose the correct multiple access technique for better communication with minimum losses.
5. Design, build, and demonstrate satellite communication link in the laboratory.

SYLLABUS

SATELLITE ORBITS

Kepler's Laws, orbital parameters, orbital perturbations, station keeping, geo stationary and non-Geo-stationary orbits – Look Angle Determination- Limits of visibility, eclipse-Sub satellite point, Sun transit outage-Launching Procedures - launch vehicles and propulsion.

SPACE SEGMENT

Spacecraft Technology- Structure, Primary power, Attitude and Orbit control, Thermal control and Propulsion, communication Payload and supporting subsystems, Telemetry, Tracking and command.

EARTH SEGMENT

The earth station - HPA – Downlink – Output back off – Satellite TWTA output – Effects of rain – Uplink rain– Fade margin – Downlink rain – Fade margin – Combined uplink and downlink C/N ratio .

SATELLITE ACCESS

Modulation and Multiplexing: Voice, Data, Video, Analog – digital transmission system, Digital video Broadcast, multiple access: FDMA, TDMA, CDMA, Assignment Methods, Spread Spectrum communication.

SATELLITE APPLICATIONS

INTELSAT Series, INSAT, VSAT, Mobile satellite services: GSM, GPS, LEO, MEO, Satellite Navigational System. Direct Broadcast satellites (DBS)- Direct to home Broadcast (DTH), Digital audio broadcast (DAB)- World space services.

Text books:

1. Satellite Communication, by Timothy Pratt, Charles Bostian, Jeremy Allnutt(Second Edition), John Wiley & Sons.
2. Satellite Technology, Principles and Applications, by Anil K. Maini, Varsha Agarwal(Second Edition), Wiley.

Reference Books:

1. Satellite Communications, by Dennis Roddy(Fourth edition),McGraw Hill.
2. Satellite Communication Systems Engineering, by Wilbur L. Pritchard, Henri G. Suyderhoud, Robert A. Nelson (Second Edition), Pearson

DIGITAL TV (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives: The student will be introduced to

1. Functionality of video principles.
2. Be aware applications of TV transport system.
3. To know need for migration to digital TV.
4. Analyse the various video compression techniques
5. Understand different video formats.

Course outcomes: After going through this course the student will be able to

1. Choose necessary components required in modern digital TV systems.
2. Design a TV transport system.
3. Design necessary formats for various applications.
4. Choose the correct compression format of available.
5. Design, build, and demonstrate digital TV transmission in the laboratory.

SYLLABUS

Fundamentals of video principals:

Principals of color vision, the CIE color system, applications of visual properties, Essential video system characteristics, the principals of video compression.

Digital TV standards:

DTV audio encoding and decoding, DTV transport system, DTV satellite transmission, DTV data broad casting.

Advanced Television (ATV) Concepts:

Why the industry is moving to DTV, Standardization efforts towards a single Standard, the ATV emergence, the digital solution, interoperability, flexibility.

DTV picture formats:

HDTV formats, Data multiplexing, HANC multiplexing.

Digital signal compression:

video compression standards, video data structure hierarchy, JPEG and motion JPEG schemes, MPEG-1, MPEG-2 video schemes.

Text books:

1. Digital TV fundamentals, by Michael Robin and Michael Poulin (second edition), McGrawHill.
2. Digital Television: A Practical Guide for Engineers, Walter Fischer and H. vonRenouard, Springer-Verlag, 2004

Reference Books:

1. R. R. Gulati, Modern Television Practice, Principles, Technology and servicing, , 2nd edition, New Age International Publishers, 2001.
2. 2. Gerald w. Collins, Fundamentals of Digital Television Transmission', John Wiley, 2001.

PROJECT PHASE-II**LAB : 9 Periods**

Sessionals	: 50
Ext. Marks	: 100
Credits	: 12

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Analyze, design and develop a technology/ process.
3. Implement and evaluate the technology at the laboratory level.
4. Write report and present it effectively.

The phase-II of the project shall consists of

Implementing, Testing and validation.
Report Writing.

Sessionals (50 Marks) will be awarded by the Project Guide based on continuous evaluation. External Evaluation (100 marks) of project report and viva voce will be conducted by a committee consisting of HOD, Guide and External Examiner.

May be carried out using in-house facilities or in an industry by specified number of students in a group.

Format for Preparation of Project Thesis for B. Tech:

1. Arrangement Of Contents: The sequence in which the project report material should be arranged and bound should be as follows:
 1. Cover Page & Title Page .
 2. Bonafide Certificate
 3. Abstract.
 4. Table of Contents
 5. List of Tables
 6. List of Figures
 7. List of Symbols, Abbreviations and Nomenclature
 8. Chapters
 9. Appendices
 10. References

*The table and figures shall be introduced in the appropriate places.